Automation - G Code Command Description

匯出日期：2025-09-27
修改日期：2023-06-23

1 G00： Rapid Linear Positioning

1.1 Command Form

G00 [P1] X_ Y_ Z_ [F1=_] [Q=_];

X、Y、Z: specified position.

P1: active constant speed command

F1: feedrate mm/min or inch/min

Q： Overlapping distance

1.2 Description

Each axles move to appointed point without interpolation, X、Y、Z are the coordinate of end position, depends on G90/G91 to move absolute or increment value.

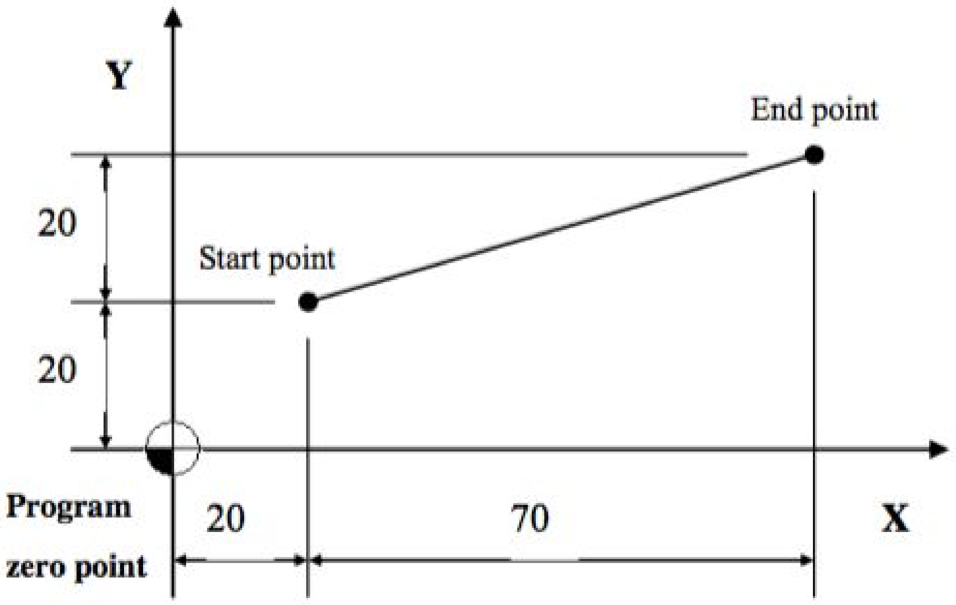
1.3 Notice

1. The movement mode can decide by parameter #411 (0: linear, 1: each axle move in max speed independently)
2. Constant feedrate is legally with both P1 and F1 argument are determined. If P1 argument is missing, system wouldn't refer F1 argument as moving speed. Moving speed is refer to Pr461~Pr480.
3. Pr411=0, Resultant speed is limited by F1 and Pr461~Pr480 if executing with P1, F1. Each axis speed is limited by Pr461~Pr480.
4. R18 Rapid Traverse Override is still valid with P1 F1=_.
   The actually highest speed wouldn't exceed Pr461~Pr480 or F1, while the override is 0~100%.
   The actually highest speed wouldn't exceed Pr461~Pr480 over five times, while the override is F0(PR3207=2, R18=1).
5. G00 command doesn't support G10 L1100 P1002 R_.
6. G70/G71 supported unit of F1 can be mm/min or inch/min.
7. Unit of F1 can be mm/min, but always remains G94. G93/G95 is invalid.
8. F1 argument support version: 10.118.12 and after.
9. When using the G00 command for axial overlapping, the front and rear blocks need to be all G00 commands or G00 command followed by G01 command or G01 command followed by G00 command.

1.4 Example

1.4.1 Example1:

PIC:



Program description:

1. first way(absolute): G90 G00 X90.0 Y40.0;
   //use difference value between appointed point and zero point to do straight interpolation to appointed point.
2. second way(increment): G91 G00 X70.0 Y20.0;
   // use difference value between appointed point and initial point to do straight interpolation to appointed point.

1.4.2 **Example2:**

Pr411 = 0、Pr461 = Pr462 = 10000

```
G71
G0 P1 X500. Y500. F1=12000.
G1 X0. Y0.
M30
```

Both axis movement are identical, the moving speed should be $\sqrt{10000^2+10000^2}=14142.1$ without F1=_.
Hence, G0 P1 F1=12000 is applied, the limit is 12000.


CNC Monitor v10.118.22F
G54
TEST2.NC N0 L2
加工監控
2019/10/3 13:58:18
管理者
絕對座標
X 259.803
Y 259.803
Z 0.000
C 0.000
剩餘距離
X 240.197
Y 240.197
Z 0.000
C 0.000
G碼狀態 G0
G17 G90 G94
G71 G40 G49
加工時間 0 : 0 : 1
累計加工 0 : 0 : 55
G00 倍率 100 %
G01 倍率 100 %
主軸倍率 100 %
F mm/min
12000.0
S
1000 RPM
總工件數 0
工件數 0
T 0 D 0 H 0
起始單節 2
G71
G0 P1 X500. Y500. F1=12000.
G0 X0. Y0.
M30
加工中..
自動執行
警報
載入程式編輯
圖形模擬顯示
MDI 輸入
加工資訊設定
磨耗設定
手輪偏置啟動
加工記錄表單
清除累計時間

2 G01: Linear Interpolation, Cutting Feed

2.1 Command Form

G01 X_Y_Z_ F__;

X、Y、Z: Specified point

F: Feed rate
In G94 mode, unit is mm/min(inch/min) <- default value of milling system boot
In G95 mode, unit is mm/rev(inch/rev) <- default value of turning system boot

2.2 Description

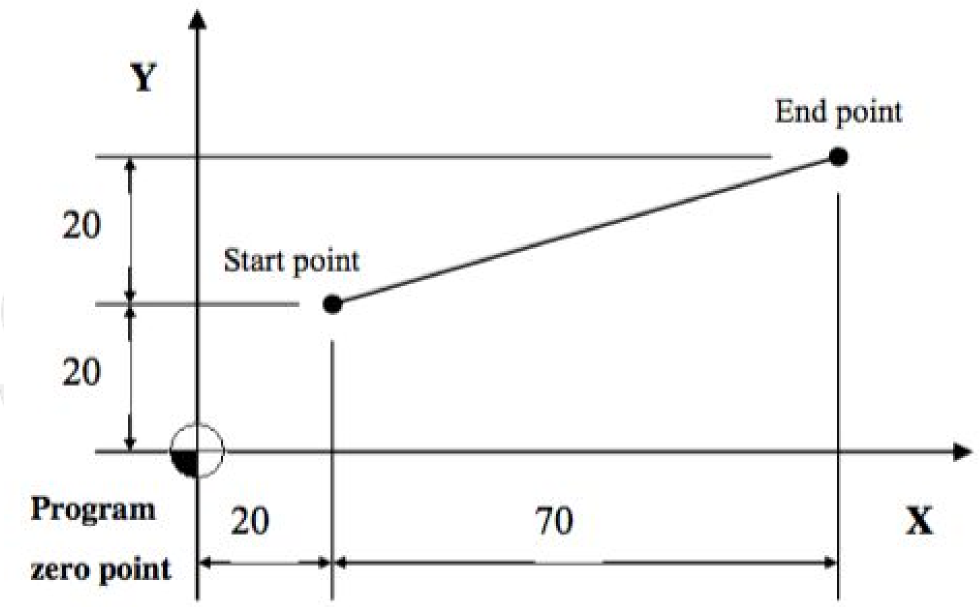
G01 executes linear interpolation, it can be used G90/G91 to decide absolute or increment mode, use feed rate provided by **F** to go to the specified position.

2.3 Notice

- The max. feed rate of G01 is defined by PR405-maximum cutting feed rate or (PR621~PR640)-each axis maximum cutting feed rate
- Default value F: 1000mm/min(inch/min) for G94 mode and 1.mm/rev(inch/rev) for G95 mode
- Default mode G94/G95 can be changed by parameter Pr3836 (reboot controller to activate setting).

2.4 Example 1

PIC:

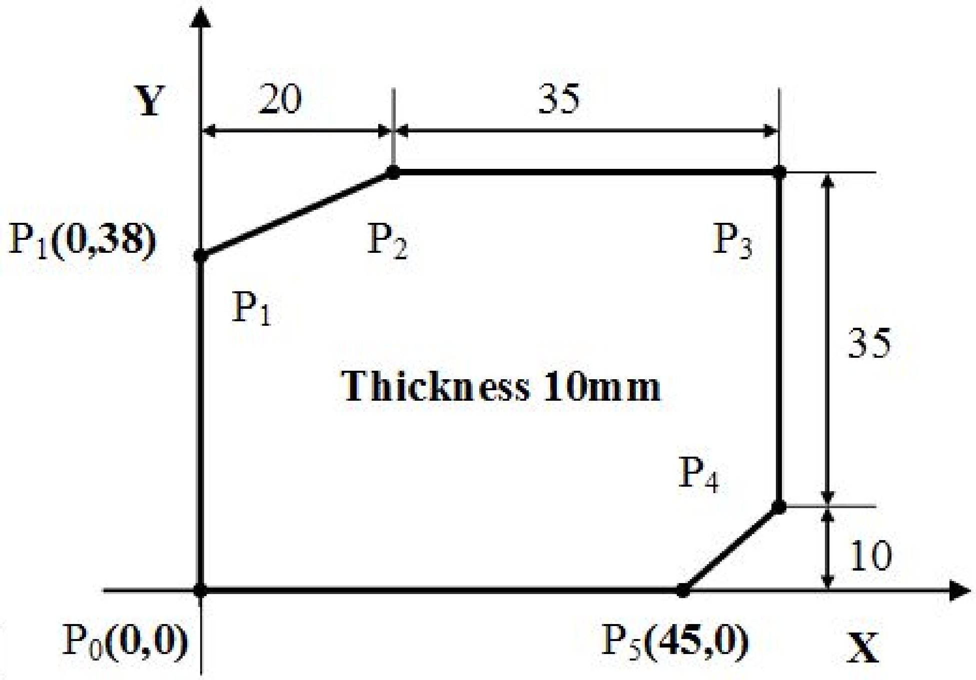


Program description:

1. Method 1 (Absolute value): G90 G01 X90.0 Y40.0;
   //Move to the specified point with the program zero pint as the relative coordinate
2. Method 2 (Increment value): G91 G01 X70.0 Y20.0;
   // Move to the specified point by the difference between the specified point and the starting point

2.5 **Example 2**

PIC:



Program description:

1. absolute way:
   N001 G00 X0.0 Y0.0 Z10.0;//positioning to above of P_0
   N002 G90 G01 Z-10.0 F1000;
   //straight interpolation to bottom of workpiece, feed rate is 1000mm/min
   N003Y38.0;//P_0 -> P_1
   N004X20.0 Y45.0; //P_1-> P_2
   N005X55.0;//P_2 -> P_3
   N006Y10.0;//P_3 -> P_4
   N007X45.0 Y0.0;//P_4 -> P_5
   N008X0.0;//P_5 -> P_0
   N009 G00 Z10.0;//positioning back to above of P_0
   N010 M30;//program ends

2. increment way:

```
N001 G00 X0.0 Y0.0 Z10.0;//positioning to above of P0
N002 G91 G01 Z-20.0 F1000;
//straight interpolation to bottom of workpiece, feed rate is 1000mm/min
N003Y38.0;//P0 -> P1
N004X20.0 Y7.0;//P1 -> P2
N005X35.0;//P2 -> P3
N006Y-35.0;//P3 -> P4
N007X-10.0 Y-10.0;//P4 -> P5
N008X-45.0;//P5 -> P0
N009 G00 Z20.0;//positioning back to above of P0
N011 M30;//program ends
```

3 G02/G03 : Helical Interpolation

3.1 Command Form

1.

$$G17 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} X\_ \; Y\_ \begin{Bmatrix} R\_ \\ I\_ \; J\_ \end{Bmatrix} Z\_ \; F\_;$$

X、Y: end position of arc;
Z: end position of straight line;
R: radius of arc;
I、J: vector from the starting point to the center of the circle;
F: feedrate;

2.

$$G18 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} X\_ \; Z\_ \begin{Bmatrix} R\_ \\ I\_ \; K\_ \end{Bmatrix} Y\_ \; F\_;$$

X、Z: end position of arc;
Y: end position of straight line;
R: radius of arc;
I、K: vector from the starting point to the center of the circle
F: feedrate;

3.

$$G19 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} Y\_ \; Z\_ \begin{Bmatrix} R\_ \\ J\_ \; K\_ \end{Bmatrix} X\_ \; F\_;$$

Y、Z: end position of arc;
X: end position of straight line;
R: radius of arc;
J、K: vector from the starting point to the center of the circle;
F: feed rate;

3.2 Description

When the 3rd axis is moving vertical to arc plane, G02/G03 is to be helical interpolation. The way to define arc plane of helical interpolation is the same as circular interpolation. Helical interpolation uses G code(G17/G18/G19) to define which plane to do circular interpolation.

G17 mode: The X-Y plane is a circular interpolation plane, and the Z-axis is linear interpolation axis.

G18 mode: The Z-X plane is a circular interpolation plane, and the Y-axis is linear interpolation axis.

G19 mode: The Y-Z plane is a circular interpolation plane, and the X-axis is linear interpolation axis.

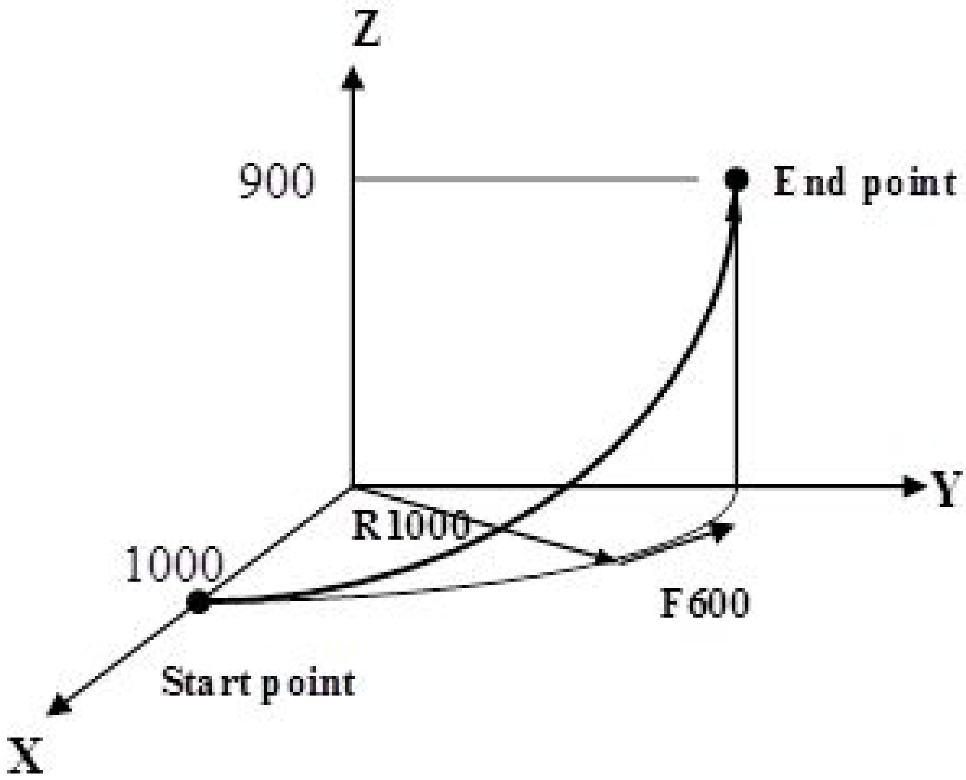
3.3 Notice

1. When G02/G03 without any R, I, J, or K argument, the block will be executed as G01.

2. When X, Y, Z, I, J, K and R argument given by G02/G03 is out of order. For example, in G17 mode, the K argument is not zero, the system would issue a [COR-006 arc end is not on the arc] alarm. This alarm can adjust the region through Pr3807.

3.4 Example

PIC:



Program description:

G17 G03 X0.0 Y1000.0 R1000.0 Z90.0 F600;

// X-Y plane arc, counterclockwise direction (CCW), Z-axis linear interpolation
// Cutting feedrate 600mm/min for spiral cutting

4 G02/G03： CW/CCW Circular Interpolation

4.1 Command Form

1. X-Y plane circular interpolation:

$$G17 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} X\_ \ Y\_ \begin{Bmatrix} R\_ \\ I\_ \ J\_ \end{Bmatrix} F\_;$$

2. Z-X plane circular interpolation:

$$G18 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} X\_ \ Z\_ \begin{Bmatrix} R\_ \\ I\_ \ K\_ \end{Bmatrix} F\_;$$

3. Y-Z plane circular interpolation:

$$G19 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} Y\_ \ Z\_ \begin{Bmatrix} R\_ \\ J\_ \ K\_ \end{Bmatrix} F\_;$$

X, Y, Z: End point position

I, J, K: the vector value that starting point of arc to the center of a circle(center of a circle—starting point)

R: Radius of arc

F: Feed rate

G90/G91 decide absolute or increment

Description

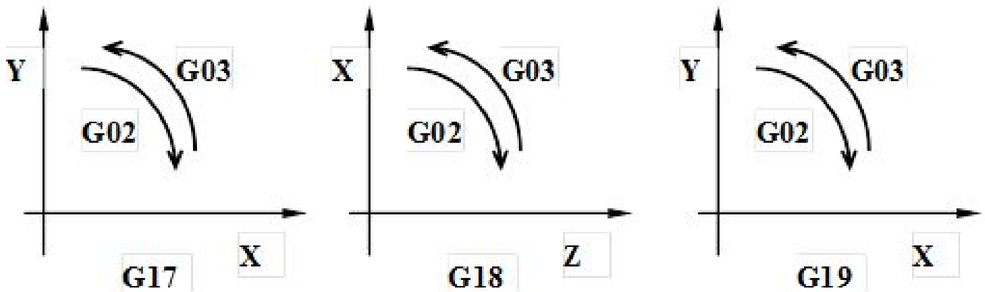
G02、G03 do circular interpolation according to appointed plane、coordinate system、size of arc and speed of interpolation, and the rotate direction decide by G02(CW)、G03(CCW). Description of the five command format as below:

Setting Data		Command	Definition
1	Plane selection	G17	X-Y plane setting
		G18	X-Z plane setting
		G19	Y-Z plane setting
2	Direction	G02	Clockwise direction (CW)
		G03	Counterclockwise direction (CCW)

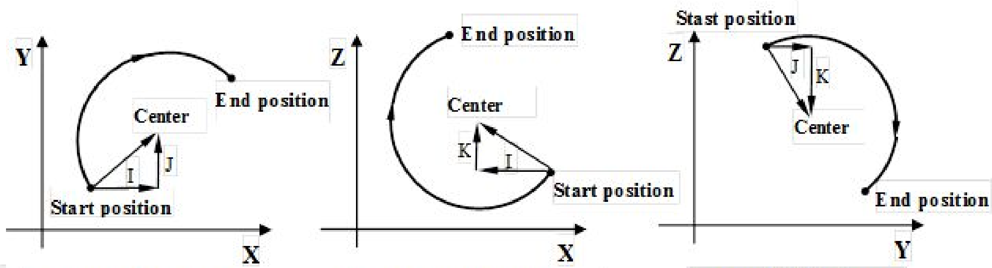
3	End position	G90	Two axes of X, Y, Z	End coordinate of arc
		G91	Two axes of X, Y, Z	Vector value from start point to end point
4	Distance from start point to center of circle		Two axes of I, J, K	Vector value from start of arc to center of circle
	Radius of arc		R	Radius of arc
5	Speed of feed (feedrate)		F	Feedrate along the arc

4.2 Notice

1. G02, G03 direction



2. I, J, K definition:



a. arc of X-Y plane
b. arc of Z-X plane
c. arc of Y-Z plane

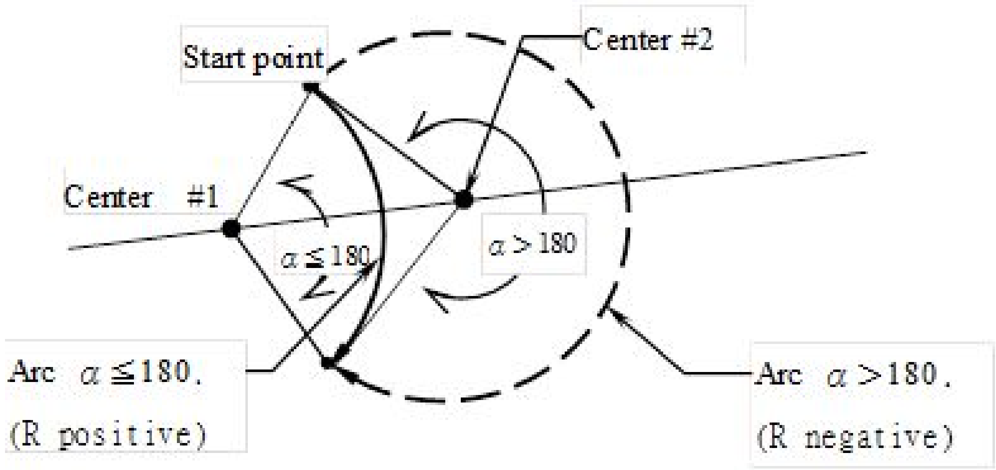
3. how to use R:
   a. When θ≦180 degree, R is positive.

$$\left\{\begin{matrix} G02 \\ G03 \end{matrix}\right\} X\_ \ Y\_ \ R25.0;$$

   b. When 180 degree < θ < 360 degree, R is negative.

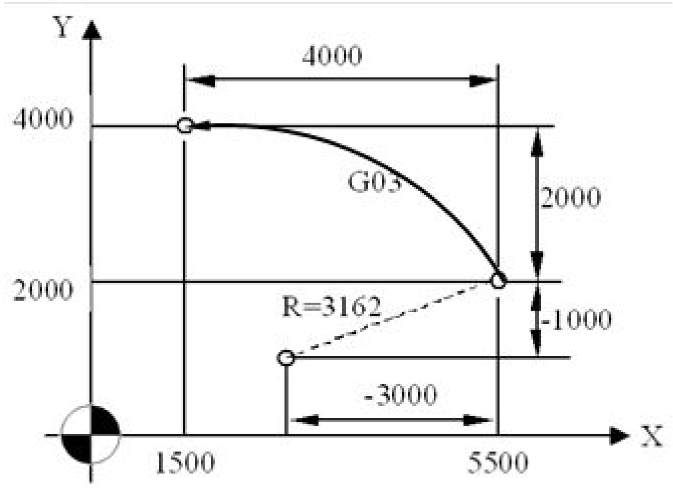
$$\left\{\begin{matrix} G02 \\ G03 \end{matrix}\right\} X\_ \ Y\_ \ R-25.0;$$

c. When θ=360 degree, only use **I**、**J**、**K**.



4.3 Example

4.3.1 Example Program 1

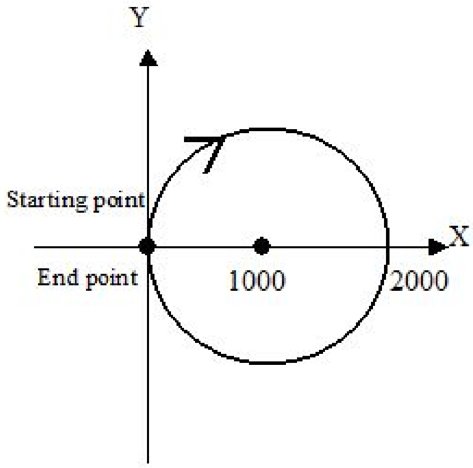


G90 G00 X5500 Y2000;//positioning to start point of arc

G17 G90 G03 X1500 Y4000 I-3000 J-1000 F200;//absolute command

(G17 G91 G03 X-4000 Y2000 I-3000 J-1000 F200;//increment command)

4.3.2 Example Program 2 (Full circular interpolation)




```
G90 G00 X0 Y0;
G02 I1000 F100; //interpolate a full circle
```

5 G02/G03： Spiral Interpolation, Conical Interpolation

5.1 Command Form

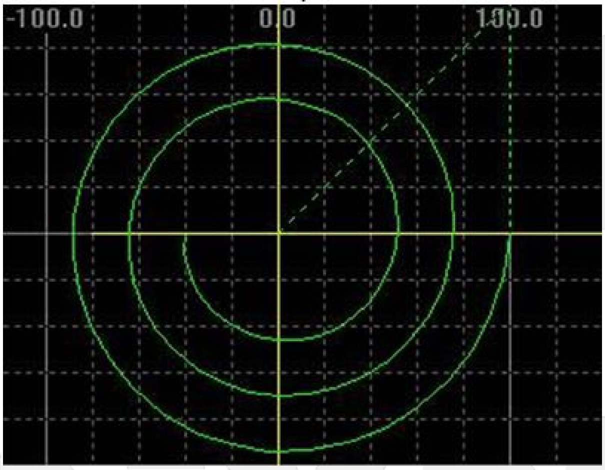
G17 G02/G03 X_ Y_ I_ J_ L_ F_;
G18 G02/G03 Z_ X_ K_ I_ L_ F_;
G19 G02/G03 Y_ Z_ J_ K_ L_ F_;
X_ Y_ Z_： end point position;
I_ J_ K_： vector from the starting point to the center of circle;
L_： number of turns
F： feedrate;

5.2 Description

The spiral interpolation syntax is similar to the circular interpolation G02/G03. The only difference is that there is one more argument L, turns of circle. L is an integer, and a part of the circle is regarded as one turn. When the G02/G03 command has the number of L turns, it is regarded as the Spiral Interpolation. The starting radius and the ending radius are allowed to be different, and the alarm that the arc end point is not at the arc will not be issued. For Spiral Interpolation, when the vertical axis command is added, it is conical interpolation.

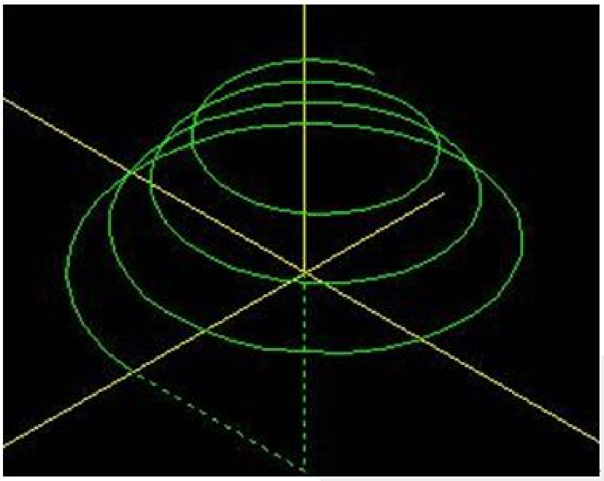
5.3 Example

G17 G00 X100. Y0 Z0;
G02 X-40. I-100. L3;



G17 G00 X100. Y0 Z0;

G02 X-40. I-100. Z80. L4;



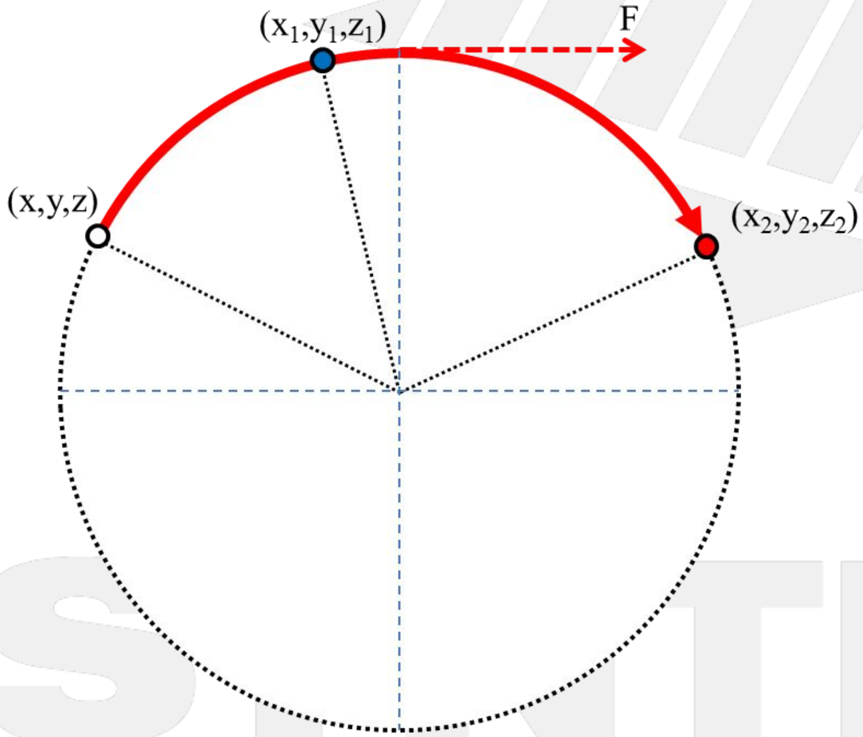
6 G02.4/G03.4： Three Point Circular Interpolation

6.1 Command Form

$$\left\{ \begin{array}{l} G02.4 \\ G03.4 \end{array} \right\} X1\_ \quad Y1\_ \quad Z1\_ \quad \alpha1\_ \quad \beta1\_ \quad F\_ ; \quad \text{// 第一程序段(圓弧中間點)}$$

$$X2\_ \quad Y2\_ \quad Z2\_ \quad \alpha2\_ \quad \beta2\_ ; \quad \text{// 第二程序段(圓弧終點)}$$

X1_ Y1_ Z1_α1_β1_： First sequence(the mid-point of arc);
X2_ Y2_ Z2_α2_β2_： Second sequence(the end point of arc);
F： Arc tangent rate;
α、β： The axial direction other than the Y and Z axis can be not specified if not required
X、Y、Z： If any of the axial specified point coordinates is omitted, it will follow the same value as the previous specified point.



(x,y,z)： Initial point of arc(the and point of previous block)
(x_1,y_1,z_1)： the mid-point of arc
(x_2,y_2,z_2)： the end point of arc
F： Arc tangent rate

6.2 Description

G02.4/G03.4 three-point circular interpolation, in order to utilize the three points given in the space, through the geometric relationship calculation, a circular interpolation function that can be connected according to the three-point sequence is obtained. When using, you only need to give the arc intermediate point and the end point coordinate value (G02.4 or G03.4 the end point of the previous block is the starting point of the arc), plus the arc tangent rate, you can draw a three-point arc in a space.

6.3 Notice

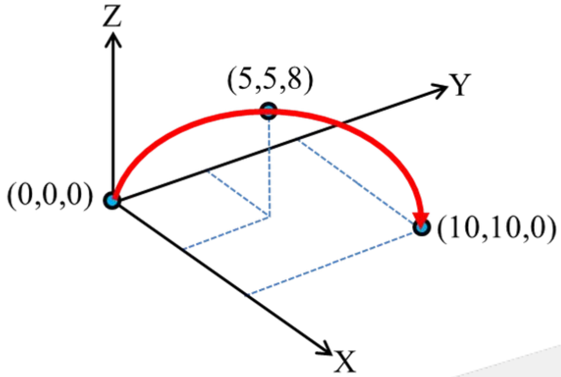
1. The valid version is 10.116.16B (inclusive) and needs to start Option19. If using G02.4/G03.4 when the G02.4/G03.4 is not enabled, the COR-100 alarm will be triggered (this model does not support this G code command).
2. The G02.4 command has the same action as the G03.4 command and can be replaced with each other and supports absolute/incremental commands.
3. The end point of the previous block of G02.4/G03.4 is the starting point of this arc.
4. The G02.4/G03.4 command is regarded as a group of two lines, which can be specified continuously. The end point of the previous arc will become the starting point of the next arc, but the F argument can only be in the odd line. If the total number of lines in the command is odd or the number of lines in the F-number is even, the COR-134 alarm will be triggered (G02.4/G03.4 command format error).
5. If the three points forming the arc are on the same line or if any two points coincide (for example, if the whole circle is specified, the end point will be the same), the interpolation will be performed in the linear interpolation mode (G01). The interpolation path starts from the start point to the middle point, and then from the middle point to the end point.
6. When a single block is executed, it will move from the start point to the end point of the arc when starting a cycle, and will not stop at the middle point of the arc.
7. When using this function, the tool radius compensation function must be turned off first, and commands such as A, C, and R are not supported. Otherwise, the COR-133 alarm will be triggered (this command is not supported in G02.4/G03.4 interpolation mode).
8. In G02.4/G03.4 interpolation mode, G53 mechanical coordinate positioning cannot be used; nor can the G02.4/G03.4 command be connected to the next block after G53 mechanical coordinate positioning. Both of the above situations will cause the arc path to go wrong.
9. If the α/β command is omitted in the first block and only specified in the second block, the unspecified α/β axis does not move from the start point to the intermediate point of the arc, but moving from the intermediate point to the end point, the α/β axis also moves to the specified position.
10. As mentioned above, if the α/β command is omitted in the second block and only specified in the first block, the α/β axis will move to the specified position when moving from the starting point of the arc to the intermediate point. When the middle point of the arc reaches the end point, the unspecified α/β axis does not move.

6.4 Example

6.4.1 Example 1:

G90 G01 X0 Y0 Z0 F5000
G02.4 X5. Y5. Z8. F1000

X10. Y10. Z0.



6.4.2 Example 2：

G90 G01 X0 Y0 Z0 F5000
G02.4 X5. Y5. F1000
Z10.

7 G04.1: Path Synchronization Waiting

7.1 Command Form

G04.1 P_ [Q_]

P: Number of the signal waited
Q: Enter the CNC path to wait for each other. If the Q argument is not specified, it means that all CNC paths wait for each other.

Q argument format description:

1. The Q argument specifies the path to wait for each other. Currently, up to 4 paths are supported. The argument of the Q for each path are listed below:

Path ID	Q Argument(decimal number)
1	1
2	2
3	3
4	4

2. For example, four paths belong to CNC main system path(Pr731=4), if 1st, 2nd, and 4th paths are waiting for each other, the Q argument must be composed of 1, 2, and 4, for example, Q124.
3. Number of the order behind Q argument is no limitation. The following six combinations mean that the 1st, 2nd, and 4th paths have to wait for each other:
Q124, Q142, Q241, Q214, Q412, Q421
4. When the waiting signal P is the same, once the path corresponding to the number of Q does exist, although the Q argument of different paths are out of order, the waiting is also valid; the following conditions all represent the same Q argument, and the axis groups 1, 2, 4 do wait for each other:
    a. Processing program of 1st path G04.1 P2 Q124
    b. Processing program of 2nd path G04.1 P2 Q241
    c. Processing program of 3th path G04.1 P2 Q412

7.2 Description

- For synchronization between cross-axis groups, G04.1 is realizable. For example, refer to the third example of the program. Change the main spindle speed of $2 at $1. If you want to change the feedrate which is corresponding to $2 in G95 mode, you should use G04.1 to stop the program pre-solution in $1 and $2 respectively. The group performs a status update, avoiding the use of the old spindle speed by $2, resulting in a feedrate error.
- Taking the dual program as an example, G04.1 P1 [Q12] in the first path and G04.1 P1 [Q12] in the second path would wait for each other until the synchronization then continues to the next block.
- Similarly, G04.1 P2 [Q12] in the first path and G04.1 P2 [Q12] in the second path will wait for each other until the synchronization then continues to the next block; the rest and so on .
- In the processing program of each path, the number of G04.1 with the same Q argument must be the same (including without Q argument), and the numbers following P must be used in order from small to large.

$1	$2	$3
G00 X0. **G04.1 P1** G01 X10. F1000 G04.1 P2 Q13 X20. G04.1 P3 Q13 X30. G04.1 P5 Q13 X40. **G04.1 P6** M30	G00 Y0. **G04.1 P1** G01. Y10. F1000 Y15. *G04.1 P4 Q23* Y20. **G04.1 P6** M99	G00 Z0. **G04.1 P1** G01 Z10. F1000 G04.1 P2 Q13 Z25. G04.1 P3 Q13 Z40. *G04.1 P4 Q23* Z55. G04.1 P5 Q13 Z70. **G04.1 P6** M99

Refer from the table above,The quantity of G04.1 without Q argument (all paths) in each path is 2, G04.1 with Q13 is 3 at first and third path and G04.1 with Q23 is 1 at the second and third path.

- When you need to automatically re-process several workpieces, please program M99 at the end of the first path program. However, please note that if you want each group program to be processed synchronously, you must program the same G04.1 P code before M99 in each path, such as G04.1 P6 at the end of $1~$3 program above.

7.3 Notice

1. When the following conditions occur, the alarm "COR-137 G04.1 P argument is in the wrong order" would be issued.
    a. When Q argument is not ordered, P argument is different.
    b. When Q arguments are the same, P arguments are different.
    c. When P arguments are the same, Q arguments are different.
2. When the following conditions occur, the alarm "COR–144 G04.1 Q argument is incorrect" would be issued.
    a. Q argument is not a positive integer.
    b. The path specified by the Q argument does not exist.
    c. Q argument specified that the path does not include the current path. For example, program G04.1 P1 Q23 in the processing program of the first path.
3. In order to avoid compatibility changes, G04.1 retains the specification without Q argument, which means that all main system paths are specified.
4. The processing program of the non-CNC main system path and the subprograms referenced by it do not support G04.1.
5. When the G04.1 wait function is executed, the system is considered to be in processing. The example below (showed the first path is processing)

Environment： C40 On, G-code execute one line each cycle start
Description： The $1 and $2 single-line execution ends, and the first path is single-block stop state.

Environment： C40 On, G-code execute one line each cycle start

Description： $1 waits for $2, the first path is in processing.. state.

7.4 Example

7.4.1 Program 1：

$1	$2
G04.1 P1; G01 X50. F2000; G04.1 P2; Z100.; G04.1 P3; X0.; G04.1 P4; Z0.; G04.1 P5; M99;	G04.1 P1; G01 X250. F3000; G04.1 P2; Z500.; G04.1 P3; X0; G04.1 P4; Z0; G04.1 P5; M99;

7.4.2 Program 2：

$1	$2	$3

G04.1 P1 Q12; // $1 and $2 synchronize
G01 X50. F2000;
Z100.;
X0.;
Z0.;
G04.1 P2; // All axis groups are synchronized and repeat processing
M99;
	G04.1 P1 Q12; // $1 and $2 synchronize
	G01 X25. F3000;
	Z50.;
	X0.;
	Z0.;
	G04.1 P2; // All axis groups are synchronized and repeat processing
	M99;
		G00 X10. Z10.;
		G04 X1.;
		X0. Z0.;
		G04 X1.;
		
		G04.1 P2; // All axis groups are synchronized and repeat processing
		M99;

7.4.3 Program 3： After two spindle are synchronized, the dual system performs the outer diameter cutting. Please pay attention to the relationship between the dual system G04.1P_ and the basic spindle S argument. If the sequence is placed incorrectly, the F argument of the second system will be worse than expected.

$1	$2
G04.1 P1 // Synchronize with $2 M03 S30 G114.1 R0 // Start spindle synchronization G04.1 P2 // Synchronize with $2 G01 U10. 1. U-10. G04.1 P3 // Synchronize with $2. Wait for the completion of $2 cutting to change the spindle speed M03 S60 // Changing the first spindle speed during spindle synchronization G04.1 P4 // Synchronize with $2 G04.1 P5 // Synchronize with $2. Wait for the completion of $2 cutting to release the spindle synchronization G113 // Release spindle synchronization M05 G04.1 P6 // Synchronize with $2. Avoid $1 executing M30 too fast, result $2 is not finished M30	G04.1 P1 // Synchronize with $1. Avoid $2 M99 back to the stall to continue execution M13 S15 G04.1 P2 // Synchronize with $1. The second spindle speed is synchronized to 30 RPM. G01 U10. F2. // Under G95, feedrate is 30*2 = 60 mm/min U-10. G04.1 P3 // Synchronize with $1 G04.1 P4 // Synchronize with $1. The second spindle speed is synchronized to 60 RPM G01 U10. // Under G95, feedrate is 60*2 = 120 mm/min U-10. G04.1 P5 // Synchronize with $1 M15 G04.1 P6 // Synchronize with $1 M99

8 G04： Dwell

8.1 Command Form

$$G04 \begin{Bmatrix} X\_ \\ P\_ \end{Bmatrix};$$

X: dwell time (With decimal point in seconds; no decimal point, in milliseconds. Use range: 0.001 ~ 9999.999 seconds)

P: dwell time (Decimal point not permitted)

8.2 Description

When performing some machining strokes where delay are required (taper pits, column pits, fisheye pits, milling corners), We can use the G04 function. To make the spindle rotate normally, and each axis pauses for a period of time to make the hole depth accurate or get a right angle before switching to the next block to achieve the desired precision.

Example

G04 X2500;//delay 2.5 sec

G04 X2.5;//delay 2.5 sec

G04 P2500;//delay 2.5 sec

G04 P2.5;//delay **2** sec (decimal point not permitted)

9 G09 : Exact Stop Check

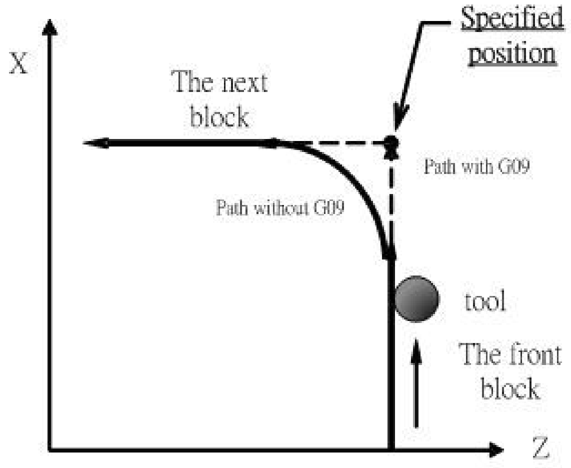
9.1 Command Form

G09 X__ Z__
X, Z: specified corner position

Description

When machining through a corner, machining tolerance can cause round corner instead of ideal sharp corner due to excessive cutting speed or servo delay. However, when perfectly sharp accuracy is required, the G09 function can be used to decelerate the tool when approaching corner until it reaches a certain range (within the wide range set by the parameter), and then the next block of command will be executed.

9.2 Illustration



10 G10： Programmable Data Input

10.1 Command Form

$$G10 \begin{Bmatrix} L10 \\ L11 \\ L12 \\ L13 \end{Bmatrix} P\_ \quad R\_ \quad ;$$

用於刀具長（H）幾何補正量
用於刀具長（H）磨耗補正量
用於刀具徑（D）幾何補正量
用於刀具徑（D）磨耗補正量

P: tool NO. ;

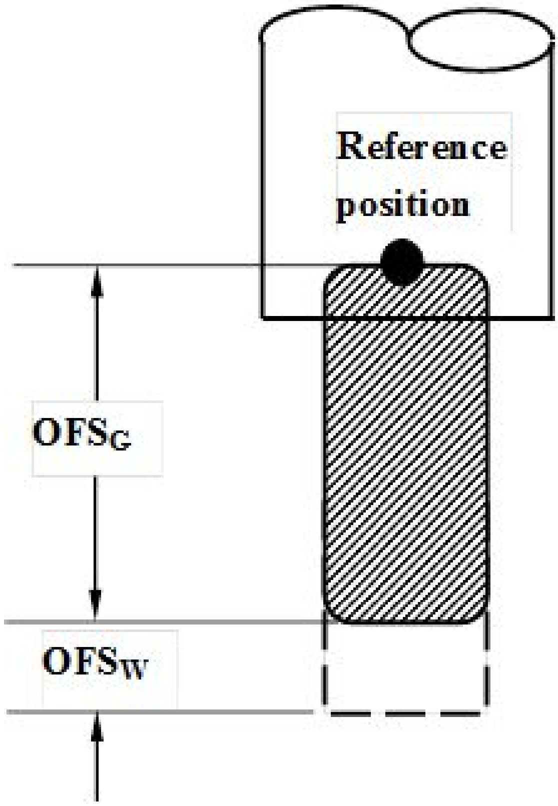
R: compensation value(data of tool length or tool diameter) ;

10.2 Description

1. G10 command can directly use program command to enter tool compensation value.
2. In absolute mode (G90), value of G10 is the new compensation value; in increment mode (G91), value of G10 is the sum of the value of the moment with the new compensation value.
3. This syntax is only available for Pr3816=0/1, inputting a single-axis compensation value.
4. PLC axis control components (PLC Axis) are not supported.

10.3 Example

PIC:


Reference
position
OFS_G
OFS_W

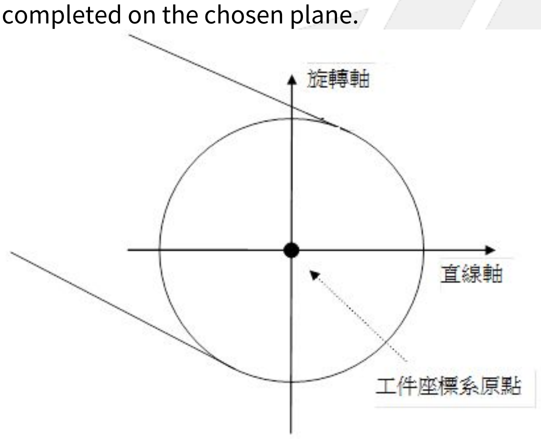
11 G12.1/G13.1： Activate/Deactivate Polar Coordinate Interpolation

Command Form

G12.1 X *C*; // enable polar coordinate interpolation function；
… // (linear or circular interpolation command in Cartesian coordinate system,
… // Cartesian coordinate system is composed of linear axes and rotary axes)
G13.1; // disable polar coordinate interpolation function
X： The X offset from the rotation center to program zero.
C： The C offset from the rotation center to program zero.

11.1 Description

1. The polar coordinate interpolation function is to transform the contour control command in Cartesian coordinate system to a linear axis exercise (tool movements) and a rotary axis exercise (workpiece movements).
2. Polar coordinate interpolation plane. Enable the polar coordinate interpolation function with G12.1 and choose a polar coordinate interpolation plane (shown as below). The polar coordinate interpolation will be completed on the chosen plane.



3. After G12.1, the absolute coordinate of C axis shows negative C axis offset; the absolute coordinate of X is affected by its diameter axis setup, below are the details：
    a. X as a radius axis and applies the radius axis programming in polar coordinate interpolation. The absolute coordinate of X shows the location after minus the X axis offset.
    G0 X50. C90. // absolute coordinate X = 50, C =90
    G12.1 X10. C5. // absolute coordinate X = 50 - 10 = 40, C = 0 - 5 = -5
    G13.1 // absolute coordinate X = 50, C = 90
    b. X as a diameter axis, it'll apply the radius axis programming in the polar coordinate interpolation. The absolute coordinate of X axis will be the location of original coordinate divided by 2 and minus the X axis offset.
    G0 X50. C90. // absolute coordinate X = 50, C =90
    G12.1 X10. C5. // absolute coordinate X = 50/2 - 10 = 15, C = 0 - 5 = -5
    G13.1 // absolute coordinate X = 50, C = 90

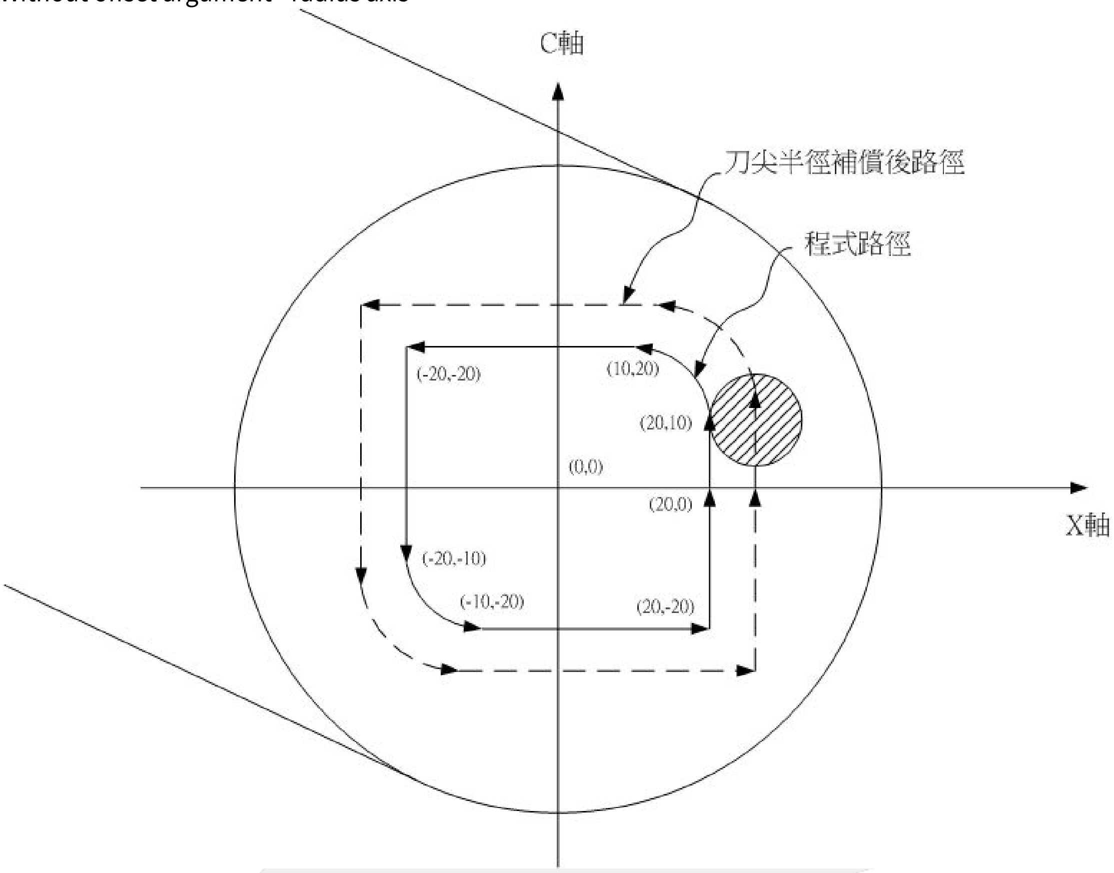
11.2 Notice

1. The offset argument is valid from version 10.116.11。

2. The polar coordinate interpolation function will be disable after reboot or system reset.
3. G codes below are available in polar coordinate interpolation：
   G01 linear interpolation
   G02/G03 circular interpolation (IJR argument is the same as normal syntax)
   G04 dwell
   G40/G41/G42 cutter radius compensation
   G65/G66/G67 call macro
4. After enable polar coordinate interpolation function, the plane selection(G17/G18/G19)(G17, if not customized) will be disabled and specified to G12.1 working plane. After the polar coordinate interpolation system is disable or reset, the system will return to the original working plane specified before the polar coordinate interpolation function is enable.
5. After enable polar coordinate interpolation function, it's unable to change the coordinate system (G50/G52/G53/G54~G59).
6. The polar coordinate interpolation function can't be enable or disable with cutter radius compensation function(G41/G42), it's only applicable when cutter radius compensation function is disable(G40).
7. After enable polar interpolation system, if needed to enable the cutter radius compensation function (G41/G42), please program an extra tool carrying block with 0 movement to ensure the contour correctness.
8. In polar coordinate interpolation mode, it's unable to select the preview mode (PR3815=1) for cutter radius compensation.
9. Program Restart： Do not restart the program section of G12.1,， or it might lead to contour error.
10. After switched to polar coordinate, the actions will be planned with hypothetical 0 degree at the instant C axis direction. Therefore, please execute the C axis orientation before executing G12.1 to make sure the continuing feeding angle is the same. (example below)
11. Currently not supporting the function with Z axis not activated, or it might lead to G02/G03 contour error.
12. The polar interpolation function can't be applied with five-axis cutter radius compensation(G43.4/G43.5).
13. In polar coordinate interpolation mode (G12.1), do not apply diameter/radius axis programming switch command (G10.9), or alarm COR-325 will be issued.
14. If given G10.9 X_ command first then enable the polar coordinate interpolation function (G12.1), the X axis will be using radius axis programming. After deactivated the polar coordinate interpolation function (G13.1), the X axis will return to the initial setup value, if wanted to return to the G10.9 setup value X, please specified G10.9 X_ again.
15. If the tool starts moving from the mechanism center of C axis, the moving way might be separated into 2 stages: first the C axis orientation then the X axis movements; it's a proper action, not asynchronous between each axes. The cause of the 2-stage action is the non-linear mechanisms transform of polar coordinate interpolation, a singular point exists at the mechanism center of C axis and the moving performance will be special at the point.
16. This feature doesn't support Multi-Axis Multi-Signal Skip Function(G31.10/G31.11).

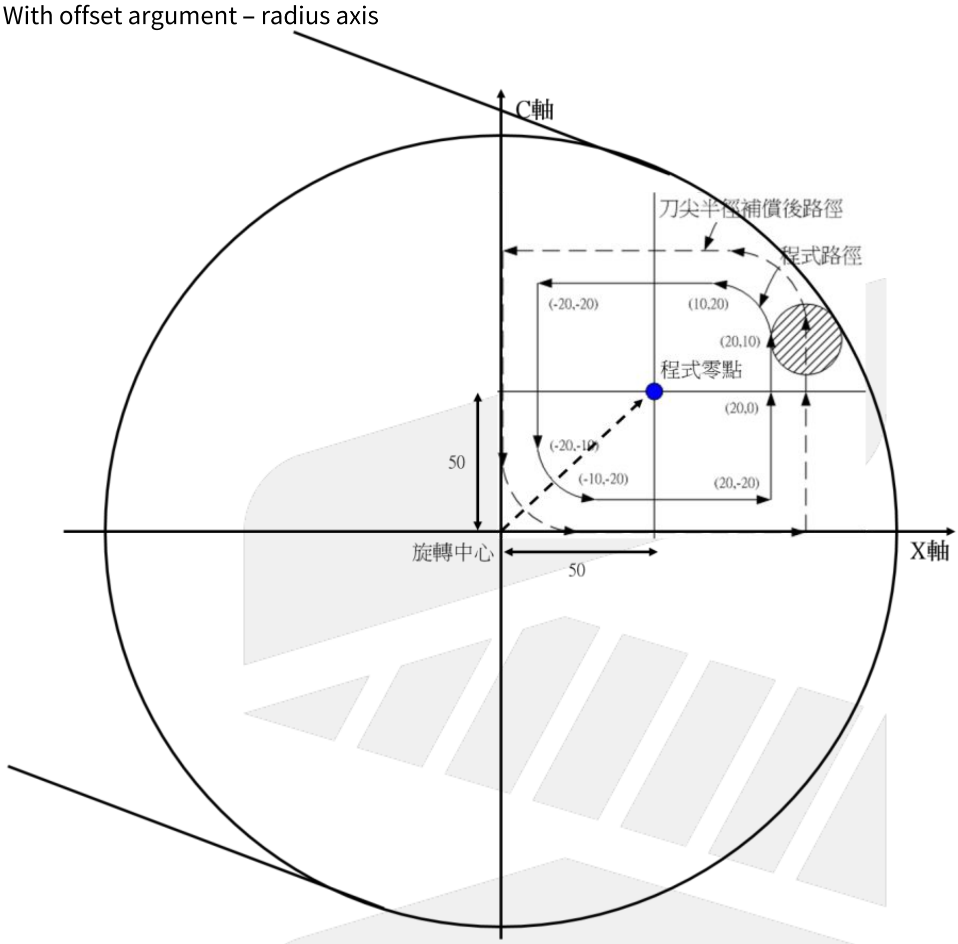
11.3 **Example**

1. Without offset argument - radius axis




```
T0101;
G00 X110. C0. Z0.; // move to the orientation point
G40 G94;
G12.1;// enable the polar coordinate interpolation
//programming with the X-C plane of Cartesian coordinate
G42 X55. ;// add an extra block with o movement
G01 X20. F100.;
C10.;
G03 X10. C20. R10.;
G01 X-20.;
C-10.;
G03 X-10. C-20. R10.;
G01 X20.;
C0;
G40 X55.;
G13.1;// disable the polar coordinate interpolation
M30
```

2. With offset argument – radius axis




```
T0101
G00 X110. C0. Z0.; // move to orrientation point
G40 G94;
G12.1 X50. C50.;// enable the polar coordinate interpolation, eccentric coordinate(50, 50)
//programming with the X-C plane of Cartesian coordinate
G42 X55.;// add a block with 0 movement
G01 X20. F100.;
C10.;
G03 X10. C20. R10.;
G01 X-20.;
C-10.;
G03 X-10. C-20. R10.;
G01 X20.;
C0;
G40 X55.;
G13.1;// disable the coordinate interpolation
M30
```

Appendix

1. Specified random axis(Customized G12.1) + cutter radius compensation
   For G12.1, X axis is the default linear axis and C axis is the default rotary axis. Due to different machine

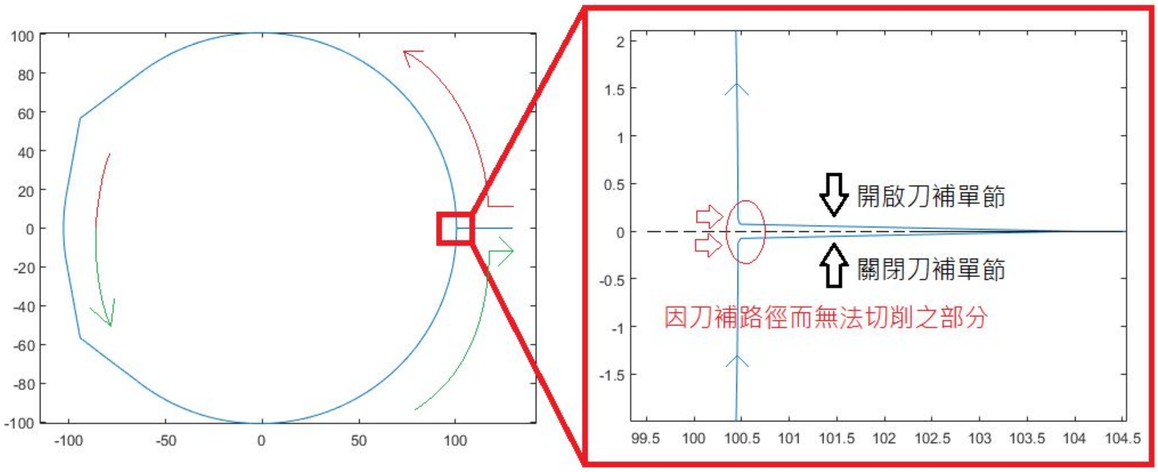
configurations, sometimes it might need to change the linear axis to Y axis, for the situation customized G12.1 is needed.

```
G10 L1301 X_ C_ R_;
X_ linear axis ID
C_ rotary axis ID
R_ 1 : Enable / 0 : Disable
```

Example (customized G012001, Y as linear axis & C as rotary axis) :

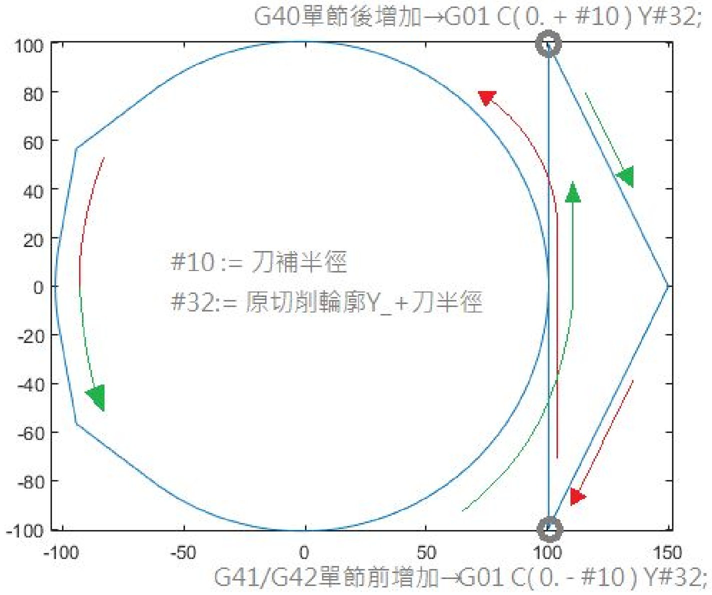
```
%@MACRO
IF (#1012<>40) THEN
   ALARM( 17 );
END_IF;
IF (#1018=96) THEN
   ALARM( 18 );
END_IF;

#30:=AXID(Y); // get Y axis ID
#31:=AXID(C); // get C axis ID
// get Y axis diameter/radius programming before G12.1 enable
#34 := ROUND( POW( 2, #30 ) );
#35 := #1814 AND #34;

IF (#25 = #0) THEN
   #25 := 0;
END_IF;
IF (#3 = #0) THEN
   #3 := 0;
END_IF;
IF ((#30=#0) OR (#30<=0) OR (#31=#0) OR (#31<=0)) THEN
   ALARM( 19 );
   M99;
END_IF;

// State Backup
#32:=#1004;
#2048:=#1002;
#2049:=#1008;
// When applying arc commands, cutter radius compensation or polar coordinate commands, please set the cutting plane with G17, G18, G19.
G91 G19 Y0 C0;
G94;
G90 G10 L1301 X#30 C#31 I#25 J#3 R1;  // Enable polar coordinate interpolation mode and assign to Y-C
IF ( #35 = 0 ) THEN
   Y( #1412 - #25 ) C-#3;
ELSE
   Y( #1412 / 2.0 - #25 ) C-#3;
END_IF;

WAIT();
G#32;
M99;
```

When enable tool compensation under polar coordinate interpolation, some area near the tool entry point might be skipped if set the cutting contour directly. (please refer to the picture below)



Suggests to design the tool carrying block in the machining file so the tool path can cut in along with the contour. Assume that both the start and finish position commands of the cutting contour are C0., then the simplest organization will be:
Add one moving block before G41/G42 and after G40 respectively to make approaching the cutting contour after cutter radius compensation. The contour of C axis should move at least the distance of cutter radius according to the contour direction (the direction is decided by the original contour direction).
Please refer to the example below for further details, Y-C polar coordinate + cutter radius compensation (tool carrying block included)

Example (using customized G12.1 + cutter radius compensation)

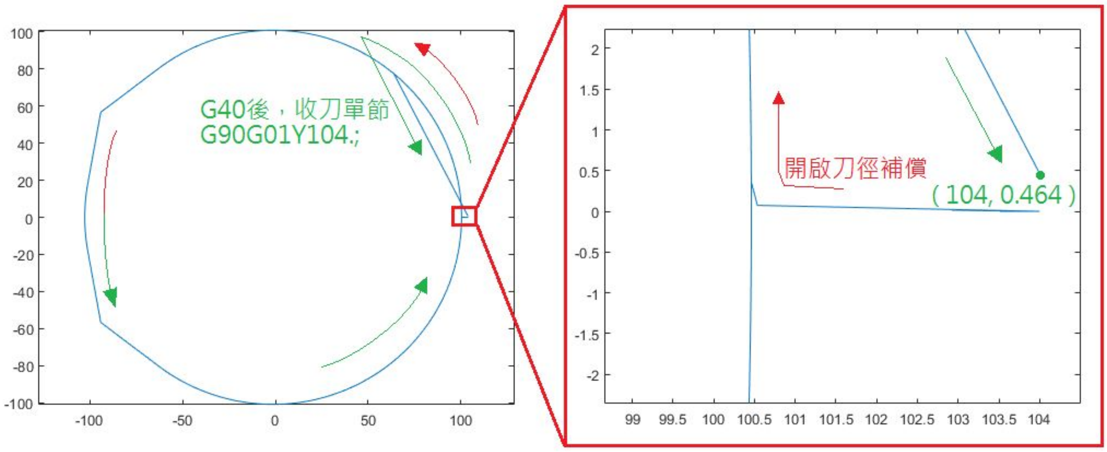


%@MACRO // the orange parts can be replaced by the blue MACRO codes
G90 G00X0.Y150. C0. Z0. F1500 ;
M03 S3000;
M08;

```
G40;

G90 G10 L12 P13 R99.86;   // #10 := 99.86; cutter radius 99.86
                          // G90 G10 L12 P13 R#10; set the cutter radius of tool No.13
G12.1;                     // enable customized Y-C polar coordinate interpolation

G90G01 F1600;
G01 C-99.86 Y100.462;      // #32 := 0.602 + #10; cutting contour + cutter radius, 99.86 + 0.602 = 100.462
                          // #G01 C( 0. - #10 ) Y#32; add a tool carrying block, Y_ moves to 100.462 and C moves
the distance of cutter radius
G42 D13 C0. Y0.602;
// cutting contour
// ====================
G03 C0.464 Y0.383 R0.602;
G03 C0.536 Y-2.952 R2.69;
G03 C-0.536 Y-2.952 R3.0;
G03 C-0.464 Y0.383 R2.69;
G03 C0. Y0.602 R0.602;
// ====================
G40;
G01 C99.86 Y100.462;      // G01 C( 0. + #10 ) Y#32; add a tool retracting block, C_ moves the distance of
cutter radius
                          // the tool retracting block is still affected by the tool compensation specifications,
should be written before G13.1
// ====================
G90 G01 C0. Y150.;         // back to orientation point
G13.1;                     // deactivate polar coordinate interpolation
M30;
```

※Note： During the tool compensation process, please not to give action commands with working plane declaration, it might trigger a temporary pause of tool compensation action and leads to the wrong moving path.

2. Overcut caused by tool retracting path
   For the problem mentioned in point 3 that some area near the tool entry point might be skipped, if wants to move the path to C≠0 with planning repeated paths, please be careful with the path of Y-C during the tool retract. In the example, the command G90 G01 Y104 is given after G40.. For instinct, the contour will be moved to 104. along Y axis, but since it's the tool retracting block and no command about C is given, C will be moving according to the command given by the previous block (G03 C0.464 Y0.383 R0.602;), and the position after disable the tool compensation will be at (104.0, 0.464). From the picture below, it's able to see the overcut caused by tool retracting path moving across the cutting contour. If wants the path to move out along Y direction, the block should be placed before G40. But please note that it's also involved with the turning of tool compensation path so the tool compensation might be functioning improperly (excessive tool radius alarm in this example). Please refer to point 3 for path modification methods.




```
%@MACRO
G90 G01X0.Y104.C0.Z0.F1500 ;
M03 S3000;
M08;
G40;

G90 G10 L12 P13 R99.86;    // set up the tool radius of tool No.13
G12.1;

G90G01 F1600;
G42D13 C0. Y0.602 ;
// interpolation contour
// =====================
G03 C0.464 Y0.383 R0.602;
G03 C0.536 Y-2.952 R2.69;
G03 C-0.536 Y-2.952 R3.0;
G03 C-0.464 Y0.383 R2.69;
G03 C0. Y0.602 R0.602;
G03 C0.464 Y0.383 R0.602;  // repeated contour, cutting the skipped parts due to cutter radius compensation.
// =====================
G40;                        // disable tool compensation
G90 G01 Y104.;                // tool retracting block, coordinate of C not specified
                           // retracted to (104.0, 0.464) and caused the overcut
// =====================
G13.1;                      // disable polar coordinate interpolation
M30;
```

12 G15/G16: Polar Coordinates Command Mode

12.1 Command Form

```
G16;          //Start polar coordinate mode
G __ X _Y _   }
  :           } //Polar coordinate command
  :           }
G15;          //Cancel polar coordinate command
```

X: polar coordinate radius

Y: polar coordinate angle(“ + ” for CW, ”—” for CCW)

12.2 Description

Enable polar coordinate mode in first line, G16 for polar coordinate command enable, G15 for polar coordinate command disable, it can use polar coordinate mode to enter position(radius and angle), G90/G91 can specify in it.

First argument is radius, second argument is angle.

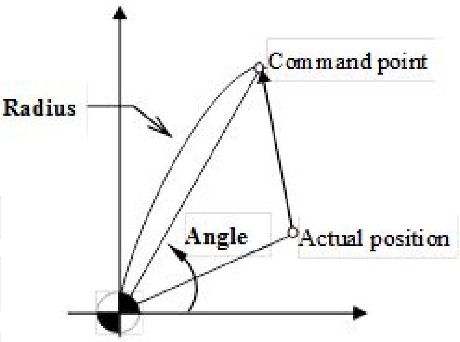
Absolute or incremental mode is specified by the G90 (absolute) command or the G91 (incremental) command.

In absolute mode, the radius or angle increases relative to the origin point.

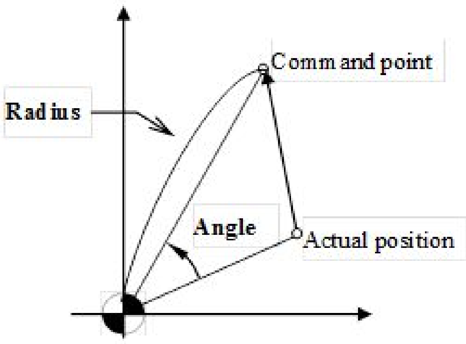
In incremental mode, the radius increases relative to the last commanded point, and the angle increases relative to the last angle.

12.3 Notice

1. when polar coordinate zero point is the same as working coordinate system

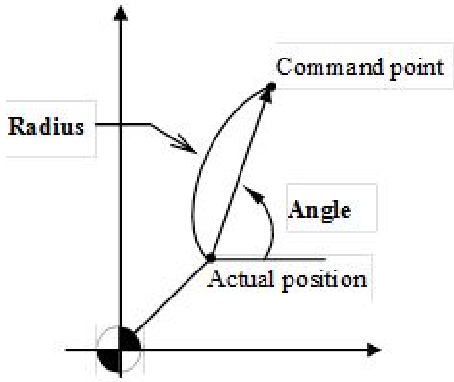


a. When angle is specified with an absolute command

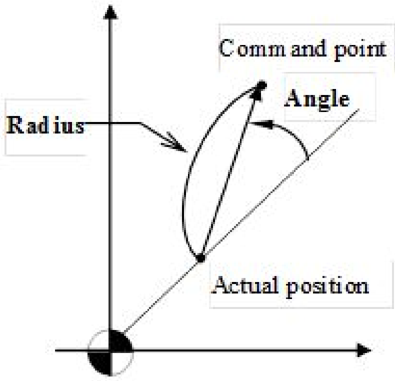


b. when angle is specifed with an increment command

2. when polar coordinate zero point is in normal position



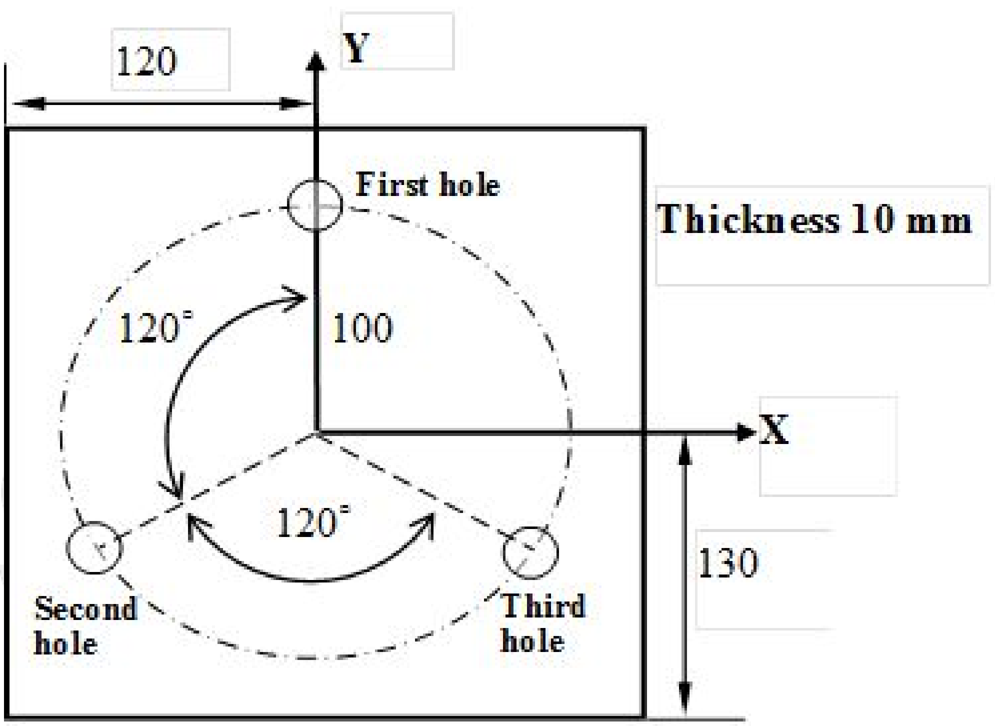
a. When angle is specified with an absolute command



b. when angle is specifed with an increment command

12.4 Example

PIC:



1. Absolute command:
   N001 T1 S1000 M03;
   //NO.1 tool(diamete r10 mm drill), spindle 1000rpm (CW)

N002 G17 G90 G16;
//X-Y plane, absolute mode, enable polar coordinate mode
N003 G99 G81 Z-12.0 R2.0 F600 K0;
//do drilling cycle, depth 12mm, feedrate 600mm/min, back to R point when finish
N004 X100.0 Y90.0;
//specified a distance 100mm, angle 90 degree(first hole)
N005 Y210.0;
//specified a distance 100mm and angle 210 degree, from the origin point(second hole)
N006 Y330.0;
//specified a distance 100mm and angle 330 degree, from the origin point(third hole)
N007 G15 G80 M05;
//polar coordinate mode disable, cycle cancel, spindle stop
N008 M30; //program ends

2. Increment command:
N001 T1 S1000 M03;
// NO.1 tool(diameter 10 mm drill), spindle 1000rpm (CW)
N002 G17 G90 G16;
// X-Y plane, absolute mode, enable polar coordinate mode
N003 G99 G81 Z-12.0 R2.0 F600 K0;
// do drilling cycle, depth 12mm, feedrate 600mm/min, back to R point when finish
N004 X100.0 Y90.0;
//specified a distance 100mm, angle 90 degree(first hole)
N005 G91 Y120.0 K2;
//increment command, angle totals 120 degree from last point (second hole)
N006 Y120.0;
//increment command, angle totals 120 degree from last point (third hole)
N007 G15 G80 M05;
// polar coordinate mode disable, cycle cancel, spindle stop
N008 M30; //program ends

13 G17/G18/G19： Plane Selection

13.1 Command Form

G17; X-Y plane selection

G18; Z-X plane selection

G19; Y-Z plane selection

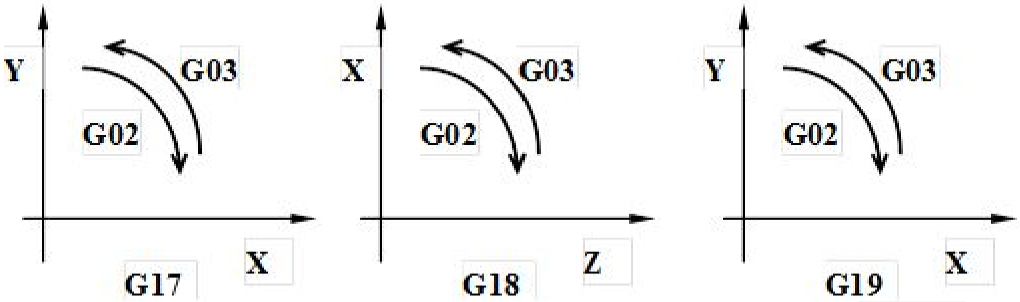
13.2 Description

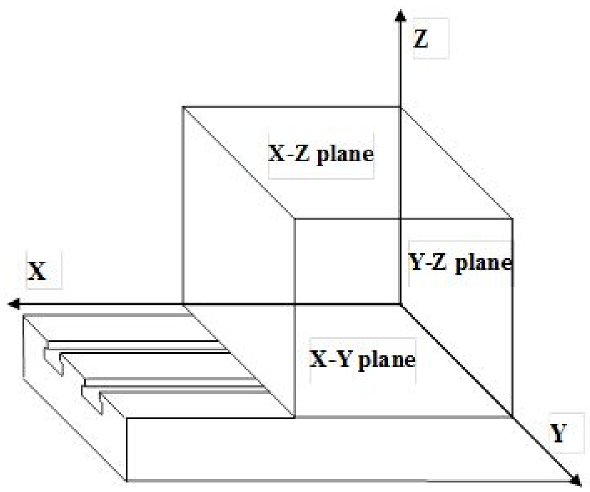
1. When using the arc interpolation, tool radius compensation command or polar coordinate command, the cutting plane must be set with G17, G18, G19 to inform the controller of the machining plane (default is G17).
2. In the X, Y, and Z directions on the cutting plane, the actual corresponding axis is called the geometry axis. The selection rules for the geometry axis are as follows：
    a. The controller divides the axis into three categories based on the axis name.
        i. X category： X， X1~X99， U， U1~U99， A， A1~A99。
        ii. Y category： Y， Y1~Y99， V， V1~V99， B， B1~B99。
        iii. Z category： Z， Z1~Z99， W， W1~W99， C， C1~C99。
    b. The axial direction of the X category is eligible to be selected as the axis in the X direction; the axial direction of the Y category is eligible to be selected as the axis in the Y direction; the axial direction of the Z category is eligible to be selected as the axis in the Z direction.
    c. In the same category, if there are multiple axes, the order described above will be referred to, and the one former, the more preferred.
    d. If there is no corresponding axis declared in a certain category, then the axis with the smallest axis ID is selected as the geometry axis of the category from the axial direction that is not selected as the geometry axis.
    e. If the number of axes declared by the system is less than three axes, there will be no geometry axis selected for a certain category. In this case, the arc interpolation, tool radius compensation command or polar coordinate command will be limited in use.
        i. Taking a two-axis lathe (Z, X-axis) as an example, only the G18 work plane can be used.
    f. An axis would not selected as two category of geometry axes at the same time.

13.3 Notice

1. When using G17、G18、G19 for machining plane switching, if the axial command is added in the same block, in addition to changing the working plane geometry axis, the movement command will also be generated. Please aware of the action of the machine to avoid danger.

13.4 Example





13.5 Example Description

The known controller parameters are set as follows：

Pr21、22、23、24、25 = [1， 2， 3， 4， 5]

Pr321、322、323、324、325 = [101， 100， 800， 302， 301]

Therefore, there are five axes in the system named X1, X, V, Z2, and Z1 (axis ID is from small to large)

According to the above rules, the three geometric axes that make up the spatial geometric coordinates are: X, V, Z1.

Example 1:

G17;

G91 G02 X5. R20. F2000;// After the G17 is specified, the arc interpolation will be displayed on the plane composed of the X and V axes.

Example 2:

G18;

G91 G02 X5. R20. F2000;// After the G18 is specified, the arc interpolation will be displayed on the plane composed of the Z1 and X axes.

13.6 Appendix

附錄

13.7 Description

1. If the user adds an axial command behind the G17/G18/G19 block, indicating that the user wants to specify the axis as the geometry axis, the priority order of the specified geometry axis is as follows (high to low):
    a. In the axial command, the axes of the axis category are X, Y, Z, U, V, and W, which are preferentially according to the axis category. The axis ID is from small to large to the geometric axis (independent of the order of declaration).
    b. The axial direction of all declared axes is from small to large to the geometric axis required by the working plane (Ex: G17 requires X, Y; G18 requires Z, X).
    c. The remaining axes in the system are assigned to the geometry axes according to preset selection rules. (not appearing in the axial direction behind the G17/G18/G19 block)

13.8 Notice

1. Not Support in 21GA-E or 6GA-E after 10.118.34(included).
2. The H command isn't C geometry axis if Pr3957 set 1.

13.9 Example Description

The known controller parameters are set as follows:

Pr21、22、23、24、25＝[1，2，3，4，5]

Pr321、322、323、324、325＝[100，200，302，301，303]

Therefore, there are five axial axes named X, Y, Z2, Z1, and Z3, and the axis ID is from small to large.

Example 1:

```
G17 G01 Y1. Z2 = 10. Z1 = 20. ;
```

The axial command contains three axes of Y, Z2 and Z1.

1. The axis that appears after the G17 block will be preferentially designated as the geometry axis: the Y axis belonging to the Y category is designated as the Y geometry axis; the Z2 axis belonging to the Z category and having the smallest axis ID is designated as the Z geometry axis.
2. G17 requires an X丶Y geometry axis, but there is no X class axis behind the G17 block. Therefore, the Z1 axis with the smallest axis ID is designated as the X geometry axis, and the X axis declared by Pr21 is designated as the geometry axis.

Example 2:

```
G18 G01 Z2 = 10. Z1 = 20. ;
```

The axial command contains two axes Z2 and Z1

1. The axis that appears after the G18 block will be preferentially designated as the geometry axis: the Z2 axis that belongs to the Z category and has the smallest axis ID is designated as the Z geometry axis.
2. The G18 requires a Z丶X geometry axis, but there is no X class axis behind the G18 block. Therefore, the Z1 axis with the smallest axis ID is designated as the X geometry axis.
3. Since the axis that appears behind the G18 block has been specified, the Y axis declared by Pr22 will be designated as the Y geometry axis.

14 G28： Return to Reference Point

14.1 Command Form

G28 X_Y_Z_;

X, Y, Z: mid-point position (absolute value in G90 mode, increment value in G91 mode)

14.2 Description

G28 is used to return to reference point or return to origin point. To avoid tool crack, it will move(G00) to the safe mid-point assigned by users first then return to machine origin point.

Note

1. The command is usually applied to auto-tool changing. Therefore, please deactivate the tool compensation function before running G28 for safety. Also, please note that the XYZ arguments are corresponded to the program coordinate.
2. If the axis type (parameter 221~236) is set to be rotary axis, please refer to Pr221~236 "Axial type" for setups.

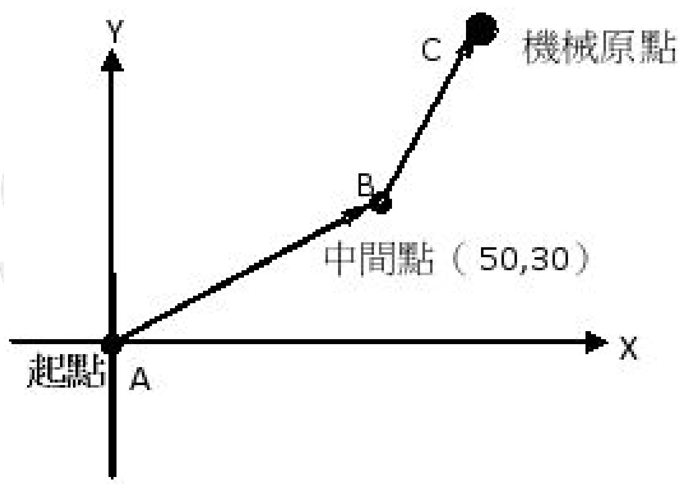
14.3 Limitations

1. G28 function only supports axes named X, Y, Z, U, V, W, A, B, C, X1, Y1, Z1, X2, Y2, Z2, the return action won't be executed if axes with other names.
2. If apply G28 function to axes with other names (ex: X3, X4), G0028 can be customized. Add Xn, Yn or Zn axial command processes.

Example

14.3.1 Problem 1:

G90 **G28** X50.0 Y30.0; //A->B->C, mid-point(50,30)



14.3.2 Problem 2:

G28 X0; //only X axis return to reference point
G28 Y0; //only Y axis return to reference point
G28 Z0; //only Z axis return to reference point

15 G29： Return From Reference Point

15.1 Command Form

G29 X_Y_Z_;

X, Y, Z: specified point； (absolute value in G90 mode, increment value in G91 mode)

15.2 Description

G29 command can make the tool rapidly moves from reference point through mid-point to the specified point base on G28 applied. Notice that G29 can't be applied alone because G29 does not specify the mid-point, it refers to the mid-point from G28. Therefore, G28 must be executed first before executing G29

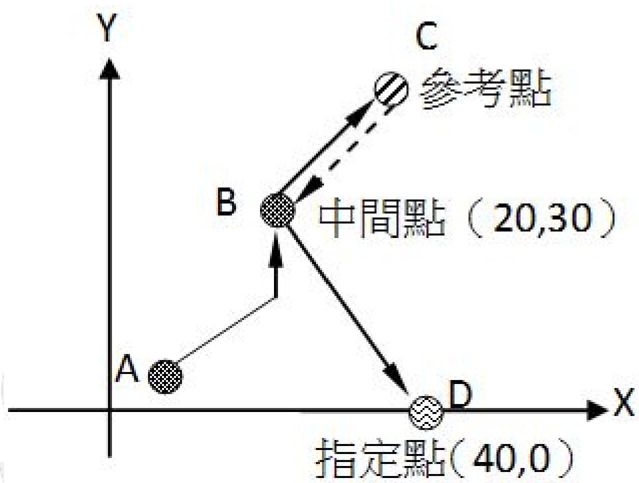
Under the absolute command (G90), the command content is the absolute coordinate; under G91, it is the increment distance from mid-point to specified point.

Limitations

1. G29 function only supports the axes named with X, Y, Z, U, V, W, A, B, C, X1, Y1, Z1, X2, Y2, Z2, the return action won't be executed if axes with other names.
2. If apply G29 function to axes with other names (ex: X3, X4), G0029 can be customize. Add Xn, Yn or Zn axial command processes.

Example

PIC:



1. Absolute command:
   N001 G90 G28 X20.0 Y30.0;
   //A>B>C, mid-point(20,30), in absolute command mode
   N002 M06;//change tool
   N003 **G29** X40.0 Y0.0;
   // C>B>D, the command content is the absolute coordinate

2. Increment command:
   N001 G91 G28 X20.0 Y30.0;
   //A>B>C, mid-point(20,30), in increment command mode
   N002 M06;//change tool
   N003 **G29** X20.0 Y-30.0;
   //C>B>D, the command content is the increment value from mid-point to the target point

16 G30： Return From Specified Reference Point

16.1 Command Form

G30 P**n** X_ Y_ Z_ ;

X, Y, Z: mid-point coordinates; (absolute value under G90, increment value under G91)

P**n**: Specified reference point (parameter #2801 ~ #2860)

P1: Mechanical origin point;

P2: Second reference point;

P_: default is P2 if omitted

16.2 Description

G30 is the Return From Specified Reference Point command, usually being applied when the location of auto-tool change is not the origin point. The action moves from current location to the user-defined safe mid-point with rapid orientation mode (G00), then return to the specified reference point (Pr2801~Pr2860).

16.3 Notice

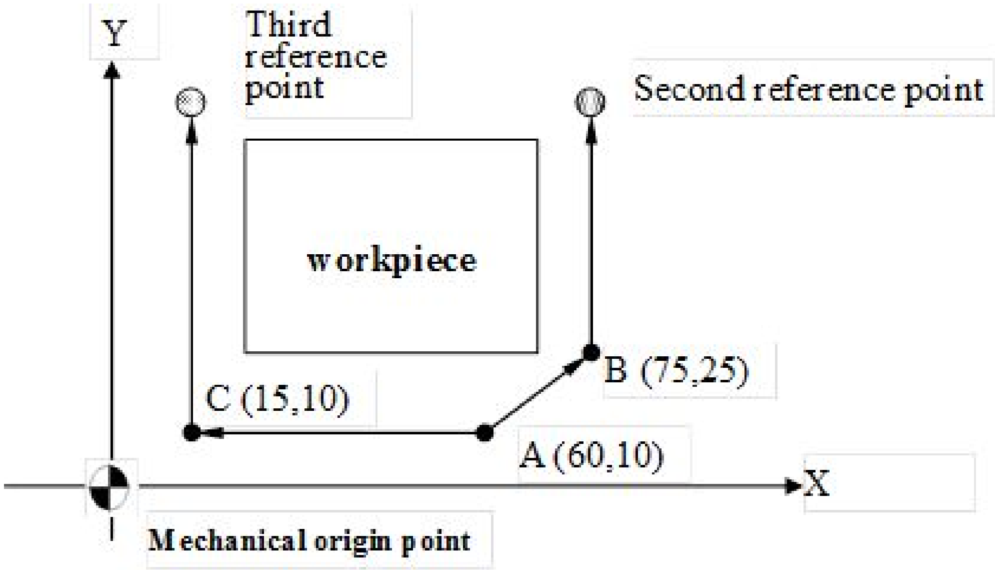
The command is usually applied in auto-tool change, for safety concerns, please deactivate the tool compensation function before executing G30. Also, please note that the XYZ arguments are corresponded to program coordinate but the final location (reference point) is corresponded to mechanical coordinate.

16.4 Limitations

1. The Return From Specified Reference Point function (G30) only supports the axes named with X, Y, Z, U, V, W, A, B, C, X1, Y1, Z1, X2, Y2, Z2, the return action won't be executed for axes with other names.
2. If wants to apply the G30 function to axes with other names (ex: X3, X4), you can customize G0030 and add Xn, Yn or Zn axial command processes.

16.5 Example

PIC:



Program description: presume the tool is on A (60,10)

1. to second reference point
   G30 P2 X75.0 Y25.0; //A>B> 2nd reference point
2. to third reference point
   G30 P3 X15.0 Y10.0; //A>C> 3rd reference point

17 G31.10/G31.11： Multi-Axis Multi-Signal Skip Function

17.1 Command Form

G31.10 X_ Y_ Z_ F_ Q_ P_

Set multi-axis multi-signal skip function

X, Y, Z: Specified Position

F: Feedrate

Q: Skip source reference

P: Deceleration time (ms)

G31.11

Execute multi-axis multi-signal skip function

17.2 Description

1. A multi-axis multi-signal skip function is composed of one G31.10 at least and one G31.11. Set first then execute.
   Both commands should be used in a multi-axis multi-signal skip function, and it is not allowed to insert other commands in between.
   a. Set multi-axis multi-signal skip function(G31.10)：
      i. Supports up to six G31.10 commands in a set of multi-axis multi-signal skip function. Skip source and deceleration time can be different in each G31.10 command.
      ii. In each G31.10, every axis specified in this command will move, stop and skip at the same time.
      iii. If the specified axis is the virtual axis set by G10 L801, all settings of F, P and Q will be applied to corresponding axis.
   b. Execute multi-axis multi-signal skip function(G31.11)：
      Execute multi-axis multi-signal skip function with previous settings of G31.10.
      When the skip signal is triggered, the axis corresponding to the skip source skips.
2. The unit of position can be mm or inch, depends on which unit setup is used(G70/G71).
3. Feedrate F：
   a. Previous feedrate will be considered while F argument is not specified.
   b. unit：
      mm/min(inch/min) in G94.
      mm/rev(inch/rev) in G95.
4. Skip source reference Q：
   a. C62 is the default skip source while Q argument is not specified
   b. The signal source of **Q101~Q132** is C-bit, corresponds to **C101~C132** respectively.
   c. The signal source of **Q201~Q218** is the **external signal (EXT) source** of serial drivers, it detect the signal of the 1st ~ 18th axis respectively. The supporting serial drivers are listed below.

Supporting Serial Drivers	External Signal (EXT) Source
M2	EXT1

Supporting Serial Drivers	External Signal (EXT) Source
M3	EXT1
RTEX	EXT1
EtherCAT	EXT1

Note: Syntec M2 doesn't support.

5. Deceleration time P：
    a. When P argument is not specified or P0 is given, there is no deceleration and the command will be suspended directly.
    b. When P argument is given, deceleration time will be considered in deceleration planning.
    Axis reaches the position specified by G31.10 when deceleration time is so long that the brake distance exceeds the destination of this block.

17.3 Notice

1. #1361~#1378, #1441~#1458, #1608 are set to 0 when CNC just powered on, RESET, system executes G31 or G31.11 or G28.1 again.
   To avoid showing the previous escape location before skip signal comes in and causes misjudgment.
2. If the final point location of G31.10 & G31.11 is too close to the skip signal triggering location, it might occasionally lead to the situations that the PLC scanned the C-bit signal after G31.11 is finished. In the situations, although the signal is triggered, but it's too late for G31.11 to execute the skip. Take the tool measuring action as example, when the situation happened, it's better to move the G31.10 & G31.11 block deeper to avoid the situation.
3. The P argument should be bigger than 0 or an integer, or alarm COR-064 will issued.
4. Alarm COR-362 issues when only G31.10 or G31.11 command is given, or inserting other commands between G31.10 and G31.11.
5. Alarm COR-362 issues when the same axis is used in different G31.10.
6. Alarm COR-362 issues when G31.10 is given more than 6 times in a multi-axis multi-signal skip function.
7. The functions below are not supported in multi-axis multi-signal skip function：
    a. G5.1 (Path Smoothing)
    b. G12.1/G13.1 (Activate/Deactivate Polar Coordinate Interpolation)
    c. G15/G16 (Polar Coordinates Command Mode)
    d. G40/G41/G42 (Cutter Radius Compensation)
    e. G43.4/G43.5 (Rotate Tool Center Point (RTCP) Type1 & 2)
    f. G10 L16 (Virtual circle radius)
8. While executing multi-axis multi-signal skip function, F(command) displays the latest F command given by NC file, F(actual) displays the value of current resultant velocity. Therefore, F(actual) may be greater than F(command). The block feedrate command of G31.11 while interpolation can be obtained by using variable K62.
   example

   sample code

   ```
   G90 G71
   ```

```
G31.10 Z1=10. F300. Q101 P100   // set Z1 goes to 10 by F300, the skip source
is C101, deceleration time is 100 ms
G31.10 Z2=20. F400. Q102 P100   // set Z2 goes to 20 by F400, the skip source
is C102, deceleration time is 100 ms
G31.11                          // execute multi-axis multi-signal skip
function
M30
```

When Z1 and Z2 reach F300 and F400 respectively, the display result：

F(command)：400 mm/min

F(actual)：500 mm/min

The block feedrate command of G31.11 while interpolation can be obtained by variable K62.

The actual compound feedrate and axis feedrate can be obtained by R700 and R701~718：

- R700：Actual compound feedrate command, unit: LIU/min.
- R701~718：Velocity of each axis, Servo On mode: according to command value; Servo Off mode: according to feedback value, unit : BLU/min.

R700：500 IU/min = 500000 LIU/min

R701：300 IU/min = 300000 BLU/min (if Z1 is the first axis)

R702：400 IU/min = 400000 BLU/min (if Z2 is the second axis)

9. If an escape signal is triggered while **cross-coordinate axis coupling** is enabled, the command of the master axis will be interrupted, **and the slave axis commands will also be interrupted accordingly** (applicable only to coupling types 2 to 5). However, note that the escape machine position recorded by the slave axis group will not be correct in this case.
   - To ensure the recorded position is correct, you can use the axis borrowing function to borrow the coupled axis into the same coordinate, and then use the G31.11 to trigger the escape signal.
10. Valid version : after 10.118.40G, 10.118.44 (included).
11. The 3rd-party EtherCAT drive support version: 10.118.82N, 10.118.86G, 10.118.90C, 10.118.94 and later versions.

17.4 Example

17.4.1 Example 1: each axis moves and skips with the settings of G31.10.(the feedrate of all G31.10 refers the previous one)

example 1

```
G90
F100.
G31.10 Z1=10. Q101 P100 // set Z1 to move to 10 by F100, skip source is C101, set
deceleration time to 100 ms
G31.10 Z2=20. Q102 P100 // set Z2 to move to 20 by F100, skip source is C102, set
deceleration time to 100 ms
G31.11                  // execute multi-axis multi-signal skip function
```

```
M30
```

The value of #1608 when all skip source are triggered.

There are axes: X, Y, Z1, Z2, Z3, Z4， the corresponding Pr21~ and Pr321~ are listed below。

When all skip source are triggered, the value of each bit of #1608 is：

bit 0： 0。(supports G31 only)
bit 1~18：

Command	**G31.10 & G31.11**					
Axis corresponding axis card port number(Pr21~)	Pr21	Pr22	Pr23	Pr24	Pr25	Pr26
Axis name(Pr321~)	X	Y	Z_1	Z_2	Z_3	Z_4
bit	1	2	3	4	5	6
value	0	0	1	1	0	0

Therefore, the value of #1608 is $2^3+2^4=24$。

17.4.2 Example 2:each axis moves and skips with the settings of G31.10.(the second feedrate of G31.10 refers the one of the first G31.10)

example 2

```
G90
G31.10 Z1=10. F100. Q101 P100   // set Z1 to move to 10 by F100, skip source is C101,
set deceleration time to 100 ms
G31.10 Z2=20. Q102 P100         // set Z2 to move to 20 by F100, skip source is C102,
set deceleration time to 100 ms
G31.11                          // execute multi-axis multi-signal skip function
M30
```

The value of #1608 when all skip source are triggered.

There are axes: X, Y, Z1, Z2, Z3, Z4， the corresponding Pr21~ and Pr321~ are listed below。

When all skip source are triggered, the value of each bit of #1608 is $2^3+2^4=24$。

Command	**G31.10 & G31.11**					
Axis corresponding axis card port number(Pr21~)	Pr21	Pr22	Pr23	Pr24	Pr25	Pr26
Axis name(Pr321~)	X	Y	Z_1	Z_2	Z_3	Z_4

17.4.3 Example 3: use multi-axis multi-signal skip function with virtual axis setting

When using G10 L801, all axes are considered as the same axis. Therefore, all axes apply the same skip function setting.

example 3

```
G90
G10 L801 P300 Q0        // cancel virtual axis Z
G10 L801 P300 Q301      // virtual axis Z corresponding to Z1
G10 L801 P300 Q302      // virtual axis Z corresponding to Z2
G10 L801 P300 Q303      // virtual axis Z corresponding to Z3

G31.10 Z10. F100 Q101   // set virtual axis Z to move to 10 by F100. i.e., Z1, Z2 and
Z3 move to 10 by resultant velocity F100
                        // skip source is C101 and there is no deceleration time
G31.11                  // execute multi-axis multi-signal skip function
M30
```

The value of #1608 when all skip source are triggered.

There are axes: X, Y, Z1, Z2, Z3, Z4， the corresponding Pr21~ and Pr321~ are listed below。

When all skip source are triggered, the value of each bit of #1608 is $2^3+2^4+2^5=56$。

Command	G31.10 & G31.11					
Axis corresponding axis card port number(Pr21~)	Pr21	Pr22	Pr23	Pr24	Pr25	Pr26
Axis name(Pr321~)	X	Y	Z_1	Z_2	Z_3	Z_4

17.4.4 Example 4: Z1 and Z2 move simultaneously, Z3 moves independently.

example 4

```
G90
G31.10 Z1=10. Z2=20. F100 Q101  // Z1 and Z2 move to 10 and 20 respectively, the
resultant velocity is 100
                                // skip source is C101 and there is no deceleration
time
G31.10 Z3=30. Q103              // set Z3 to move to 30 by F100, skip source is C101
and there is no deceleration time
G31.11                          // execute multi-axis multi-signal skip function
M30
```

The value of #1608 when all skip source are triggered.

There are axes: X, Y, Z1, Z2, Z3, Z4， the corresponding Pr21~ and Pr321~ are listed below。

When all skip source are triggered, the value of each bit of #1608 is $2^3+2^4+2^5=56$。

Command	**G31.10 & G31.11**					
Axis corresponding axis card port number(Pr21~)	Pr21	Pr22	Pr23	Pr24	Pr25	Pr26
Axis name(Pr321~)	X	Y	Z_1	Z_2	Z_3	Z_4

18 G31：Skip Function

18.1 Command Form

G31 X_ Y_ Z_ F_ Q_ P_;

X, Y, Z: Specified Point

F: Feedrate

Q: Skip source reference

P: Deceleration time (ms)

18.2 Description

1. If Q argument is not specified, corresponds to **C62**。
2. The signal source of **Q101~Q132** is C-bit, corresponds to **C101~C132** respectively.
3. The signal source of **Q201~Q218** is the **external signal (EXT) source** of serial drivers, it detect the signal of the 1st ~ 18th axis respectively. The supporting serial drivers are listed below.

Supporting Serial Drivers	External Signal (EXT) Source
M2	EXT1
M3	EXT1
RTEX	EXT1
EtherCAT	EXT1

Note: Syntec M2 doesn't support. Can not trigger with drive parameter【Pn-F01】(Output Software Trigger)

4. When P argument is not specified, deceleration function missing, the command will be suspended.
5. When specified **P0**, deceleration function is missing, the command will be suspended.
6. when P argument is specified, the deceleration according to the planned deceleration time. The program will stop at the block if deceleration time given is not enough.
7. The 3rd-party EtherCAT drive support version: 10.118.82N, 10.118.86G, 10.118.90C, 10.118.94 and later versions.

Note：If the signal source is C62, C101~ or external signal (EXT) source of serial drivers, please contact the machine manufacturer for wiring connection.

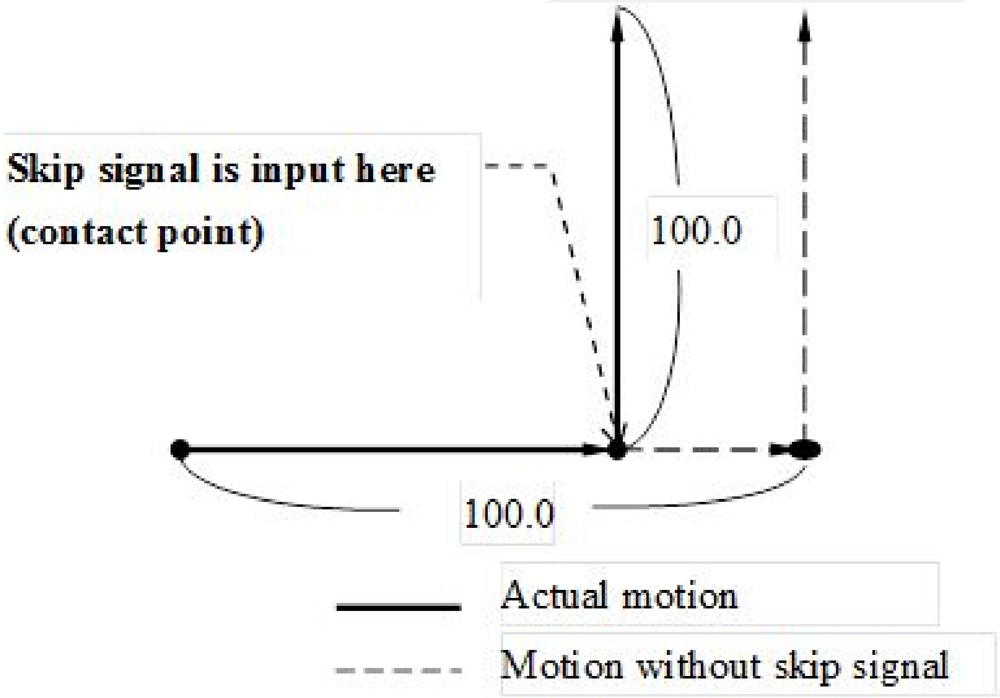
18.3 Description

1. If the final point location of G31 is too close to the skip signal triggering location, it might occasionally lead to the situations that the PLC scanned the C-bit signal after G31 is finished. In the situations, although the signal is triggered, but it's too late for G31 to execute the skip. Take the tool measuring action as example, when the situation happened, it's better to move the G31 block deeper to avoid the situation.
2. If an escape signal is triggered while **cross-coordinate axis coupling** is enabled, the command of the master axis will be interrupted, **and the slave axis commands will also be interrupted accordingly** (applicable

only to coupling types 2 to 5). However, note that the escape machine position recorded by the slave axis group will not be correct in this case.
  - To ensure the recorded position is correct, you can use the axis borrowing function to borrow the coupled axis into the same coordinate, and then use the G31 to trigger the escape signal.
3. The P argument should be bigger than 0 or an integer, or alarm COR-64 will issued.

18.4 Example

18.4.1 Example 1: Incremental command(G91)

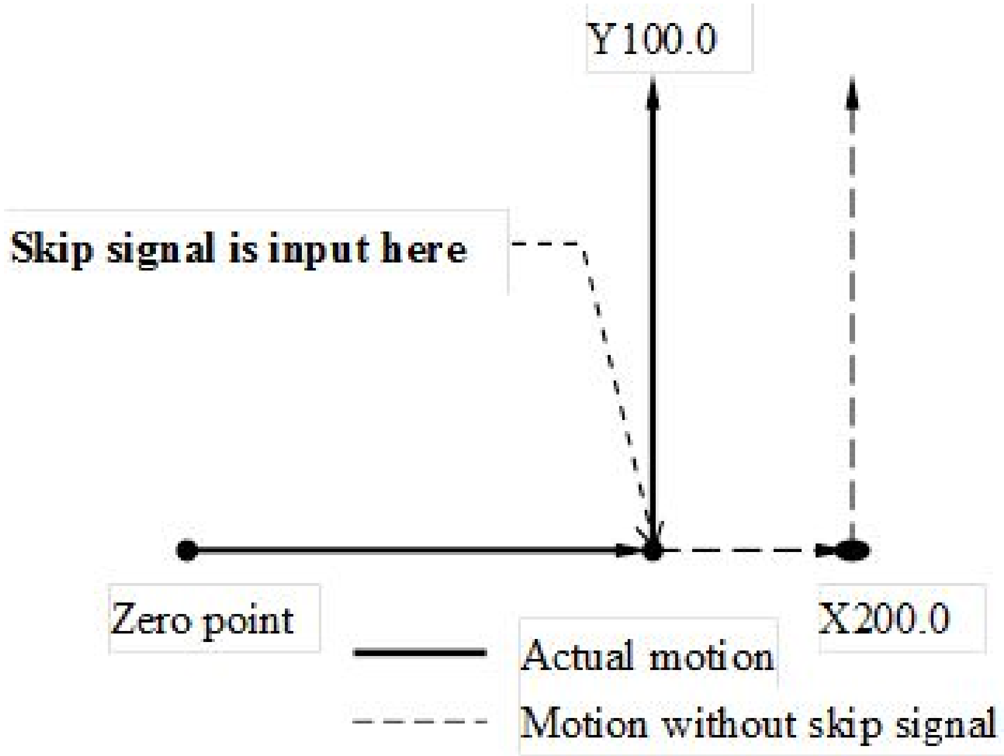


Program description:

N001 G31 G91 X100.0 F100; //original contour until run into impede

N002 Y100.0;//not waiting for the completion of former block, take the contact point as relative coordinate and change the contour to specified position

18.4.2 Example 2: absolute command for single axis(G90)

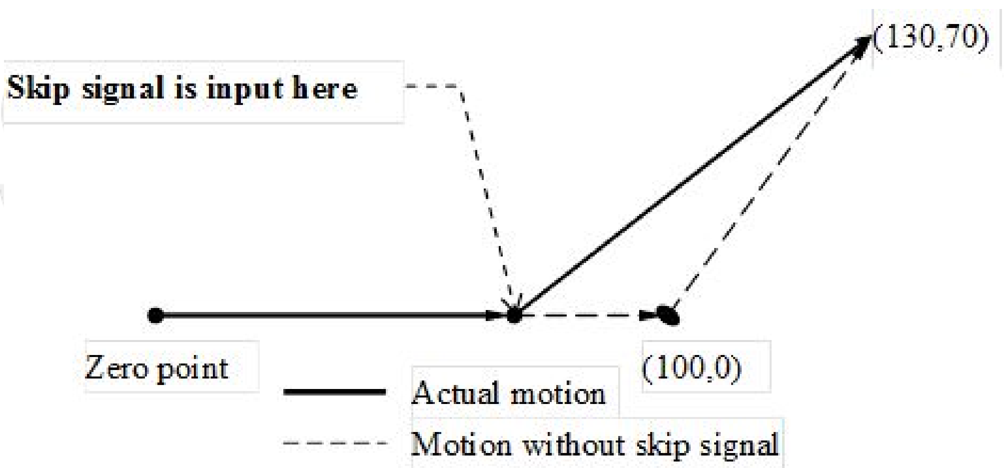


Program description:

N001 G31 G90 X200.0 F100; //original contour until run into impede

N002 Y100.0; //not waiting for the completion of former block, take the zeor point as relative coordinate and change the contour to specified position

18.4.3 Example 3: absolute command for 2 axes(G90)



Program description:

N001 G31 G90 X100.0 F1000; //original contour until run into impede

N002 X130.0 Y70.0; //not waiting for the completion of former block, take the origin point as relative coordinate and change the contour to specified position

19 G43.4/G49：Rotate Tool Center Point (RTCP)

19.1 Command Form

G43.4 H_ Q_ I_;
G49;

G43.4 : Enable Rotate Tool Center Point Type1 (RTCP Type1)
H : Tool number
Q : Tool attitude reference coordinate system
I : Maximum compensation speed of the linear axes
G49 : Disable Rotate Tool Center Point Type1

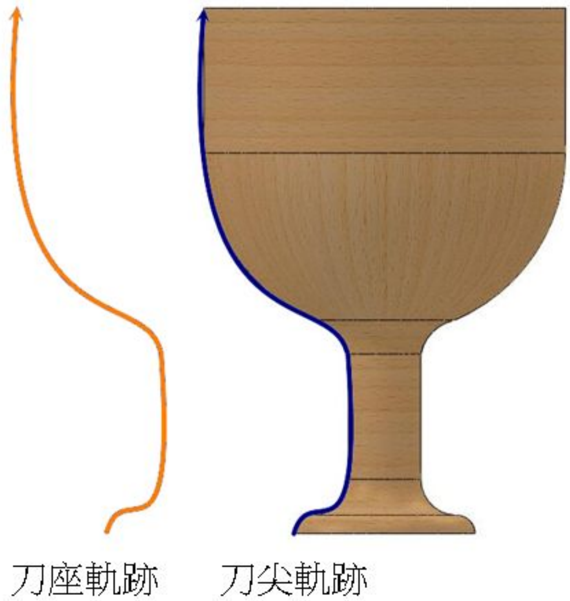
The applying method of RTCP is the same as tool length compensation (G43), only needs to give the G43.4 command before the machining starts. After giving the command to assign the tool number, it'll be able to apply RTCP with the specified tool length.

19.2 Description

Rotational Tool Center Point :

RTCP is the Rotate Tool Center Point function. For general machines, moving commands from the controller are given to the tool holder or the spindle nose; after activating the RTCP function, the moving commands will be controlling the tool center point. The RTCP function is the specified function of five-axis machining centers. There are 2 machining contour in the picture below, the orange one is the contour of general machining state, the controller controls the contour of spindle nose, so there will be a one tool length gap between the machining contour and workpiece surface; the blue one is the contour of tool center point after activated RTCP, which is able to directly edit the machining program with workpiece surface coordinates.

The change of tool length and the mechanism differences between machines can all be ignored by using this kind of machining programs and the programs can also be applied more efficiently.



Maximum Compensation Speed of the Linear Axes :

Under the RTCP (Rotational Tool Center Point) function, the tool tip is constrained to move along a specified trajectory.
When the rotary axes generate additional displacement and speed during motion, the linear axes must compensate accordingly to ensure that the tool tip remains on the intended path.
This additional linear axes speed used to maintain the tool tip position is referred to as the compensation speed of the linear axes.
However, if the posture change of the rotary axes is too large, the compensation speed of the linear axes may become excessively high.
Therefore, the compensation speed of the linear axes must be set to prevent abnormal machine movement.
This limit is defined by parameter Pr3091 Maximum compensation speed of the linear axes

19.3 Note

1. G41, G42 cutter radius compensation functions can't be applied together.
2. G43, G44, G43.4 tool length compensation functions can't be applied together.
3. The tool length should be set positive.
4. Before applying G53, G28, G29, G30, please remember to disable RTCP Type1 with G49 to avoid abnormal machine actions.
5. In RTCP Type1 mode, if activates the HPCC function with G05 P10000, alarm [COR-140 G05 is banned in RTCP mode] will be issued.
6. Do not apply with the polar axis interpolation function (G12.1).
7. This feature doesn't support Multi-Axis Multi-Signal Skip Function(G31.10/G31.11).
8. Things about settings of tool attitude reference coordinate system
    a. Tool attitude referenced coordinate system will be reset to value of Pr3057 while reset or reboot.
    b. If the workpiece coordinate system rotates, and the tool attitude reference coordinate system is specified as workpiece coordinate system, the tool axis direction will be controlled in workpiece coordinate system.
        i. If the direction of the tool is aligned with the rotation axis, the rotation axis will not move.
    c. No matter the tool attitude reference coordinate system is, description of the tool direction follows: rotate the first rotation axis and then the second rotation axis. Definition of the rotation axis is determined according to the configuration of the mechanism.
    d. In the case of workpiece coordinate system rotates, it is recommended to specify workpiece coordinate system as the tool attitude reference coordinate system.
    e. If manual RTCP is executed while tool attitude take workpiece coordinate system as reference, the rotary axis will rotate according to the mechanical coordinate system, and tool axis direction will be described in workpiece coordinate system and displayed in the absolute coordinate column.
    f. Valid version is 10.118.86.
9. About the Maximum Compensation Speed of the Linear Axes
    a. Applicable Condition : This parameter is effective only when Pr3803 (Motion control mode for cutting) is set to Smooth acceleration mode.
    b. Configuration Method
        i. The maximum compensation speed of the linear axes can be specified either by the I parameter of G43.4 command or by the setting of Pr3091.
           If the I parameter is not specified, the maximum compensation speed of the linear axes will follow the value of Pr3091.
           If the I parameter is specified, its value takes precedence.
        ii. The unit, effect, and valid range of the I parameter are identical to those of Pr3091, and the usage method can be directly referenced accordingly.
    c. Configuration Recommendation :
        i. Set the I parameter equal to the F parameter.
           With this configuration, even when a motion block involves large posture changes or a large

rotation radius, the linear axis feedrate will be nearly equal to the I parameter. This helps to effectively avoid the risk of excessive machine speed.

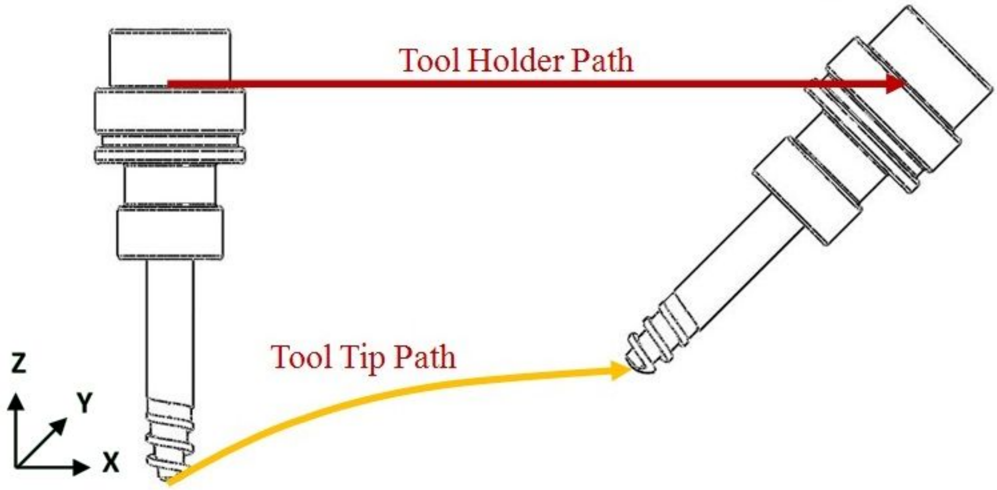
ii. If the efficiency of setting the I parameter equal to the F parameter is considered too low, it is also acceptable to set the I parameter greater than the F parameter. This can accelerate posture transitions and improve overall motion efficiency.

d. Valid Versions : 10.118.86U, 10.120.16V, 10.120.24C, 10.120.28 and and subsequent versions.

19.4 Program Example

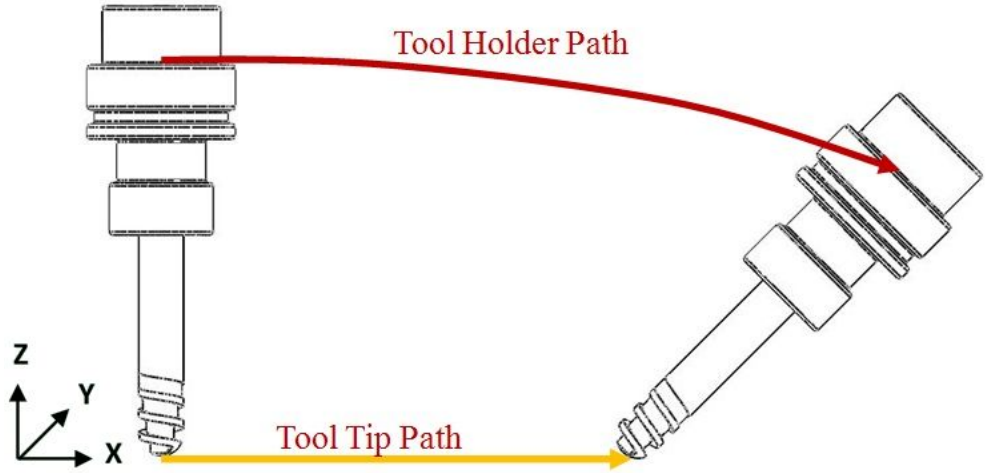
Example - Route before RTCP on and off

RTCP Type1 disable:
G00 X0. Y0. Z0. B0. C0.
G01 X50. Y0. Z0. B45. C0.
Machine motion:



RTCP Type1 enable:
G43.4 H1
G00 X0. Y0. Z0. B0. C0.
G01 X50. Y0. Z0. B45. C0.

Machine motion：



Before enable RTCP Type1, the motion of linear and rotary axes are independent; after activating RTCP Type1, the linear motion commands of tool center point will take the priority and the rotary axis will be rotating with the tool center point.

Example - switch tool direction reference coordinate system during machining process

```
%@MACRO
// Spindle Type, tool direction is in X direction
// Pr3057=0, set default tool attitude reference coordinate system to machine coordinate system
G90 G49 G54 P2;            // switch workpiece coordinate system
G00 X0. Y0. Z0. A0. C0.;   // tool attitude is in (1,0,0) direction of machine coordinate system now
#30:=#1848;                // back up tool attitude reference coordinate system
G43.4 Q1;                  // tool attitude take workpiece coordinate system as reference
G00 C90.;                  // tool attitude is in (0,1,0) direction of workpiece coordinate system now
G43.4 Q#30;                // restore origin tool attitude reference coordinate system
G00 C0.;
M30;
```

20 G51.2/G50.2 : Polygon cutting

20.1 **Command Form**

Enable polygon cutting

G51.2 P_ Q_ [R_] [K_] ;

- P: Base spindle (workpiece axis) rotation speed rate or the number of blades. Default is P=1 (range from integer 1to 999)
- Q: Synchronous spindle (tool axis) rotation speed rate or the number of polygon edges. Default is Q=1(range from integer 1to 999).
- R: Synchronous phase shift (range from 0°to 359.999°).
- K: Synchronous group number 1~3, multiple sets of synchronization, can use up to 3 sets at the same time. When K is not specified, the first set of synchronization is preset. Multiple sets of synchronization function start at version 10.116.24M & 10.116.32.

Disable polygon cutting

G50.2 [K_] ;

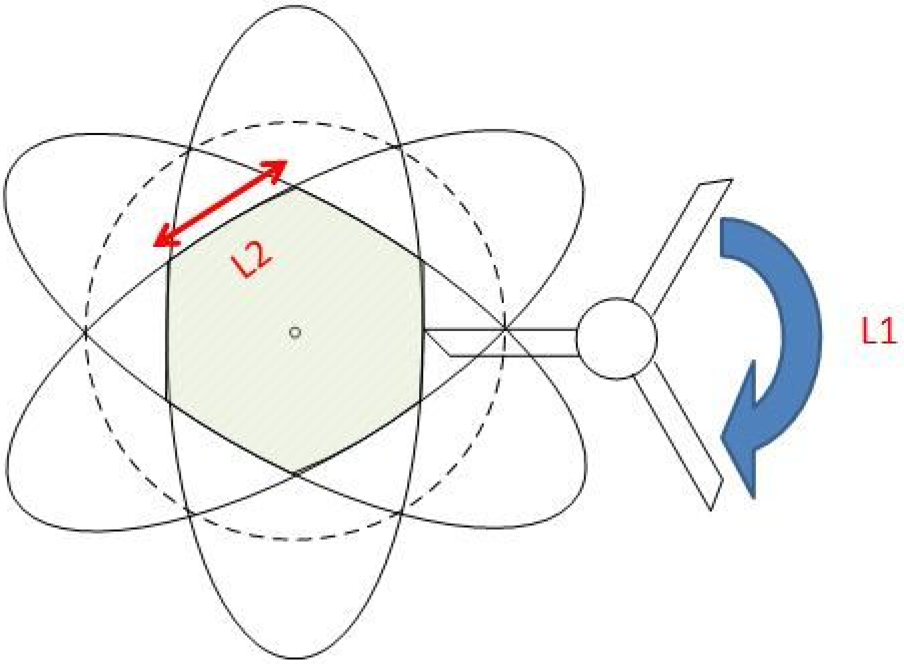
20.2 **Description**

1. G51.2 command synchronizes workpiece axis and tool axis at a fixed phase shift and RPM ratio to perform polygon cutting.
2. Synchronous spindle RPM: Base spindle RPM * Q / P
3. Synchronous phase shift: The clockwise angle difference between Synchronous spindle and Base spindle. Phase synchronization will not be enabled if no R argument given.
4. G50.2 disables polygon cutting.
5. G51.2 is available since version 10.113.0, unavailable in earlier versions (9.0 and 10.0).

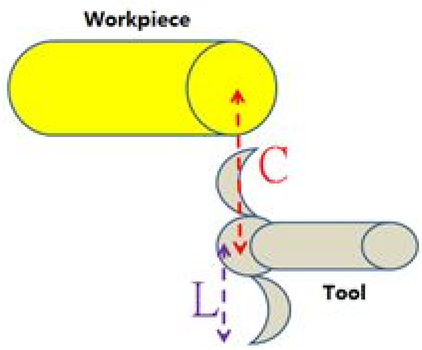
20.3 **Note**

1. **Spindle status description:**
    a. When the synchronization complete signal is On, pressing Reset will cancel G114.1 synchronization status (synchronization complete signal Off) after both spindles stop.
    b. When the synchronization complete signal On, G50.2 will cause the system to directly cancel the synchronization status (synchronization completion signal Off).
    c. The Base Spindle cannot use the synchronous function under position control mode (C63), and Synchronous Spindle is not recommended to use the synchronous function under position control mode.
2. **Multiple sets of synchronization:**
    a. The synchronous start command (G114.1) can be repeated (but the K value is not repeatable).
    b. A Base Spindle can have multiple Synchronous Spindle at the same time.
    c. The Synchronous Spindle can no longer be a Base Spindle of other spindles. (COR102)
    d. The value of System Data 45/46 is the angle difference between the Base Spindle and the Synchronous Spindle of the last synchronization command. The set is displayed as the angle difference of the second-to-last set in the same period after sync cancel, and so on.
3. **Precaution when machining:**
    a. When giving synchronous phase shift, the R value equals to the wanted angle difference between tool and workpiece multiply Q and divided by P (see example).
    b. P and Q values must be integers. For non-integer ratio 1:2.5, user shall use equivalent integer ratio instead, e.g. G51.2 P2 Q5.

c. To ensure that the absolute position of the workpiece is correct, a tool zero teach is necessary when the tool is first installed (see example preparation).
d. In actual cutting, notice that the perimeter between the blades should be longer than the polygon edge. See the following figure, L1 must be longer than L2 to ensure that the polygon shape is correct.



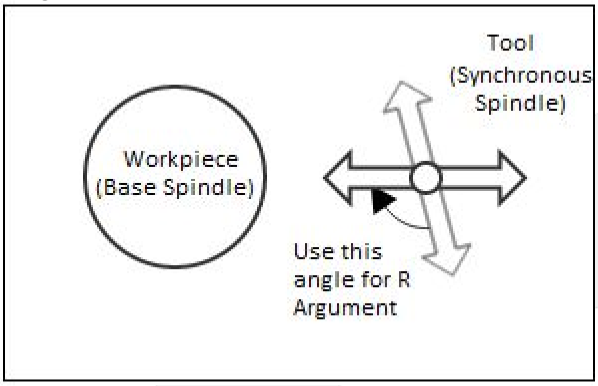
e. G51.2 is a new processing technique that uses the speed difference between the tool and the workpiece to quickly machine a polygonal workpiece, but the cutting mechanism often causes non-planar workpiece surface due to the change of machining conditions. The following table is used to calculate the resulting flatness of workpiece for reference by the first-line colleagues.

<table>
<tr><th>RPM Ratio (i)</th><th colspan="4">G51.2 Result of Workpiece</th></tr>
<tr><td rowspan="3">>2</td><td>K>L</td><td>K=L</td><td>K<L</td><td rowspan="5">

L=Tool radius; i = RPM Ratio (Q/P); C= Distance from Tool center to Workpiece center</td></tr>
<tr><td>Convex</td><td>Flat</td><td>Concave</td></tr>
<tr><td colspan="3">Formula: K = C/(i-1)^2</td></tr>
<tr><td>=2</td><td colspan="3">Convex</td></tr>
<tr><td><2</td><td colspan="3">Convex</td></tr>
</table>

20.4 **Example**

- **Preparation**
  To assure the absolute position of the workpiece (predictable absolute angle of the finished workpiece), tool's home position needs to be setup. There are three different ways to setup the tool's home position:
  a. Measure the angle between the tool zero point and tool setting position, phase difference defined via R argument.



  b. Adjust the tool to the tool setting position and set the tool position to the home position.
  c. Adjust the tool to the tool setting position and then perform the main synchronous angle teaching (F4->F4->F3). After this action, the reference angles of the Base and Synchronous axes will be stored in the Registry Table.
  (Note) Tool setting position: Align the tool blade to the direction from tool center to workpiece center (see schematic above). Performing any of the above three tool setting above can ensure the workpiece at 0 degree is aligned (refer to the schematic). To shift the workpiece by an angle, just add (the angle to be shift) x Q /P) to existing angle shift value (R).

- **Sample Command**
  Ex1. Hexagon with 3 flute (blade) tool : G51.2 P3 Q6 (or G51.2 P1Q2)
  Ex2. Pentagon with 2 flute (blade) tool : G51.2 P2 Q5

Example 1

```
S1 = 1000                //Workpiece axis (basic spindle) rotate speed 1000 RPM
M03                         //Workpiece axis (basic spindle) spindle rotate CW
S2 = 500                  //Tool axis (Synchronous spindle) rotate speed 500
                                //RPM
M204                        //Tool axis (Synchronous spindle) spindle rotate CCW
G51.2 P1 Q2 R60     //Tool axis (Synchronous spindle)
                                //synchronizes to 2000RPM and the phase
                                //difference is 30 degree. Cut for quadrangle
M81                         //Reading S62, check the synchronization success.
G01 X50                   //Start cutting
G04 X5
G01 X0                    //return
G50.2;                      // cancel polygon cutting
G51.2 P1 Q3 R180   //Tool axis (Synchronous spindle)
                                 //synchronize to 3000RPM and the
                                 //phase difference is 60 degree. Cut for hexagon.
M81; // reading S62. Check the synchronization success.
G01 X50                   //start cutting
G04 X5
```

```
G01 X0                     //return
G50.2                       //cancel polygon cutting
M05                          //Workpiece axis (basic spindle) stop
M205                        //Tool axis (Synchronous spindle) stop
M30                          //program finish
```

Note: After version 10.116.1, the core will automatically wait for synchronization. Non eed for the M code (M81).

Example 2 (Multiple sets of simultaneous use)

Scenarios:

Pr4021 = 1 (K1: 1st spindle)
Pr4022 = 2 (K1: 2nd spindle) // The first and second spindles are synchronized in other processing areas.
Pr4023 = 3 (K2: 3rd spindle)
Pr4024 = 4 (K2: 4th spindle) // The third and fourth spindles clamp the workpiece while rotating
Pr4025 = 3 (K3: 3rd spindle)
Pr4026 = 5 (K3: 5th spindle) // The fifth spindle follows the third spindle for polygon cutting

```
M03 S1000 // spindle 1 CW on
M203 S2=1500 // spindle 2 CW on
M303 S3=2000 // spindle 3 CW on
M403 S4=300 // spindle 4 CW on
M503 S5=100 // spindle 5 CW on
G04 X3. // wait

G114.1 K1 // enable 1st spindle syncronization
G04 X3. // wait
G114.1 R90 K2 // enable 2nd spindle syncronization
G04 X3. // wait
G51.2 P1 Q2 R60 K3 // enable 3rd spindle syncronization
G04 X3. // wait
S1500 // change spindle target speed
G04 X3. // wait
S500 // change spindle target speed
G04 X3. // wait

G113 K2 // diable 2nd spindle syncronization
G50.2 K3 // diable 3rd spindle syncronization
G113 K1 // diable 1st spindle syncronization
G04 X3. // wait

M05 // stop spindle 1
M205 // stop spindle 2
M305 // stop spindle 3
M405 // stop spindle 4
M505 // stop spindle 5
M30 // end
```

20.4.1 Synchronization error

Since the RPM of the two spindles of G51.2 can be different, the synchronous angle error is not simple subtraction but calculated using the following formula (will be displayed in System Data 45 and 46):
Synchronous angle error = (synchronous axis feedback angle - synchronous axis reference angle) -synchronous ratio* (base axis feedback angle - base axis reference angle) - phase difference

among them:
The synchronous ratio is Q / P, and the reference angle is the setting value of the Registry Table (set by F4->F4->F3)
The phase difference is R, and the feedback angle is the read value from encoder (equivalent to R761~R776)

21 G52： Local Coordinate System Setup

21.1 Command Form

G52 X__ Y__ Z__ ;

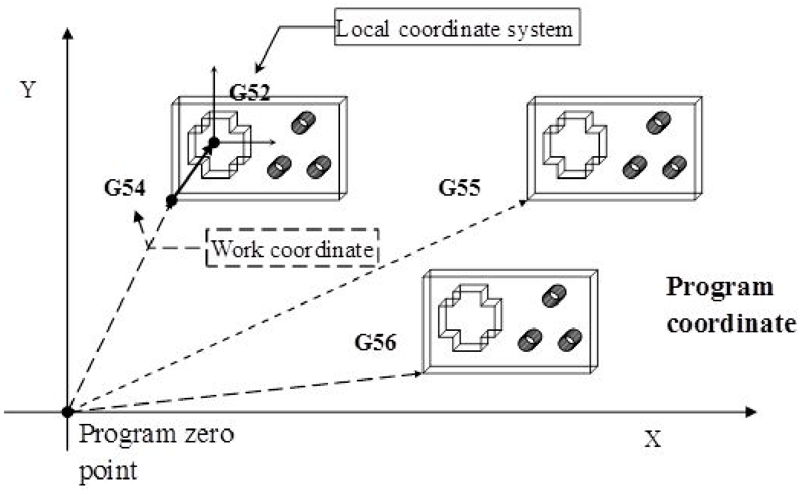
X, Y, Z: coordinate value setup

21.2 Description

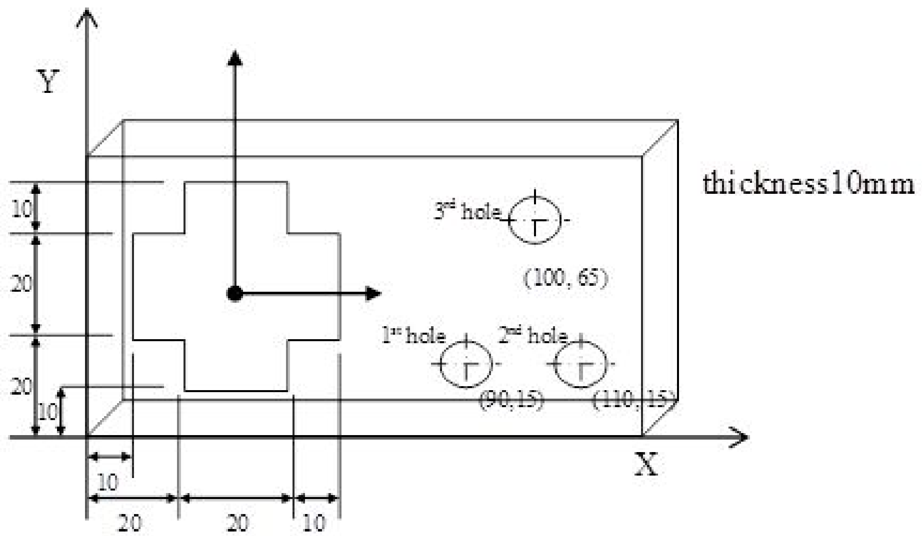
When a working coordinate system (G54~G59) is specified, if another sub-coordinate system is required for the machining workpiece, the sub-coordinate system is the local coordinate system.

G52 X0.0 Y0.0 Z0.0: cancel the local coordinate system

21.3 Coordinate System



21.4 **Program Example**



Program description:

N001 T1 S1000 M03; // tool No.1 (diameter 10mm), spindle 1000rpm (CW)

N002 G54 X0.0 Y0.0 Z0.0; // specify the working coordinate (G54)

N003 G00 X90.0 Y15.0 Z10.0; // rapid orientation to above the specified position

N004 G43 H01; // activate tool length compensation (tool No.1)

N005 G99 G81 Z-15.0 R2.0 F1000; // execute the drilling cycle, stop at point R when return, feedrate 1000mm/min, drill hole 1

N006 X110.0; // drill hole 2

N007 X100.0 Y65.0; // drill hole 3

N008 G80; // cancel cycle

N009 M05; // spindle stop

N010 G28 X0.0 Y0.0 Z10.0; // return to reference point, center point X0.0, Y0.0, Z10.0

N011 T2 M06 S1000 M03; // execute the tool change (tool No.2, diameter 10mm), spindle rotation start after tool exchange, 1000rpm (CW)

N012 G52 X30.0 Y30.0 Z0.0; // specify the zero point of local coordinate to X40.0, Y40.0, Z0.0 on the working coordinate (G54) (geometry center of workpiece)

N013 G00 X0.0 Y0.0 Z10.0; // rapid orientation to X0.0, Y0.0, Z10.0 on local coordinate (above the hole)

N014 G01 Z-12.0; // linear cutting to bottom of the hole

N015 G17 G41 D02; // activate left cutter radius compensation (tool No.2)

N016 G91 X20.0; //specify the cutting movements to be executed with increment value

N017 Y10.0;

N018 X-10.0;

N019 Y10.0;

N020 X-20.0;

N021 Y-10.0;

N022 X-10.0;

N023 Y-20.0;

N024 X10.0;

N025 Y-10.0;

N026 X20.0;

N027 Y10.0;

N028 X10.0;

N029 Y10.0;

N030 G90 G00 Z10.0; // specify the rapid orientation to be executed with absolute value (tool retract)

N031 G52 X0.0 Y0.0 Z0.0; // cancel the local coordinate system

N032 G40 M05; // deactivate the compensation, spindle stop

N033 M30; // program end

22 G53.1： Tool Alignment for Tilted Working Plane Machining

22.1 Command Form

G68.2 X_ Y_ Z_ I_ J_ K_;
G53.1 [P_];

G68.2： enable the tilted working plane coordinate system;
G53.1： tool alignment function;

P: select the moving direction of rotary axis,

0 : the 1st rotary axis (Master axis) moves along with the shortest contour;(default value)

1 : the 1st rotary axis rotates towards positive direction;

2 : the 1st rotary axis rotates towards negative direction;

Before execute the machining process, please apply G53.1 or G53.6 after G68.2 so the tool can be aligned to the tilted working plane coordinate system.

22.2 Description

After the tilted working plane coordinate system is set, G53.1 is required for the tool alignment. The G53.1 command is attached to G68.2, they must be applied at the same time.

22.3 Note

1. Do not specified G53.1 before G68.2.
2. Please apply positive tool length (G43 after G53.1)
3. After G43 is executed, the program coordinate would represent the position of tool center point. User should apply G49 after tilted working plane machining to cancel tool center point control.
4. The P argument will be 0 (default value) if it's not specified.
5. If specified arguments that are not P0, P1, P2, alarm 【COR-149 Tilted working plane machining tool alignment P argument over range】 will be issued.
6. When the argument is set 0, the system will search for the shortest moving contour for 1st rotary axis (Master axis) first . If the target angle or the route to it is out of range (Pr3009~), the another angle will be selected instead; if both target angles or routes to them are out of range(Pr3009~), alarm 【COR-153 No solution for the tool direction】 will be issued.
7. When the argument is set 1 or 2, if the target angle or the route to it is out of range(Pr3009~), alarm 【COR-153 No solution for the tool direction】 will be issued.
8. For rotary axis definitions corresponding to different mechanisms, please refer to 1.3 旋轉軸定義 and 1.4 參數說明。

	0(default)	**1**	**2**
Spindle/ Table/Mixed	1st rotary axis (Master axis)/ shortest path	1st rotary axis (Master axis)/positive	1st rotary axis (Master axis)/negative

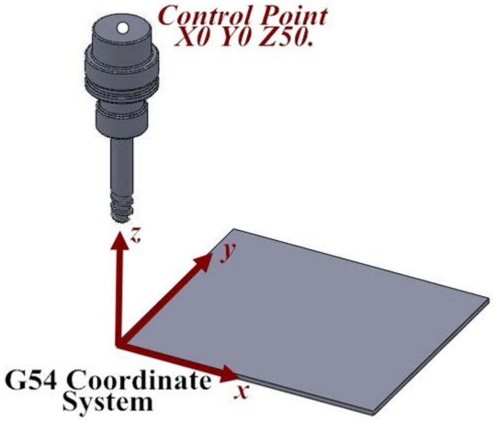
22.4 Prorgam Example

Take the program below as example, below explains the basic actions of tilted working plane coordinate

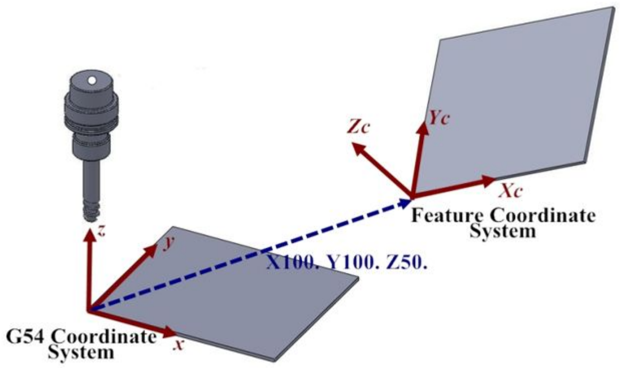
```
N1 G90 G54 G01 X0 Y0 Z50. F1000;
N2 G68.2 X100. Y100. Z50. I30. J15. K20.;
N3 G01 X0 Y0 Z50. F1000;
N4 G53.1;
N5 G43 H1;
N6 G01 X0 Y0 Z0;

... //  tilted working plane machining

N98 G49;
N99 G69;
N100 G01 X0. Y0. Z50.;
```

The example will be explain line by line below:

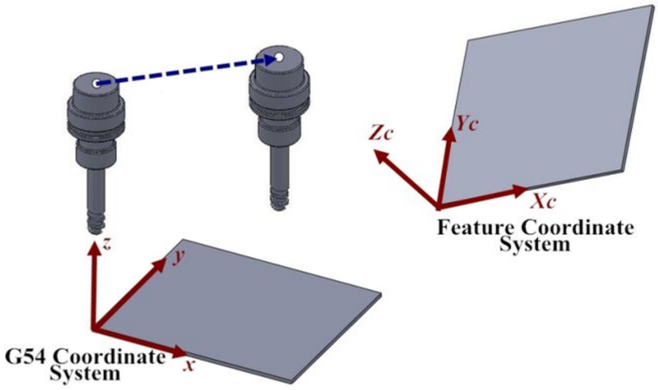
```
N1 G90 G54 G01 X0 Y0 Z50. F1000;
// interpolation to Z50 on G54 coordinate system in the speed at F1000 speed
```



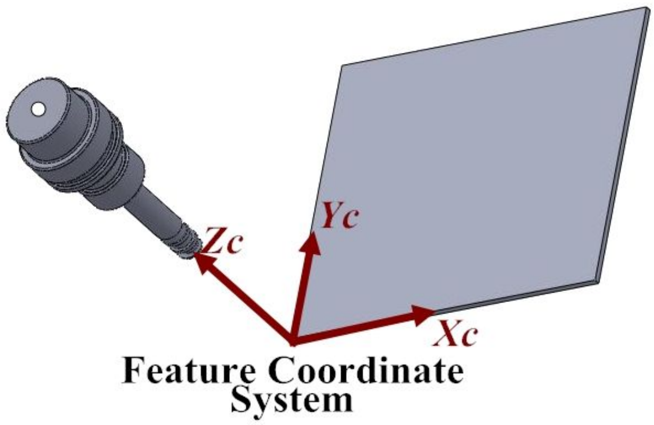

```
N2 G68.2 X100. Y100. Z50. I30. J15. K20.;
// specify X100. Y100. Z50. on G54 coordinate system as the zero of tilted working
plane coordinate system, and set the Euler angle I30. J15. K20. The program
coordinate system will change to the tilted working plane coordinate system after
command G68.2 is executed.
```



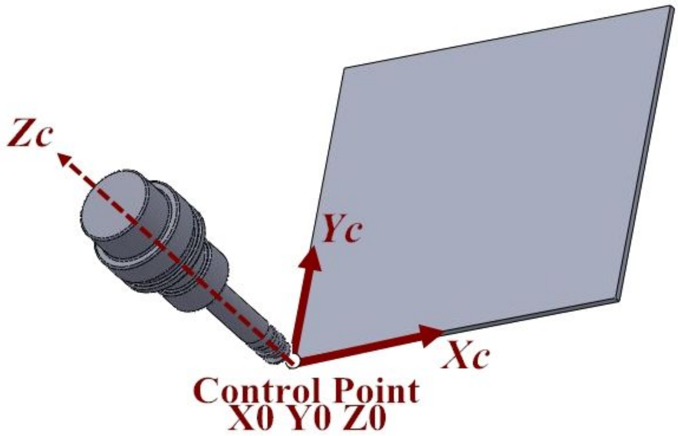

```
N3 G01 X0 Y0 Z50. F1000;
// interpolation to Z50. of tilted working plane coordinate system at F1000 speed,
but the tool direction remains the same
```



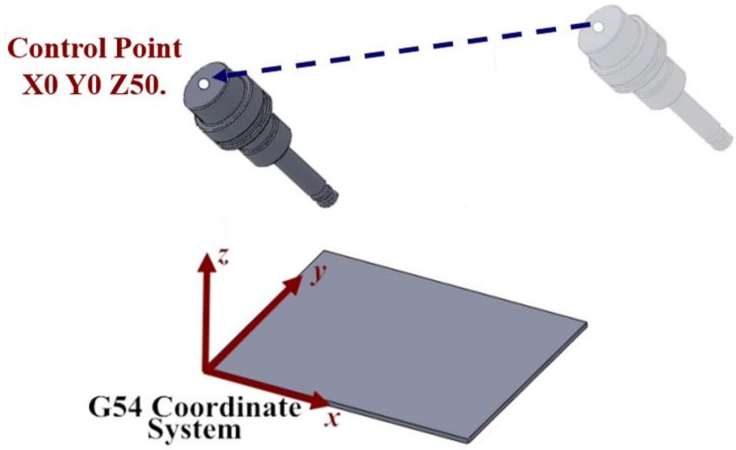

```
N4 G53.1;
// the tool direction automatically aligns to the Z axis of tilted working plane
coordinate system
```



```
N5 G43 H1;
// tool length compensation, the control point changes to the position of tool center point.

N6 G01 X0 Y0 Z0;
// interpolation to the X0 Y0 Z0 position of tilted working plane coordinate system.
```




```
N98 G49;
// cancel tool center point control
N99 G69;
// cancel tilted working plane coordinate system
N100 G01 X0. Y0. Z50.;
// interpolation to X0. Y0. Z50. of G54 coordinate system
```



22.5 Appendix

Singular point situation

When a tool vector can be achieved by "multiple position" of one rotary axis, this rotary axis is said to be in a **singular point**.

After the working plane coordinate tilting(G68.2, G68.3), there's a chance that a rotary axis is in a singular point.

Rotary axis in singular point will be fixed at its current program position after a tool alignment. And the new working plane coordinate will be based on this fixed rotary position, too.

> ⚠ Because of singular points, one tool vector can map to multiple rotary angle answers, depending on your current rotary position.
> This might cause different results of absolute position from one machine position under different mechanical chain settings.

EX1:

Assume a five-axis milling machine with +Z direction has work piece coordinate setting and position as follow:

G54P1(G54)		**Machine**		**Absolute**		**Fig.**
X	0.000	X	0.000	X	0.000	
Y	0.000	Y	0.000	Y	0.000	
Z	0.000	Z	0.000	Z	0.000	
A	0.000	A	0.000	A	0.000	
C	0.000	C	0.000	C	0.000	

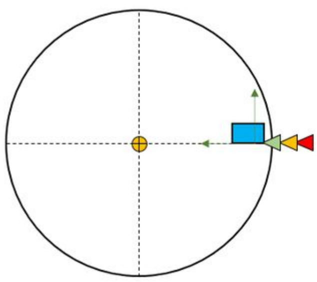
Now, execute commands below:

```
1 G68.2 X0. Y0. Z0. I0. J0. K0.
2 G53.1
```

According to working plane coordinate tilting at L1, tool direction should be +Z still.

To make tool direction +Z, axis A should be fixed at 0.000 degree, whereas axis C can be at ANY angle. That is, axis C is in a singular point.

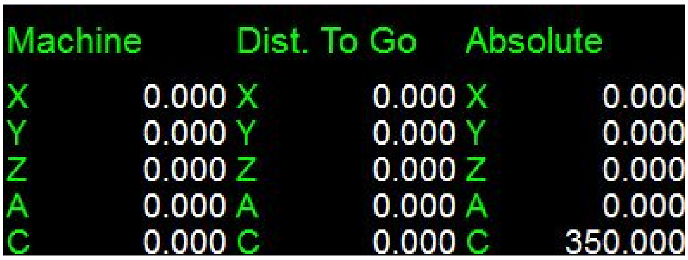
Since axis C is in a singular point, it will be fixed at 0.000 degree(in absolute coordinate).



Rotary position in Tool alignment coord. &
in Absolute coord. &
in Machine coord. = 0 °

EX2:

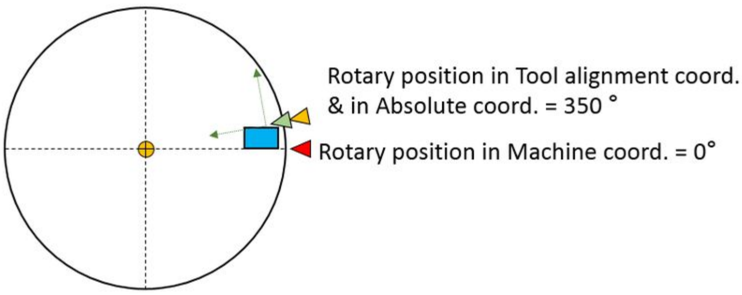
Assume a same situation like EX1, except the work piece coordinate offset of C axis is 10.000 degree:

G54P1(G54)		Machine		Absolute		Fig.
X	0.000	X	0.000	X	0.000	
Y	0.000	Y	0.000	Y	0.000	
Z	0.000	Z	0.000	Z	0.000	
A	0.000	A	0.000	A	0.000	
C	10.000	C	0.000	C	350.000	

Execute the same commands as EX1:

```
G68.2 X0. Y0. Z0. I0. J0. K0.
G53.1
```

C axis is in a singular point; thus, it will be fixed at 350.000 degree(in absolute coordinate).



23 G53： Machine Coordinate Orientation

23.1 Command Form

G53 [P1] X___ Y___ Z___ [F1=_] [Q=_];

X、Y、Z: specified position.

P1： active constant speed command

F1： feedrate mm/min or inch/min

Q： Overlapping distance

23.2 Description

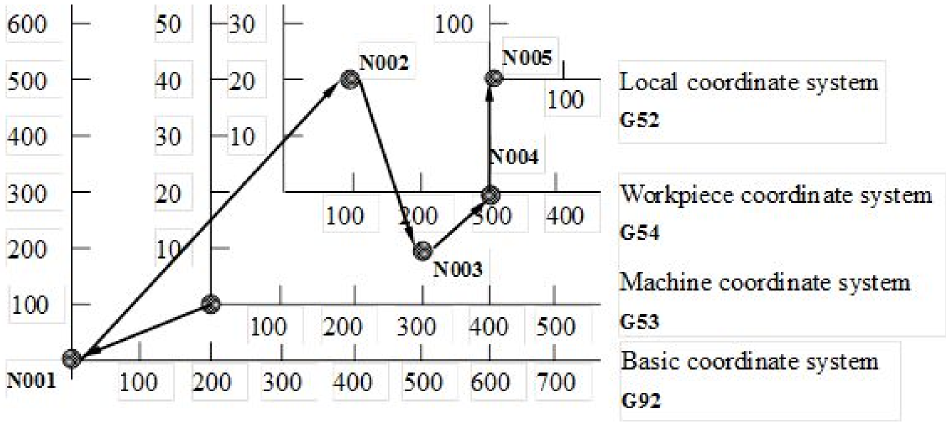
The machine zero point is a fixed point set by the machine manufactures while building the CNC machine, it's a fixed and unchangeable coordinate system. When G53 coordinate command is specified, the tool will move to the specified position on machine coordinate, when it returns to the machine zero point (0,0,0), the point is the zero point of machine coordinate system.

23.3 Note

1. G53 only affects in the specified block (returns to program coordinate system if only the coordinate arguments are given in the next block);
2. G53 only works in the absolute mode (G90), it'll simply run the increments in increment mode(G91);
3. If Pr3809 set 1 and G53 is specified together with the UVW command it won't be take as the XYZ axis inc. command.
4. Before specifying G53, please cancel the related cutter radius compensation, tool length compensation or position compensation;
5. Before setting up the coordinate system with G53, please build the coordinate system manually according to the reference point return.
6. If the axis type(Pr221~236) is set to be rotary axis, please refer to "Syntec CNC Parameter" Pr221~236 "Axis type".
7. G53 default feedrate is feedrate of G00.
8. Constant feedrate is legally with both P1 and F1 argument are determined. If P1 argument is missing, system wouldn't refer F1 argument as moving speed.
9. G70/G71 supported unit of F1 can be mm/min or inch/min.
10. Unit of F1 can be mm/min, but always remains G94. G93/G95 is invalid.
11. F1 argument support version are 10.118.41P, 10.118.48A, 10.118.49 and after.
12. When using the G53 command for axial overlapping, both the front and rear blocks are G53 commands.
13. The last G53 must have no Q parameter when at the end of the continuous G53 overlapping.
14. Q argument support version are 10.118.41P, 10.118.48A, 10.118.49 and after.

23.4 **Program Example**

23.4.1 Example1:



Program description:

N001 G92 X-200.0 Y-100.0; // specify the basic coordinate system
N002 G54 G90 X100.0 Y200.0; // move to the specified position on workpiece coordinate system
N003 G53 X300.0 Y100.0; // move to the specified position on machine coordinate system
N004 X300.0 Y0; // since G53 only affects in the specified block, this block continues the G54 command and moves to the specified position on workpiece coordinate system
N005 G52 X300.0 Y200.0; // set local coordinate system to the specified position on workpiece coordinate system

N006 X0.0 Y0.0;

23.4.2 Example2:

```
G71;
G53 X100. Y100.; // G53 feedrate is feedrate of G00
G53 P1 X50. Y50. F1=1000.; // G53 feedrate is F1=1000
G01 X0. Y0.;
M30;
```

24 G54~G59.9： Workpiece Coordinate System Setup

24.1 Command Form

$$
\left\{
\begin{array}{c}
G54\ P1(G54) \\
G54\ P2(G55) \\
G54\ P3(G56) \\
G54\ P4(G57) \\
G54\ P5(G58) \\
G54\ P6(G59) \\
G54\ P7(G59.1) \\
G54\ P8(G59.2) \\
\vdots \\
G54\ P15(G59.9) \\
G54\ P16 \\
\vdots \\
G54\ P100
\end{array}
\right.
X\_Y\_Z\_;
$$

G54 P1(G54): 1st workpiece coordinate system

:

:

G54 P6(G59): 6th workpiece coordinate system

G54 P7(G59.1): 7th workpiece coordinate system

:

:

G54 P15(G59.9): 15th workpiece coordinate system

G54 P16: 16th workpiece coordinate system

:

:

G54 P100: 100th workpiece coordinate system

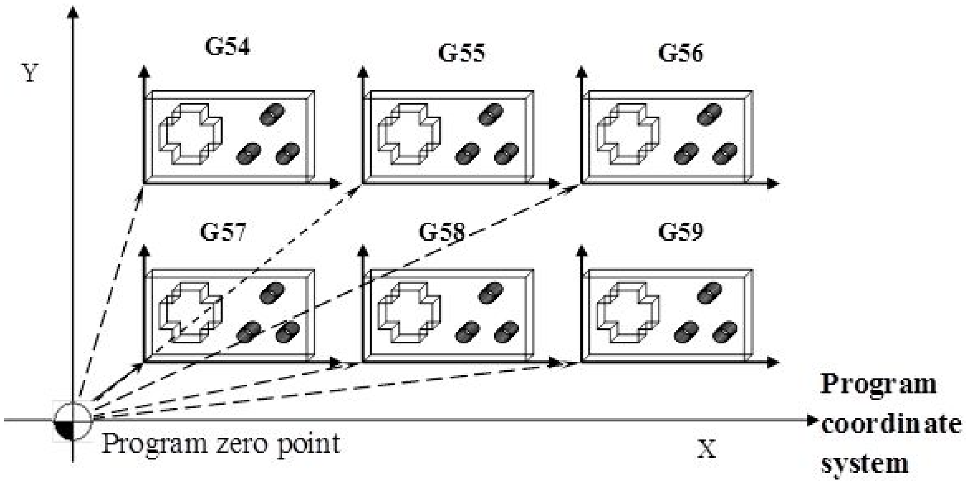
X, Y, Z: move to the specified position on the set workpiece coordinate system；

24.2 Description:

If there are multiple workpieces on the machine platform when operating. By applying workpiece coordinate systems, it's able to define their position on the machine coordinate with G54~G59, G59.1~G59.9, G54 P16~G54 P100 are 100 different coordinate systems. Therefore, the machining process can be executed one by one.

The workpiece coordinate system can be set by Pr3229, 0: activate; 1 deactivate.

24.3 Example


Y
G54
G55
G56
G57
G58
G59
Program zero point
X
Program coordinate system

25 G63： Cutting Mode

25.1 Notice

eHMC only provides command G63.

G61/G62/G64 are not supported.

25.2 Command Form

G61; // exact stop examination mode

G62; // curved surface cutting mode

G63; // tapping mode

G64; // curved surface cutting mode

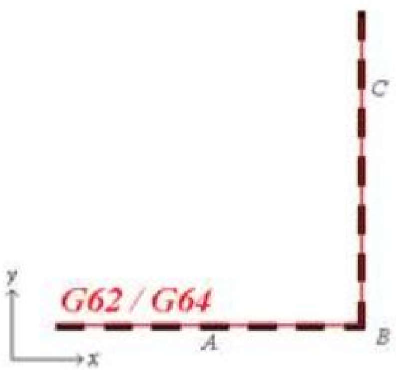
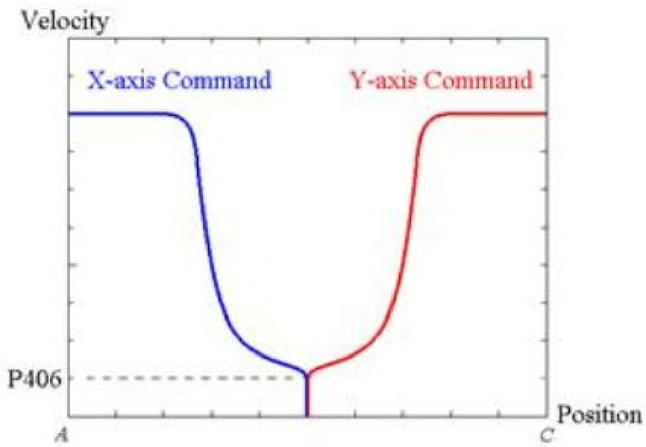
25.3 Description

The differences between each mode are listed below, the default mode is G64 cutting mode. After execute a certain mode, it'll be effective till another mode is assigned.

Command name	G code	Valid region	Description
Exact stop	G09	Only effective in block with G09.	When tool decelerates at the end of contour, The precision error occurs at the corner when the tool direction turns. G09 is used to control the precision error.
Exact stop mode	G61	Enable after execute G61, disable til G62, G63 or G64	The tool decelerates at the end of cutting contour, when it arrives at the end point, a feedback signal is sent to make sure the position is in the set range. The next contour will be executed after confirmation.
Curved Surface Cutting Mode	G62	Enable after execute G62, disable til G61, G63 or G64	Applicable to curved surface cutting. The tool does not decelerate at the end of contour(refer to the speed command curve shown below) and continues to execute next contour. Able to carry the P argument and select a high speed high accuracy parameter. (Note 2)
Tapping mode	G63	Enable after execute G63, disable til G61, G62 or G64	Applicable to tapping. The synchronous between spindle and feeding axis is determined by the ratio of spindle rotation speed S and feedrate F. During tapping, feed overrode and feed hold cannot be adjusted.

Command name	G code	Valid region	Description
Cutting mode	G64	Enable after execute G64, disable til G61, G62 or G63	Applicable to curved surface cutting. The tool does not decelerate at the end of path (refer to the speed command curve shown below) and continues to execute next path. Able to carry the P argument and select a high speed high accuracy parameter. (Note 2)

Figure: The action of G62/G64 while cutting over the corner.

G Code	Cutting Path	Speed Command Curve
G62/G64		

Explanation:

G62/G64 corner speed control mode will slow the speed down to the corner speed with Pr406 while interpolation over the corner, so there will be no command contour error at the corner. For the machining requires repeating cutting process such as mold machining, this mode provides better corner precision and reappearance.
For the corner, the vibration caused by the JERK of speed command can be improved by Pr404, set Pr404 to 10~20 can make effective improvements.

25.4 Note

1. G62 / G64 mode are more suitable for mold machining.
2. G62 Pn/G64 Pn, n = 0 ~ 9, it's able to choose a high speed high accuracy parameter, n = 21 ~ 23, it's able to choose high speed high accuracy parameter for product. Valid version is 10.118.78B, 10.118.81.
3. For multiple high speed high accuracy parameter sets, the command later one will overwrite the former one. It'll return to the default parameter (P0) after reset.
    a. Before version 10.118.42T, 10.118.48E, 10.118.52, the default parameter (P0) is Standard Parameter.
    b. After version 10.118.42T, 10.118.48E, 10.118.52, the default parameter (P0) is Pr3835 Initial Machining Condition.
4. Using G62/G64 P_ or G62/G64 R_ to choose invalid precision channel, post COR-103 alarm.
5. The G63 mode requires the use of M03/M04 to specify the tapping rotation direction. M03 indicates right-hand tapping, while M04 indicates left-hand tapping.

26 G65： Call Single Macro

26.1 Command Form

G65 P_ L_ ;

P: number of the calling program;

L: repeating times;

26.2 Description

After calling macro, the program specified by P will be called and executed, L specifies the repeating times. But it is only effective in the block with G65.

Example

G65 P10 L20 X10.0 A10.0 Q10.0;

// Call and execute sub-program O0010 20 times continuously and assign the X, A, Q value into the sub-program.

// Therefore, it's able to do the calculations with #24, #1, #17 in the sub-program.

// The arguments are not limited to be X, Y, Z, it only needs to follow the macro writing rules.

27 G66/G67： Call Modal Macro

27.1 Command Form

G66 P_ L_ ; // macro calling

G67 ; // cancel macro calling

P: number of the calling program;

L: repeating times;

27.2 Description

After the macro command G66 is called, the sub-program specified by P will be called and executed, and L specifies the repeating times of G66. If there is moving instruction in the block, the G66 block will be executed again after the moving block is completed. The mode won't stop till being cancelled by G67 (system first calculates the number of blocks with moving instruction inside between G66 and G67, and does the required repeating times while executing the G66 block)

Program Example

N001 G91
N002 G66 P10 L2 X10.0 Y10.0; // repeat the calling of sub-program O0010 twice and assign value X0.0 Y10.0 to execute
N003 X20.0; // move X axis to the position of 20.0 and call G66 P10 L2 X10.0 Y10.0
N004 Y20.0; // move Y axis to the position of 20.0 and call G66 P10 L2 X10.0 Y10.0
N005 G67 // cancel the macro calling

28 G68/G69 : Coordinate Rotation

28.1 Command Form

(G17) G68 X_ Y_ R_;

(G18) G68 Z_ X_ R_;

(G19) G68 Y_ Z_ R_;

X_, Y_, Z_: absolute coordinate of rotation center

R_: rotation angle

G69; // cancel coordinate rotation

28.2 Description

After the coordinate rotation starts, all movement commands will rotate with the rotation center, so the geometric figure will be rotated by an angle. The rotation center is only effective in absolute command, if all the commands are increment value, the actual rotation center will be the starting point of the contour.

28.3 Note

Please use G17, G18, and G19 to set the geometry axis of the working plane first. If the axial address of the G68 command is not the working plane geometry axis, the coordinate rotation function will not work.

28.4 Program Example

program 1:

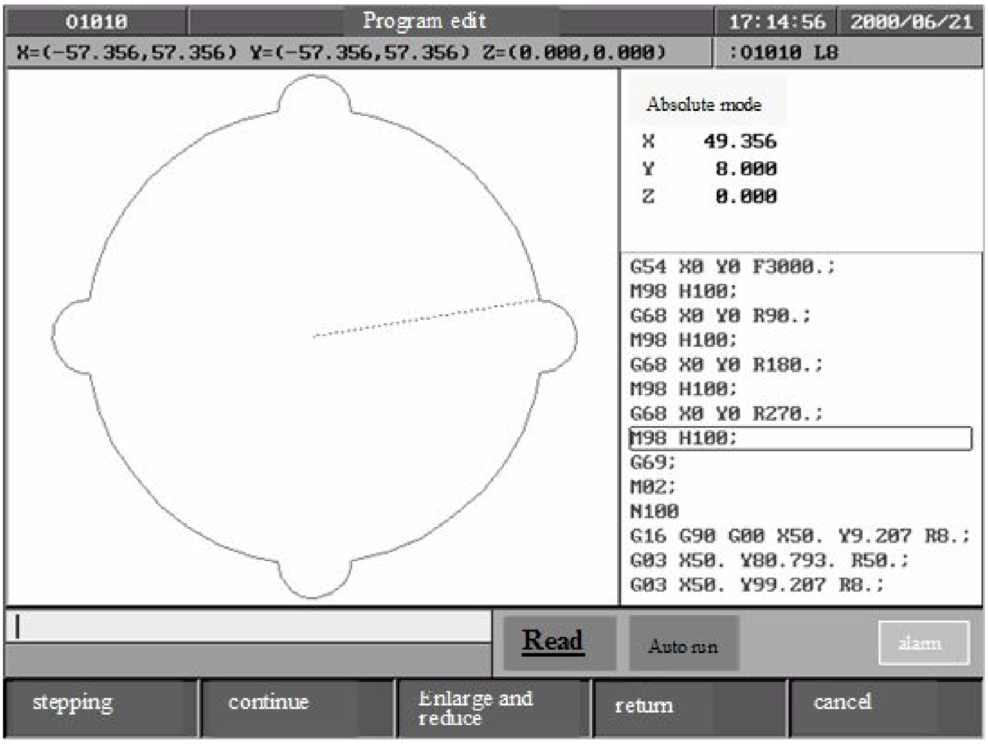
```
G54 X0 Y0 F3000.;
G16;                        // enable polar coordinates
G90 G00 X50. Y9.207 R8.;    // orientate to the starting point
M98 H100;                   // first machining process
G68 X0 Y0 R90.;             // the coordinate rotates by 90 degrees
M98 H100;                   // second machining process
G68 X0 Y0 R180.;            // the coordinate rotates by 180 degrees

M98 H100;                   // third machining process
G68 X0 Y0 R270.;            // the coordinate rotates by 270 degrees
M98 H100;                   // fourth machining process
G69;                        // disable coordinate rotation
G15;                        // disable polar coordinate
M02;                        // main program end
N100                        // contour sub-program start
```

G90 G01 X50. Y9.207 R8.;

G03 X50. Y80.793. R50.;

G03 X50. Y99.207 R8.;

M99; // contour sub-program return



program 2:

G54 X0 Y0 F3000.;

G16; // enable polar coordinate system

G90 G00 X50. Y9.207 R8.; // orientate to the starting point

M98 H100; // first machining process

G68 X0 Y0 R45.; // the coordinate rotates by 45 degrees

M98 H100; // second machining process

G68 X0 Y0 R90.; // the coordinate rotates by 90 degrees

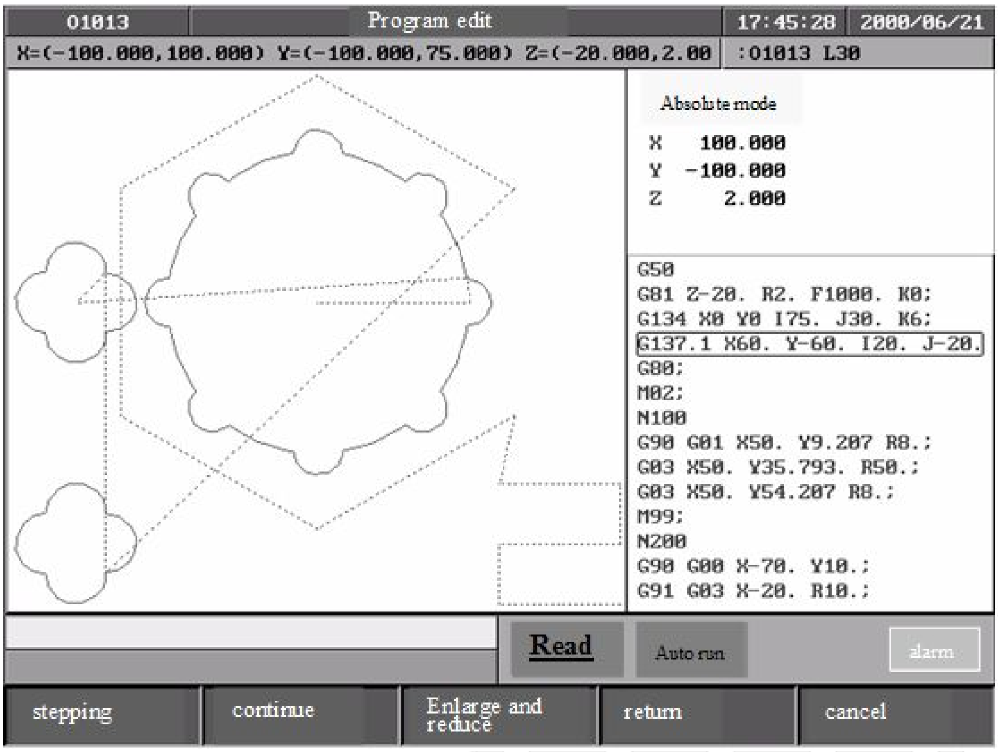
M98 H100; // third machining process

G68 X0 Y0 R135.; // the coordinate rotates by 135 degrees

M98 H100; // fourth machining process

G68 X0 Y0 R180.; // the coordinate rotates by 180 degrees

```
M98 H100;                          // fifth machining process
G68 X0 Y0 R225.;                   // the coordinate rotates by 225 degrees
M98 H100;                          // sixth machining process
G68 X0 Y0 R270.;                   // the coordinate rotates by 270 degrees
M98 H100;                          // seventh machining process
G68 X0 Y0 R315.;                   // the coordinate rotates by 315 degrees
M98 H100;                          // eighth machining process
G69;                               // disable coordinate rotation
G15;                               // disable polar coordinate system
G00 X-80. Y0.
M98 H200;                          // process first “flower”
G51.1 Y-40.;                       // symmetry axis Y-40.
M98 H200;                          // process second “flower”
G50;                               // cancel mirror image
G90 G81 Z-20. R2. F1000. K0;       // start G81 drilling cycle
G134 X0 Y0 I75. J30. K6;           // circumference hole cycle
G137.1 X60. Y-60. I20. J-20. P3 K3; // chess type hole cycle
G80;                               // cancel drilling cycle
M02;                               // main program end
N100                               // cdontour sub-program
G90 G01 X50. Y9.207;
G03 X50. Y35.793 R50.;
G03 X50. Y54.207 R8.;
M99;                               // sub-program return
N200                               // sub-program start (flower)
G90 G00 X-70. Y10.;

G91 G03 X-20. R10.;
G03 Y-20. R10.;
G03 X20. R10.;
G03 Y20. R10.;

M99;                               // sub-program return(flower)
```


O1013
Program edit
17:45:28
2000/06/21
X=(-100.000,100.000) Y=(-100.000,75.000) Z=(-20.000,2.00
:O1013 L30
Absolute mode
X 100.000
Y -100.000
Z 2.000
G50
G81 Z-20. R2. F1000. K0;
G134 X0 Y0 I75. J30. K6;
G137.1 X60. Y-60. I20. J-20.
G80;
M02;
N100
G90 G01 X50. Y9.207 R8.;
G03 X50. Y35.793. R50.;
G03 X50. Y54.207 R8.;
M99;
N200
G90 G00 X-70. Y10.;
G91 G03 X-20. R10.;
Read
Auto run
alarm
stepping
continue
Enlarge and reduce
return
cancel

29 G68.2： Tilted Working Plane Machining

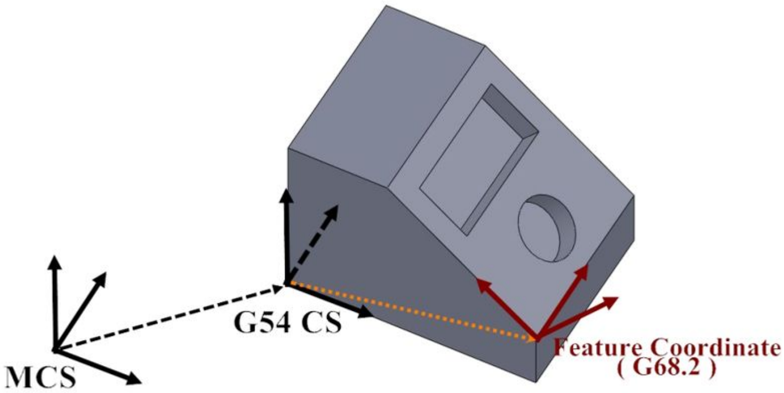
Command Form

```
G68.2 X_ Y_ Z_ I_ J_ K_;
G69 ;
```

G68.2： enable the tilted working plane coordinate function;
G69： disable the tilted working plane coordinate function;
X_ Y_ Z_： zero point of tilted working plane coordinate (relative to the zero point of G54 coordinate system);
I_ J_ K_： Euler angle of tilted working plane coordinate;

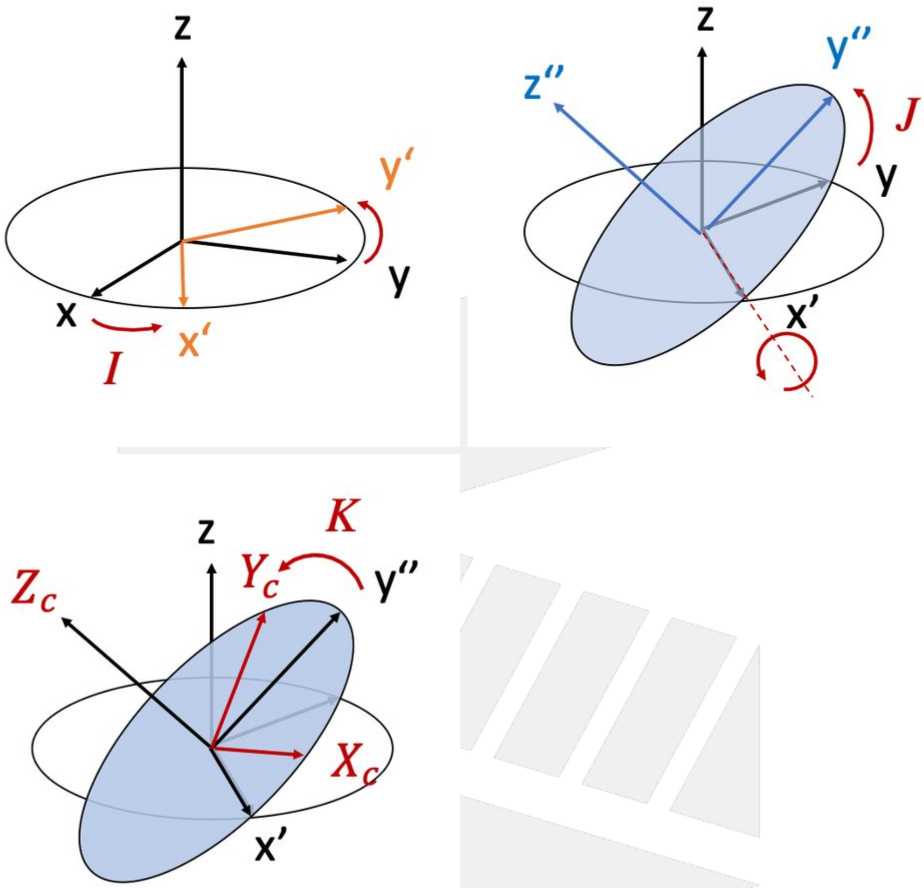
29.1 Description

1. Tilted Working Plane Machining, also called Feature Coordinate, is able to rotate the program coordinates to any specified angle and execute the machining process with the normal 3-axis machining program.
2. The zero point of tilted working plane coordinate is set related to the G54 coordinate system, shown as the orange dotted line in the picture below; and the tilting angle is set by the Euler angle.



3. For the zero point setup of tilted working plane coordinate, which are XYZ arguments after G68.2, only needs to specified the program zero point and the distance of each axis from the G54 coordinate system zero point.
4. The Euler angle is required for the setup of tilted working plane coordinate angle, the Euler angle has a clear definition. The IJK arguments after G68.2 represent for 3 rotating angle, respectively Z axis-X axis- Z axis in order.
   a. First, rotate the original XYZ coordinate system around Z axis by a certain angle, we get a new coordinate system X'Y'Z', the rotated angle is defined as angle I;
   b. Then rotate the original X'Y'Z' coordinate system around X' axis by another certain angle, we get a new coordinate system X''Y''Z'', the rotated angle is defined as angle J,
   c. Last, rotate the original X''Y''Z'' coordinate system around Z'' axis by another certain angle, we get a new coordinate system XcYcZc, the rotated angle is defined as angle K,

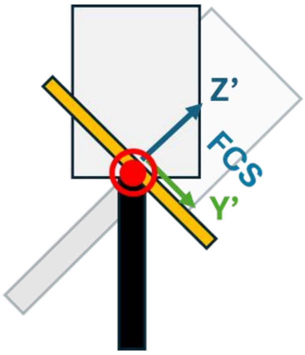
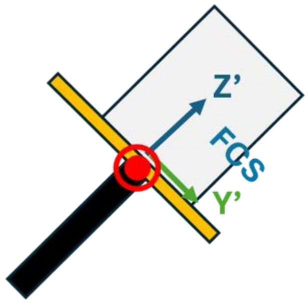
d. XcYcZc is the final tilted working plane coordinate system.



5. Once the feature coordinate function is activated, the fixed-angle RTCP function will take effect when G53.1, G53.3, or G53.6 is subsequently called.
   After the fixed-angle RTCP function is enabled, the control point position will be calculated based on the machine's kinematic chain.
   For example:

Example

```
N1 G01 A-45. // move rotation axis before launch feature
coordinate
N2 G68.2 X0. Y0. Z0. I0. J-45. K0.
N3 G53.1 // launch Fix-Angle RTCP
N4 G43 H1
```

Before G53.1	After G53.1
	

6. Related Parameters: Pr3014 Tilted working plane coordinate system retaining state

Number	Name	Range	Unit	illustration
3014	Tilted working plane coordinate system retaining state	[0,2]		0: Not retaining the tilted working plane coordinate set by G68.2 after reset or power off/on. 1: Retain the tilted working plane coordinate set by G68.2 after reset but clear the coordinate system after reboot. 2: Always retain the tilted working plane coordinate set by G68.2 after reset or reboot.

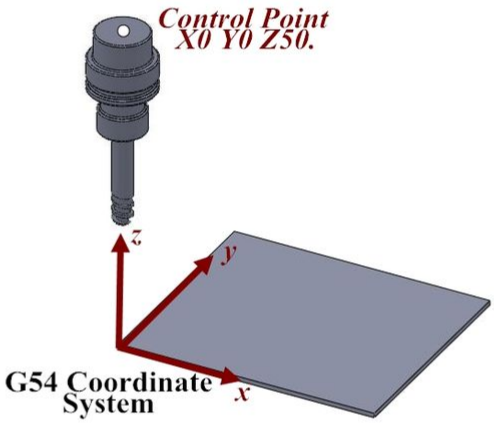
29.2 Note

1. G68.2 can be executed for multiple times.
2. Each setting is relative to G54 coordinate.
3. Do not specified command G53.1 before G68.2.
4. Do not move the rotary axes individually after activating G68.2. If you need to align the tool direction to the feature coordinate for machining, use the tool alignment function to align the tool accordingly before starting the operation.

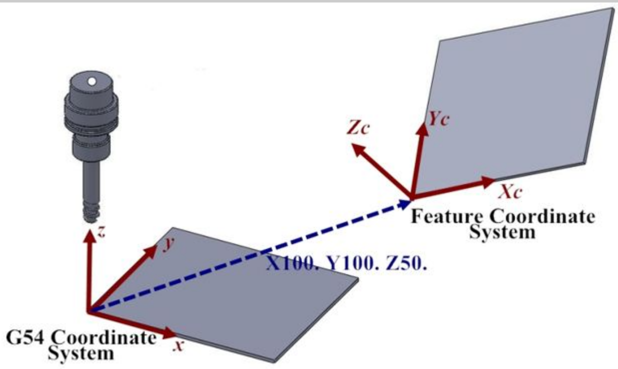
29.3 Program Example

```
N1 G90 G54 G01 X0 Y0 Z50. F1000;
N2 G68.2 X100. Y100. Z50. I30. J15. K20.;
N3 G01 X0 Y0 Z50. F1000;
N4 G53.1;
N5 G43 H1;
N6 G01 X0 Y0 Z0;
```

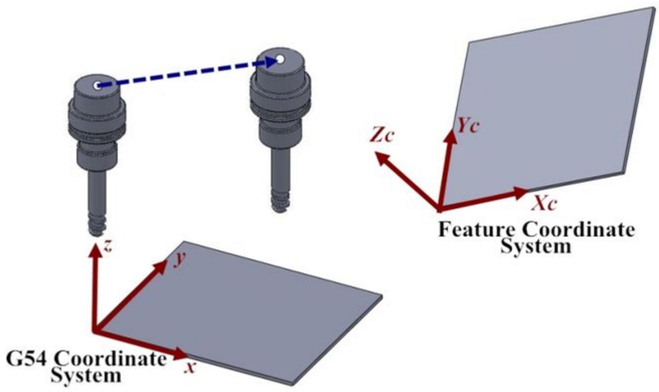
```
N1 G90 G54 G01 X0 Y0 Z50. F1000;     //interpolation to Z50 on G54 coordinate system at F1000 speed
```



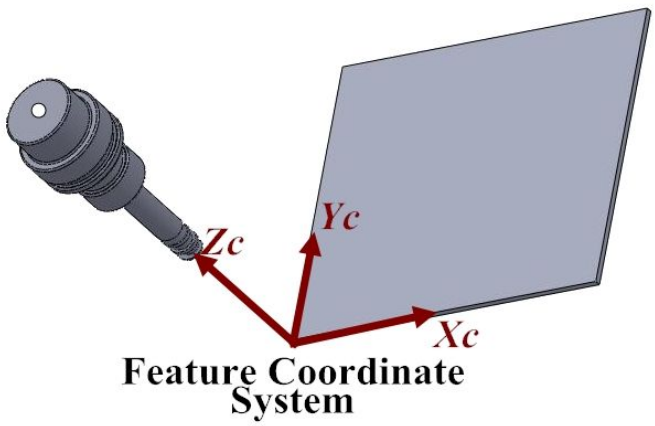

```
N2 G68.2 X100. Y100. Z50. I30. J15. K20.;     //specify X100. Y100. Z50. on G54 coordinate system as the zero point of tilted working plane coordinate, and set the Euler angle I30. J15. K20.. The program coordinate system will change to the tilted working plane coordinate after G68.2 is executed.
```



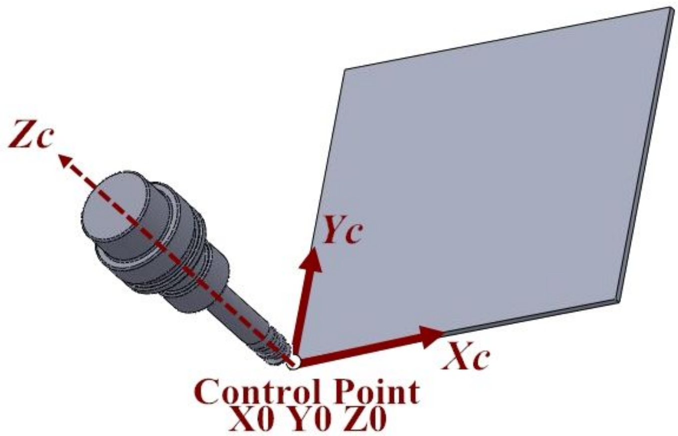

```
N3 G01 X0 Y0 Z50. F1000;     //interpolation to Z50. of tilted working plane coordinate at F1000 feedrate, but the tool direction remains the same
```




```
N4 G53.1;     //the tool direction automatically aligns to the Z axis of tilted
working plane coordinate system
```




```
N5 G43 H1;     //tool length compensation, the controlling point changes to the
position of tool center point
N6 G01 X0 Y0 Z0;     //interpolation to the X0 Y0 Z0 position of tilted working plane
coordinate.
```


Zc
Yc
Xc
Control Point
X0 Y0 Z0

30 G70/G71： Imperial/SI Unit Setup

30.1 Command Form

G70; Imperial Unit Setup

G71; SI Unit Setup

30.2 Description

The origin offset value of workpiece coordinate, tool data, system parameter, and reference point will still be correct after switching the unit system. The system will deal with the unit transform automatically.

After changing the unit system, the operation units below will be changed:

- the coordinate, unit of speed
- increment JOG unit
- MPG JOG unit

30.3 Note

The rotary axis has no imperial unit, therefore when executing the synchronous moving command of linear axis and rotary axis, the command value of linear axis is divided by 25.4 but the command value of rotary axis will remain the same. Thus, for the synthesis speed, the percentage of rotary axis will rise massively and make the linear axis speed decreases rapidly, please be extra careful with the situation.

31 G73： High Speed Peck Drill Cycle

31.1 Command Form

G73 X_ Y_ Z_ R_ Q_ F_ K_ ;

X(U) or Y(V): hole position(absolute/increment, please set parameter 3809 is 1 when using increment value)

Z:

G91: the distance from the initial point to Z point (directional)

G90: program coordinate of Z point

R:

G91: the distance from the initial point to R point (directional)

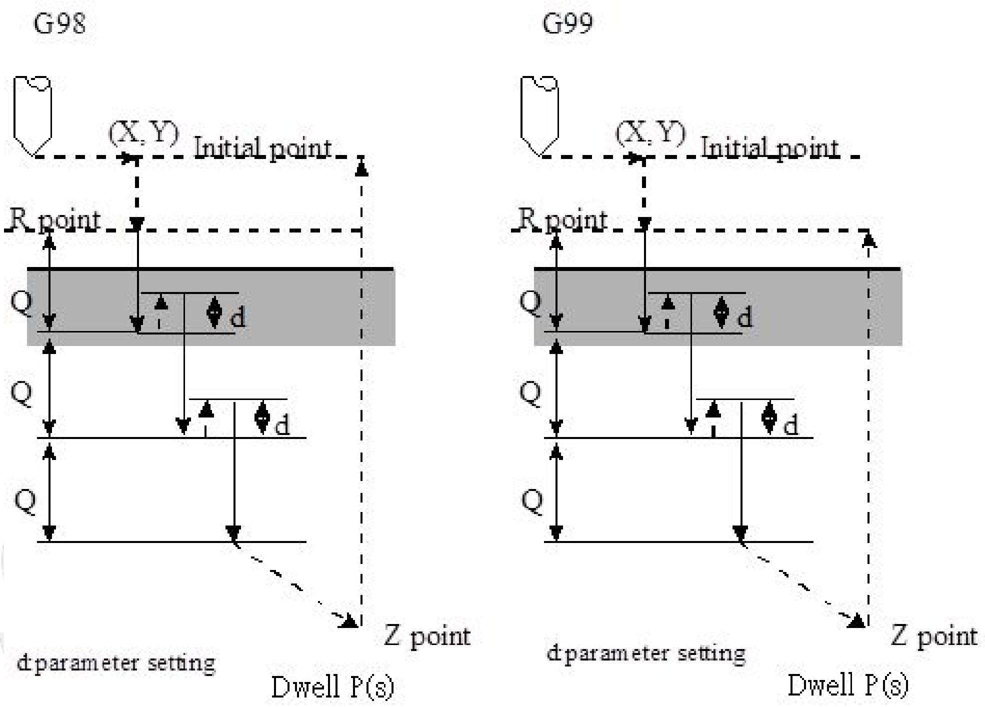
G90: program coordinate of R point

Q: depth of each feed (increment and positive, the minus will be ignored)

F: feedrate

K: repeating times (repeating the moving and drilling, G91 increment input is effective)

Absolute/Increment coordinate of X, Y, Z, R can be specified by G90/G91.



31.2 Description

The process of the action is shown below:

1. Start the machining process, move the tool to specified (X,Y) with G00.
2. Move the tool to the specified R point with G00.

3. Downwards a distance Q relative to the current drilling depth with G01.
4. Lift up a retract depth d with G01 (Pr4002).
5. Repeat drilling above till reaches the bottom Z point.
6. Lift up to initial point(G98) or R point with G00 (G99).

31.3 Note

1. Please start spindle rotation with M code before applying G73.
2. If M Code and G73 are specified in the same block ,the M Code will only be executed once in the block.
3. When specified to repeat K times, M Code will only be executed at first cycle, it won't be executed for others.
4. G73 is a module G Code, effective after being executed. If an (X,Y) coordinate is the only argument given in the next line ,the controller will start the tapping at (X,Y).
5. G73 can be cancelled by G80, or being cancelled automatically when the program executes G00, G01, G02, G03 or other G code cycles.
6. Before changing the drilling axis, G73 cycle cycle must be cancelled first.
7. If no axis (X, Y, Z) moving command is included in the block, the drilling action won't be executed.
8. The argument specified by Q and R will only be set when executing the block with drilling action, it won't be set in the blocks with no drilling actions.
9. G codes G00, G01, G02, G03 can't be specified in the same block with G73, or the G73 cycle command will be cancelled.
10. In G73 cycle, the tool radius compensation (G41/G42/G40) will be ignored.
11. Arguments X,Y,Z,A,B,C,U,V,W are valid when the parameter 3809 is 0.
12. The increment command for drilling axis is ignored when the increment and absolute command for drilling axis are commanded simultaneously.

31.4 Program Example

N001 F1000. S500;

N002 M03; // start the drill, CW rotation

N003 G90;

N004 G00 X0. Y0. Z10.; // move to the initial point

N005 G17;

N006 G90 G99; //specify the coordinate of R point, Z point and hole NO.1, cutting depth 2.0

N007 G73 X5. Y5. Z-10. R-5. Q2.;

N008 X15.; // hole NO.2

N009 Y15.; // hole NO.3

N010 G98 X5.; // hole NO.4, return to the initial point

N011 X10. Y10. Z-20.; // hole NO.5, set new Z point as -20

N012 G80;

N013 M05; // stop the tool

N014 M30;

32 G81/G82/G83/G85/G86： Regular Drilling Cycles

32.1 Notice

eHMC only provides G81/G82/G83/G85/G86.

G84 and G87~G89 are not supported.

Description

Regular Drilling Cycles simplifies the drilling process from multiple blocks into a specified G code in one block.
Regular Drilling Cycles:

Code	Name		Drilling Direction & Action	Whether Pause at the Bottom of the Hole	Spindle Action at the Bottom of the Hole	Break Away Action	Usage
G80							Cancel the Cycle
G81	Normal Drilling (Pr4008 = 0)		-Z, Cutting Feed	No	-	Rapid Feeding	Drilling Cycle
	Rapid Drilling (Pr4008 = 1)		-Z, Cutting Feed	No	-	Rapid Feeding (unspecified F2)/ Cutting Feed F2	
G82	Drilling Cycle with Dwelling at Hole Bottom		-Z, Cutting Feed	Yes	-	Rapid Feeding	Drilling Cycle
G83	Peck Drilling		-Z, Cutting Feed	No	-	Rapid Feeding	Drilling Cycle
G84	Tapping Cycle (without Q)	Normal	-Z, Cutting Feed	Yes	Pause + CCW Rotation	**to Point R:** Cutting Feed **Point R to Initial Point:** Rapid Feeding	Tapping Cycle

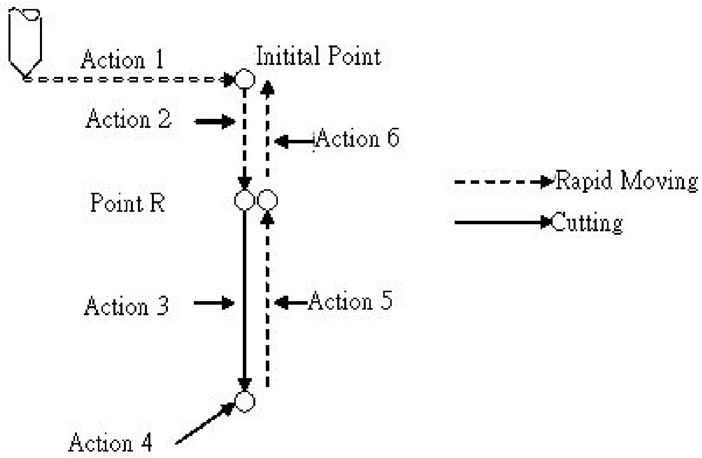
Code	Name		Drilling Direction & Action	Whether Pause at the Bottom of the Hole	Spindle Action at the Bottom of the Hole	Break Away Action	Usage
		Rapid	-Z, Cutting Feed	No	CCW Rotation	**to Point R:** Cutting Feed **Point R to Initial Point:** Rapid Feeding	
	Tapping Cycle (with Q)	Rapid Peck Tapping	-Z, Cutting Feed	Yes	Pause + CCW Rotation	**to Point R:** Cutting Feed **retract depth d:** Cutting Feed **Point R to Initial Point:** Rapid Feeding	
		Normal Peck Tapping	-Z, Cutting Feed	Yes	Pause + CCW Rotation	**to Point R:** Cutting Feed **Point R to Initial Point:** Rapid Feeding	
G85	Drilling Cycle		-Z, Cutting Feed	No	-	**to Point R:** Cutting Feed **Point R to Initial Point:** Rapid Feeding	Drilling Cycle
G86	Rapid Drilling Cycle		-Z, Cutting Feed	No	Pause	Rapid Feeding	Drilling Cycle
G87	Fine Boring Cycle on Back		+Z, Cutting Feed	Yes	Spindle locates and stops	Rapid Feeding	Boring Cycle

Code	Name	Drilling Direction & Action	Whether Pause at the Bottom of the Hole	Spindle Action at the Bottom of the Hole	Break Away Action	Usage
G88	Semiautomatic Fine Boring Cycle	-Z, Cutting Feed	Yes (Include M0 program command)	Pause	Manual	Boring Cycle
G89	Boring Cycle with Dwelling at Hole Bottom	-Z, Cutting Feed	Yes	-	**to Point R:** Cutting Feed **Point R to Initial Point:** Rapid Feeding	Boring Cycle

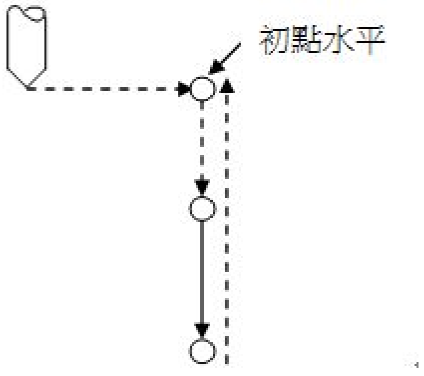
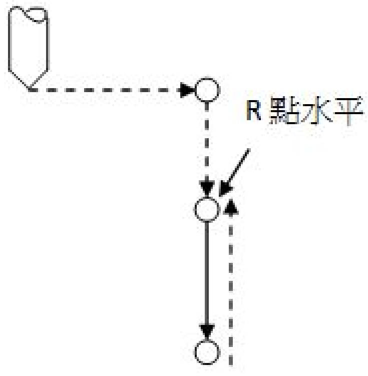
Note 1: Execute drill CCW rotation with M04 in G code.
Note 2: Triggering conditions of G84 Rapid Tapping Cycle: (No Q Argument) and (Serial Spindle or P-Type Spindle) and (No P Argument)

In general, the drilling cycle is composed of the 6 continuous actions below:

1. Rapid orientate to the specified point on X, Y axes.
2. Rapid moving to point R.
3. Drilling process.
4. Actions at the bottom of the hole.
5. Leave and move to point R.
6. Rapid moving to the initial point.



For the return action, specified the tool to return to R point or initial point with G98/G99. (Please refer to the picture below)
The initial point won't change even the drilling action is executed in the way of G99. If the last return position is the initial point, the starting point will be the initial point; if it's R point, then will be the R point.

G98	G99
	

33 G81： Drilling Cycle

33.1 Command Form

G81 X_Y_Z_R_F_F2=_K_Q_D_

X(U) or Y(V): hole position(absolute/increment, please set parameter 3809 is 1 when using increment value)

Z:

G91: the distance from the R point to Z point (directional)

G90: program coordinate of Z point

R:

G91: the distance from the initial point to R point (directional)

G90: program coordinate of R point

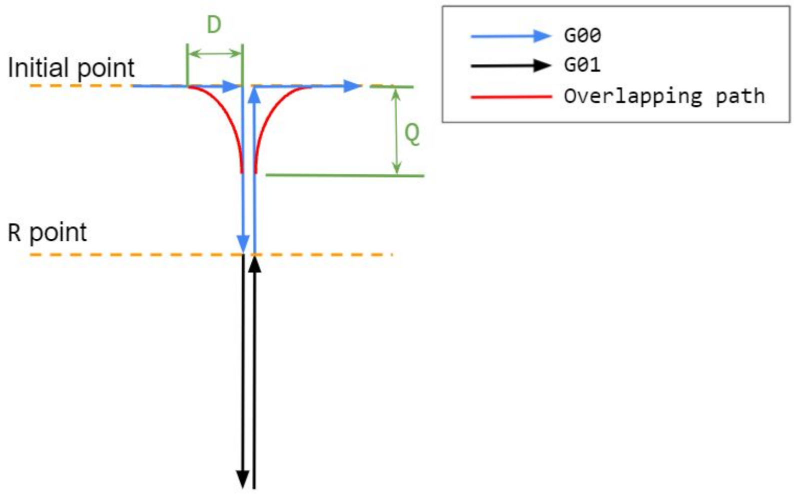
F: feedrate

F2=: Lifting rate (when Pr4008 is set 0 or the drilling mode is switch to normal drilling mode, F2 command is invalid)

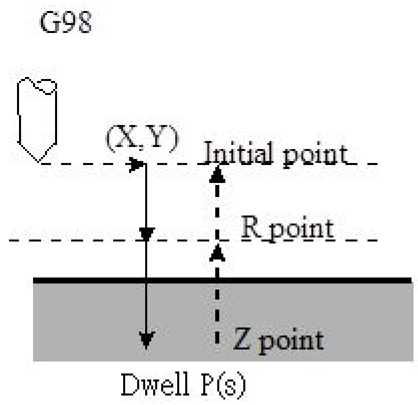
K: repeating times (repeating the moving and drilling actions, G91 increment input is effective)

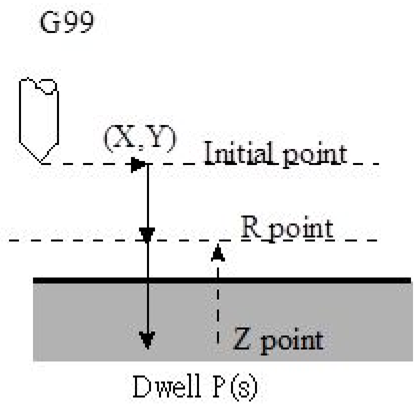
Q: Overlapping distance of the drilling axis (When the drilling finishes, it is the overlapping distance from exit the hole along the drilling axis to the next G00.). It is valid when Pr4008(setting for drilling/tapping mode) value is 1. Unit: LIU.

D: Overlapping distance of the positioning axis (After the previous drilling finishes, it is the overlapping distance from G00 positioning to the current position.). It is valid when Pr4008(setting for drilling/tapping mode) value is 1. Unit: LIU.



Absolute/Increment coordinate of X, Y, Z, R is decided by G90/G91.





33.2 <u>**Description**</u>

1. The process of G81
    a. Start the machining process, move the tool to specified (X,Y) with G00.
    b. Move the tool to the specified R point with G00.
    c. Move the tool to the Z point at the bottom of the hole with G01.
    d. Lift up to the initial point (G98) or the R point (G99) with F2 speed . If F2 is not specified or smaller than 0, the lifting speed default is G00 speed.
2. .The action of overlapping
    a. When the program has two consecutive blocks of G81 or the consecutive blocks of G81 and G00, the second block will start when the first block remains a certain distance from the finish. The distance is called "overlapping distance", see Q & D in the figure above.
    b. Overlapping is suitable for the continuous drilling cycle. It does not need Q/D argument in every line of the program but 'overlaps' by a constant distance. In Program example 2, the overlapping distance of the drilling axis and the positioning axis of each drilling command is 2 and 3 respectively. And the overlapping distance will not reset to zero until the disable command G80 is executed.
    c. In order to enable overlapping, set Pr4008 to 1, otherwise the Q and D parameters have no effect; If Pr4008 is set to 1 but the Q and D arguments are not given, the overlapping still won't be executed.

33.2.1 TYPE I: Normal Drilling

1. Effective condition: Pr4008 = 0。
2. Not supporting F2 argument, thus unable to control the lifting speed.
3. MPG simulation, Feedhold, Reset and G01 feedrate switching are valid during drilling process.

33.2.2 TYPEll: Rapid Drilling

1. Effective condition: Pr4008 = 1 and G01 feedrate before drilling should be 100%.
2. Better precision at the bottom of the hole.
3. The machine action is smoother when Z axis reverse at the bottom of the hole.
4. Not able to activate MPG simulation, Feedhold, Reset and change the G01 feedrate during the drilling process.
5. Do not modify the G01 feedrate during the drilling process of multiple holes, it might be switched to the normal drilling and it's unable to switch back.

33.3 Note

1. Please start spindle rotation with M code before applying G81.
2. If M Code and G81 are specified in the same block ,the M Code will only be executed once in the block.
3. When specified to repeat K times, M Code will only be executed at first cycle, it won't be executed for others.
4. Before changing the drilling axis, G81 cycle command must be cancelled first.
5. If no axis (X, Y, Z) moving command is included in the block, the drilling action won't be executed.
6. The argument specified R will only be set when executing the block with drilling action, it won't be set in the blocks with no drilling actions.
7. G codes G00, G01, G02, G03 can't be specified in the same block with G81, or the G81 cycle command will be cancelled.
8. In G81 cycle, the tool radius compensation (G41/G42/G40) will be ignored.
9. If the F2 argument is not an integer, the system alarm [COR-045 L Code must be an integer] will be issued.
10. It's unable to apply the overlapping function when C40 is On.
11. The valid version for Pr4008 in drilling mode: 10.116.10J, 10.116.16B
12. Arguments X,Y,Z,A,B,C,U,V,W are recognized as axis command when the parameter 3809 is 0.
13. The increment command for drilling axis is ignored when the increment and absolute command for drilling axis are commanded simultaneously.

33.4 Program Example

33.4.1 Example 1

```
N001 F1000. S500;
N002 G90;
N003 G00 X0. Y0. Z10.; // move to the initial point
N004 G17;
N005 M03; // start drill CW rotation
N006 G90 G99; // specify the coordinate of R point, Z point and hole NO.1
N007 G81 X5. Y5. Z-10. R-5. F2=1000; // set the lift up speed to 1000
N008 X15.; // hole NO.2
N009 Y15.; // hole NO.3
N010 G98 X5.; // hole NO.4, and return to the initial point
N011 X10. Y10. Z-20.; // hole NO.5, set new z point as -20
N012 G80;
N013 M05; // stop the tool
N014 M30;
```

33.4.2 Example 2

```
N001 F1000. S500;
N002 G90;
N003 G00 X0. Y0. Z10.; // move to the initial point
N004 G17;
N005 M03; // start drill CW rotation
N006 G90 G99; //  specify the coordinate of R point, Z point and hole NO.1
N007 G81 X5. Y5. Z-10. R-5. Q2. D3.; // set the overlapping distance of the drilling axis(Q) and the positioning axis(D)
to 2 and 3 respectively
N008 X15.; // hole NO.2
N009 Y15.; // hole NO.3
N010 G98 X5.; // hole NO.4, and return to the initial point
```

```
N011 X10. Y10. Z-20.; // hole NO.5, set new z point as -20
N012 G80;
N013 M05; // stop the tool
N014 M30;
```

34 G82： Drilling Cycle with Dwelling at Hole Bottom

中文版 Mandarin Version: G82： 孔底暫停鑽孔循環

34.1 Command Form

G82 X_ Y_ Z_ R_ P_ F_ K_ ;

X(U) or Y(V): hole position(absolute/increment, please set parameter 3809 is 1 when using increment value)

Z:

G91: the distance from the initial point to Z point (directional)

G90: program coordinate of Z point

R:

G91: the distance from the initial point to R point (directional)

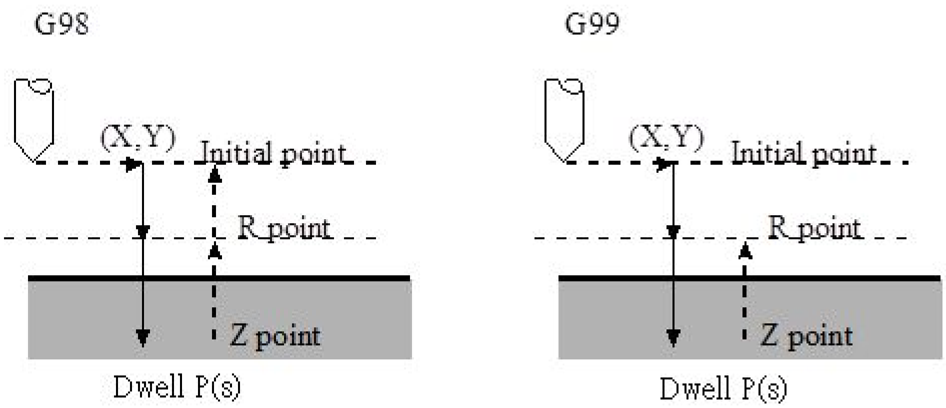
G90: program coordinate of R point

P: dwelling time(sec)

F: feedrate

K: repeating times (repeating the moving and drilling, G91 increment input is effective)

Absolute/Increment coordinate of X, Y, Z, R can be specified by G90/G91.



34.2 Description

1. Start the machining process, move the tool to specified (X,Y) with G00.
2. Move the tool to the specified R point with G00.
3. Move the tool to the Z point at hole bottom with G01.
4. Dwell for P seconds.
5. Lift up to the initial point (G98) or the R point of the program (G99) with G00.

34.3 Notes

1. Please start spindle rotation with M code before applying G82.
2. If M Code and G82 are specified in the same block ,the M Code will only be executed once in the block.

3. When specified to repeat K times, M Code will only be executed at first cycle, it won't be executed for others.
4. Before changing the drilling axis, G82 cycle must be cancelled first.
5. If no axis (X, Y, Z) moving command is included in the block, the drilling action won't be executed.
6. The argument specified by R will only be set when executing the block with drilling action, it won't be set in the blocks with no drilling actions.
7. G codes G00, G01, G02, G03 can't be specified in the same block with G82, or G82 command will be cancelled.
8. In G82, the tool radius compensation (G41/G42/G40) will be ignored.
9. Arguments X,Y,Z,A,B,C,U,V,W are recognized as axis command when the parameter 3809 is 0.
10. The increment command for drilling axis is ignored when the increment and absolute command for drilling axis are commanded simultaneously.

34.4 Example

```
N001 F1000. S500;
N002 G90;
N003 G00 X0. Y0. Z10.; // move to the initial point
N004 G17;
N005 M03; // start drill CW rotation
N006 G90 G99;
// specify the coordinate of R point, Z point and hole NO.1, dwelling time 2 seconds
N007 G82 X5. Y5. Z-10. R-5. P2.;
N008 X15.; // hole NO.2
N009 Y15.; // hole NO.3
N010 G98 X5.; // hole NO.4, return to the initial point
N011 G80;
N012 M05; // stop the tool
N013 M30;
```

35 G83： Peck Drill Cycle

Command Form

G83 X_Y_Z_R_Q_F_K_;

X(U) or Y(V): hole position(absolute/increment, please set parameter 3809 is 1 when using increment value)

Z:

G91: the distance from the initial point to Z point (directional)

G90: program coordinate of Z point

R:

G91: the distance from the initial point to R point (directional)

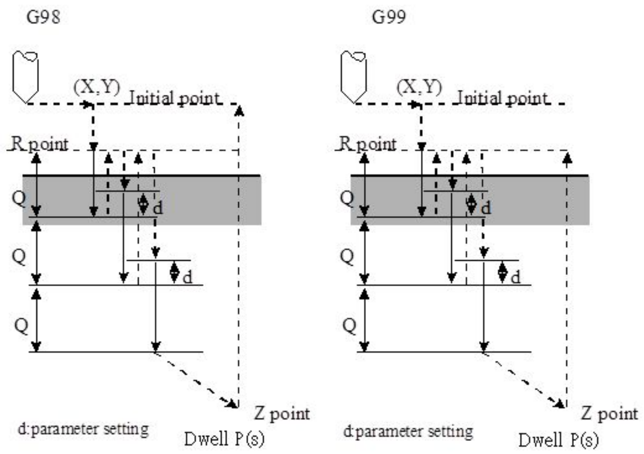
G90: program coordinate of R point

Q: depth of each feed (increment and positive, the minus will be ignored)

F: feedrate

K: repeating times (repeating the moving and drilling, G91 increment input is effective)

Absolute/Increment coordinate of X, Y, Z, R can be specified by G90/G91.



35.1 Description

1. Start the machining process, move the tool to specified (X,Y) with G00.
2. Move the tool to the specified R point with G00.
3. Downwards a distance Q relative to the current drilling depth with G01.
4. Lift the tool up to point R on workpiece surface with G00.
5. Move the tool to the position with a retracing distance d above the current cutting depth. (set by Pr4002)
6. Move the tool for a distance of a cutting feed Q from the current cutting depth with G01 again.
7. Lift the tool up to point R on workpiece surface with G00.
8. Repeat the steps above till reaching point at the bottom of the hole.
9. Lift the tool up to the initial point (G98) or point R of the program (G99).

35.2 Note

1. Please start spindle rotation with M code before applying G83.
2. If M Code and G83 are specified in the same block ,the M Code will only be executed once in the block.
3. When specified to repeat K times, M Code will only be executed at first cycle, it won't be executed for others.
4. Before changing the drilling axis, G83 cycle command must be cancelled first.
5. If no axis (X, Y, Z) moving command is included in the block, the drilling action won't be executed.
6. The argument specified by Q and R will only be set when executing the block with drilling action, it won't be set in the blocks with no drilling actions.
7. G codes G00, G01, G02, G03 can't be specified in the same block with G83, or the G83 cycle command will be cancelled.
8. In G83 cycle, the tool radius compensation (G41/G42/G40) will be ignored.
9. Arguments X,Y,Z,A,B,C,U,V,W are recognized as axis command when the parameter 3809 is 0.
10. The increment command for drilling axis is ignored when the increment and absolute command for drilling axis are commanded simultaneously.

Program Example

```
N001 F1000. S500;
N002 M03; // start the drill, CW rotation
N003 G90;
N004 G00 X0. Y0. Z10.; // move to the initial point
N005 G17;
N006 G90 G99;
// specify the coordinate of R point, Z point and hole NO.1, cutting depth 3.0
N007 G83 X5. Y5. Z-10. R-5. Q3.;
N008 X15.; // hole NO.2
N009 Y15.; // hole NO.3
N010 G98 X5.; // hole NO.4, return to the initial point
N011 G80;
N012 M05; // stop the tool
N013 M30;
```

36 G85： Drilling Cycle

Command Form

G85 X_Y_Z_R_F_K_;

X(U) or Y(V): hole position(absolute/increment, please set parameter 3809 is 1 when using increment value)

Z:

G91: the distance from the initial point to Z point (directional)

G90: program coordinate of Z point

R:

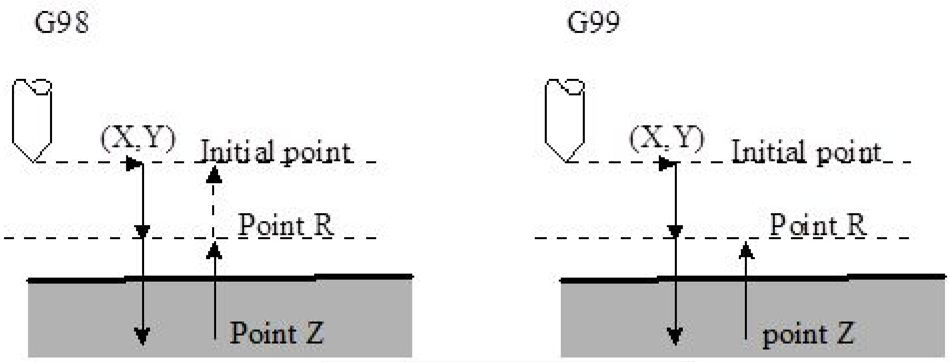
G91: the distance from the initial point to R point (directional)

G90: program coordinate of R point

F: feedrate

K: repeating times (repeating the moving and drilling, G91 increment input is effective)

Absolute/Increment coordinate of X, Y, Z, R can be specified by G90/G91.



36.1 Description

1. Start the machining process, move the tool to specified (X,Y) with G00.
2. Move the tool to the specified R point with G00.
3. Move the tool to the Z point at the bottom of the hole with G01.
4. Move the tool to the R point with G01.
5. Return to the initial point (G98) or the R point (G99) with G00.

36.2 Note

1. Please start spindle rotation with M code before applying G85.
2. If M Code and G85 are specified in the same block ,the M Code will only be executed once in the block.
3. When specified to repeat K times, M Code will only be executed at first cycle, it won't be executed for others.
4. Before changing the drilling axis, G85 cycle command must be cancelled first.
5. If no axis (X, Y, Z) moving command is included in the block, the drilling action won't be executed.
6. The argument specified R will only be set when executing the block with drilling action, it won't be set in the blocks with no drilling actions.

7. G codes G00, G01, G02, G03 can't be specified in the same block with G85, or the G85 cycle command will be cancelled.
8. In G85 cycle, the tool radius compensation (G41/G42/G40) will be ignored.
9. Arguments X,Y,Z,A,B,C,U,V,W are recognized as axis command when the parameter 3809 is 0.
10. The increment command for drilling axis is ignored when the increment and absolute command for drilling axis are commanded simultaneously.

36.3 Program Example

```
N001 F1000. S500;
N002 G90;
N003 G00 X0. Y0. Z10.; // move to the initial point
N004 G17;
N005 M03; // start the drill, CW rotation
N006 G90 G99; // specify the coordinate of R point, Z point and hole NO.1
N007 G85 X5. Y5. Z-10. R-5.;
N008 X15.; // hole NO.2
N009 Y15.; // hole NO.3
N010 G98 X5.; // hole NO.4, and return to initial point
N011 G80;
N012 M05; // stop the tool
N013 M30;
```

37 G86： High Speed Drilling Cycle

Command Form

G86 X_Y_Z_R_F_K_;

X(U) or Y(V): hole position(absolute/increment, please set parameter 3809 is 1 when using increment value)

Z:

G91: the distance from the initial point to Z point (directional)

G90: program coordinate of Z point

R:

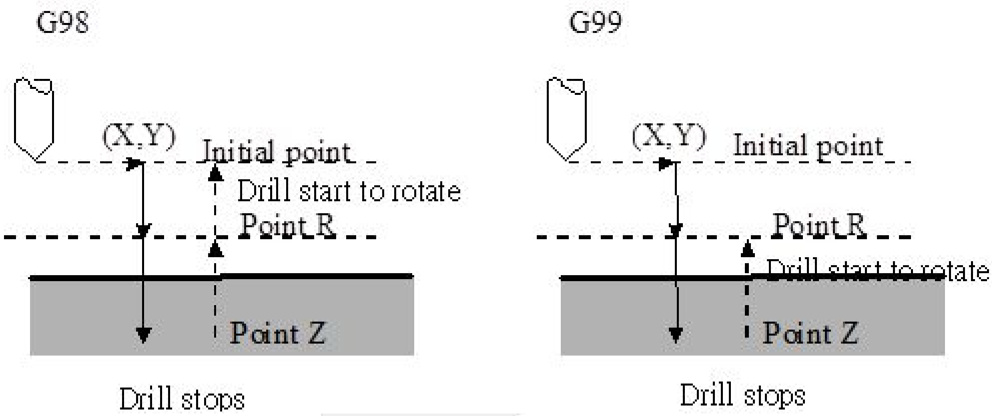
G91: the distance from the initial point to R point (directional)

G90: program coordinate of R point

F: feedrate

K: repeating times (repeating the moving and drilling, G91 increment input is effective)

Absolute/Increment coordinate of X, Y, Z, R can be specified by G90/G91.



37.1 Description

1. Start the machining process, move the tool to specified (X,Y) with G00.
2. Move the tool to the specified R point with G00.
3. Move the tool to the Z point at hole bottom with G01.
4. Return to the initial point (G98) or the R point of the program (G99) with G00.

37.2 Note

1. Please start spindle rotation with M code before applying G86.
2. If M Code and G86 are specified in the same block ,the M Code will only be executed once in the block.
3. When specified to repeat K times, M Code will only be executed at first cycle, it won't be executed for others.
4. Before changing the drilling axis, G86 cycle cycle must be cancelled first.
5. If no axis (X, Y, Z) moving command is included in the block, the drilling action won't be executed.
6. The argument specified by R will only be set when executing the block with drilling action, it won't be set in the blocks with no drilling actions.

7. G codes G00, G01, G02, G03 can't be specified in the same block with G86, or the G86 cycle command will be cancelled.
8. In G86 cycle, the tool radius compensation (G41/G42/G40) will be ignored.
9. Arguments X,Y,Z,A,B,C,U,V,W are recognized as axis command when the parameter 3809 is 0.
10. The increment command for drilling axis is ignored when the increment and absolute command for drilling axis are commanded simultaneously.

37.3 Program Example

```
N001 F1000. S500;
N002 G90;
N003 G00 X0. Y0. Z10.; // move to the initial point
N004 G17;
N005 M03; // start drill CW rotation
N006 G90 G99; //specify the coordinate of R point, Z point and hole NO.1
N007 G86 X5. Y5. Z-10. R-5.;
N008 X15.; // hole NO.2
N009 Y15.; // hole NO.3
N010 G98 X5.; // hole NO.4, return to initial point
N011 G80;
N012 M05; // stop the tool
N013 M30;
```

38 G90/G91： Absolute/Increment Command

Command Form

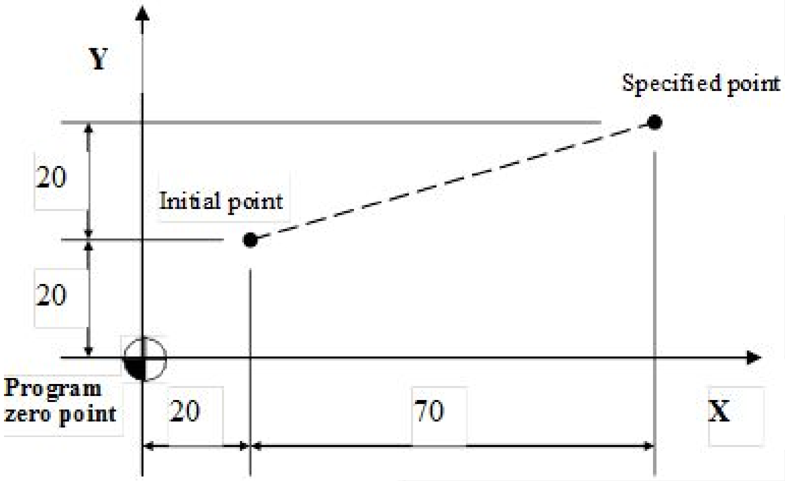
G90;

G91;

38.1 Description

G90: absolute command.

G91: incremental command.



38.2 Program Example

1. 1st way(absolute): G90 G00 X90.0 Y40.0；
   // execute the linear cutting with the distance from program zero point to the specified point
2. 2nd way(increment): G91 G00 X70.0 Y20.0；
   // execute the linear cutting with the distance from the starting point to the specified point

39 G92.1： Absolute Zero Point Coordinate System Setup II

39.1 Command Form

G92.1 X_ Y_ Z_ I_ J_ K_ R_
X, Y, Z： specified the coordinate as the zero point of program coordinate system;

I： take X axis as the rotation center and rotates the YZ plane.

J： take Y axis as the rotation center and rotates the XZ plane.

K： take Z axis as the rotation center and rotates the XY plane.

R： the rotation angle of the coordinate

39.2 Description

G92.1 is similar to G92, both are used to build new coordinate systems. This command sets the specified point (assigned by the command) of current coordinate system as the program zero point of the new sub-coordinate system.

After setup, the tool will start the machining at the specified point and all the absolute commands will be computed with this coordinate system.

39.2.1 Comparing between G92 and G92.1

Command	Description
G92 X20. Y15. Z20.	Set the current position as the **X20. Y15. Z20 of new coordinate system**
G92.1 X20.Y15. Z20.	Set the **X20. Y15. Z20. of current coordinate system** as the zero point of new coordinate system

39.3 Note

1. The Machine Coordinate of controller is obtained from the formula： Machine Coordinate = Workpiece Coordinate (G54~) + Program Coordinate + G92.1 Offset + External Offset + MPG Offset + Tool Length Compensation.
2. G92.1 Offset is equal to the argument X_, Y_, Z_ in command of G92.1.
3. Argument I_, J_, K_ in G92.1 = The axes of G92.1 coordinate system's rotation center.
4. Argument R in G92.1 = G92.1 coordinate system's rotation angle.
5. The axial macro-variables of G92.1 coordinate system's offset are #1901~1918.
6. The macro-variable of G92.1 coordinate system's rotation angle is #1930.
7. The axial macro-variables of G92.1 coordinate system's rotation center are #1931~#1933.
8. Please do not use G92 and G92.1 at the same time.

Program Example

39.3.1 Example 1, comparing between G92 and G92.1 (no external offset, tool length, tool wear compensation)

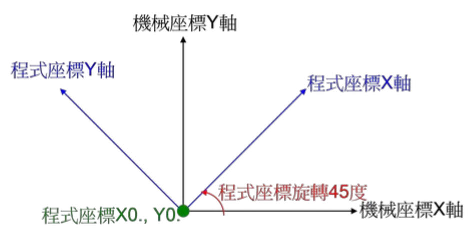
G92	G92.1

N1 G90 X10. Y10.
//machine coordinate X10. Y10.
//program coordinate X10. Y10
//#1901 #1902 coordinate X0. Y0.
N2 G92 X20. Y20.
//machine coordinate X10. Y10.
//program coordinate X20. Y20.
//#1901 #1902 coordinate X-10. Y-10.
N3 X50.
//machine coordinate X40. Y10.
//program coordinate X50. Y20.
//#1901 #1902 coordinate X-10. Y-10.
N4 M30
	N1 G90 X10. Y10.
	//machine coordinate X10. Y10.
	//program coordinate X10. Y10.
	//#1901 #1902 coordinate X0. Y0.
	N2 G92.1 X20. Y20.
	//machine coordinate X10. Y10.
	//program coordinate X-10. Y-10.
	//#1901 #1902 coordinate X20. Y20.
	N3 X50.
	//machine coordinate X70. Y10.
	//program coordinate X50. Y-10.
	//#1901 #1902 coordinate X20. Y20.
	N4 M30

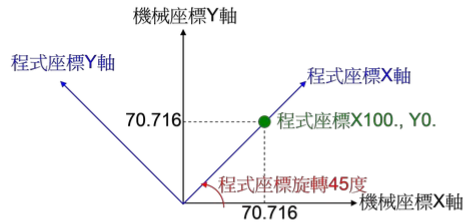
39.3.2 Example 2

Program Content	Figure

N1 G90 G0 X0. Y0.
//machine coordinate X0. Y0.
//program coordinate X0. Y0.
//#1901 #1902 coordinate X0. Y0.	

N2 G92.1 X0. Y0. K1. R45.
//machine coordinate X0. Y0.
//program coordinate X0. Y0.
//#1901 #1902 coordinate X0. Y0.
//XY plane rotates 45° with rotation center Z axis on program coordinate system, #1930 = 45°
	

Program Content	Figure

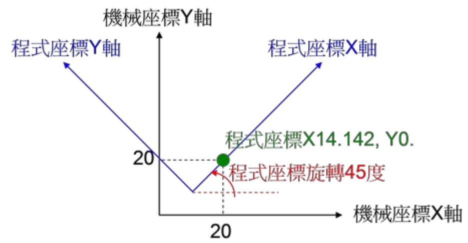
N3 G01 X100.
//machine coordinate X70.711 Y70.711
//program coordinate X100.000 Y0.000
//#1901 #1902 coordinate X0.000 Y0.000
	

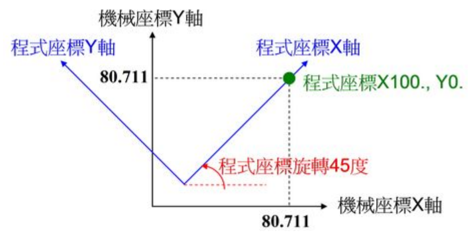
N4 M30	

39.3.3 Example 3

Program Content	Figure

N1 G90 G0 X20. Y20.
//machine coordinate X20. Y20.
//program coordinate X20. Y20.
//#1901 #1902 coordinate X0. Y0.	

N2 G92.1X10. Y10.K1.R45.
//machine coordinate X20. Y20.
//program coordinate X14.142 Y0.
//#1901 #1902 coordinate X10. Y10.
//XY plane rotates 45° with rotation center Z axis on program coordinate system, #1930 = 45°
	

N3 G01 X100. machine coordinate X80.711 Y80.711 program coordinate X100. Y0. #1901 #1902 coordinate X10. Y10.	
N4 M30	

40 G92： Absolute Zero Point Coordinate Setup I

Command Form

G92 X_ Y_ Z_;

X, Y, Z: set the program coordinate of current position

For example: The current program coordinate is X10, Y20, Z30, after running G92 X0 Y0 Z0, the program coordinate will be changed to X0, Y0, Z0.

40.1 Description

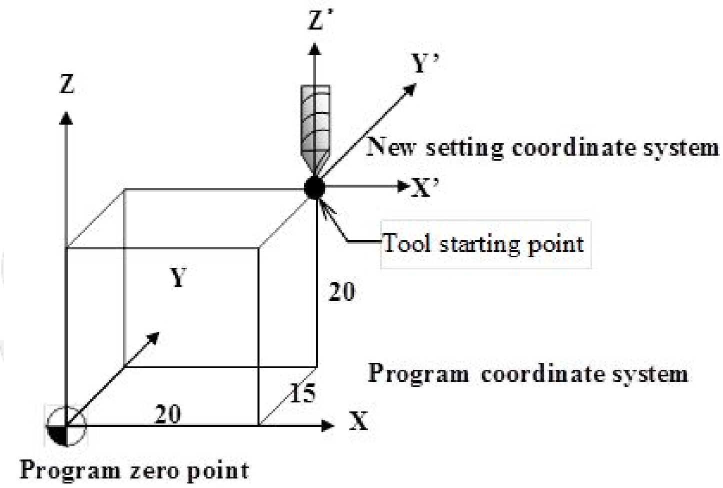
When designing a program, sometimes a new program zero point is required for special situations, thus the G92 command can be used to build a new coordinate system.

This command sets the current tool position as the specified point of the new sub-coordinate system. After setup, the tool will start the machining from this point, absolute commands will be computed with this new coordinate system.

40.2 Note

1. The Machine Coordinate of controller is obtained from the formula： Machine Coordinate = Workpiece Coordinate (G54~) + Program Coordinate + G92 Offset + External Offset + MPG Offset + Tool Length Compensation.
2. G92 Offsets are equal to Program Coordinate before running G92 minus Program Coordinate after running G92.
3. The axial macro-variables of G92 coordinate system's offset are #1901~1918.
4. Please do not use G92 and G92.1 at the same time.

40.3 Program Example



G90 // Absolute command

G01 X20.0 Y15.0 Z20.0 // machining to X20.0 Y15.0 Z20.0, the current program coordinate is X20.0 Y15.0 Z20.0

G92 X5.0 Y10.0 Z15.0; // change the program coordinate of current position to X5.0 Y10.0 Z15.0. G92 coordinate system offset value X=15, Y=5, Z=5. The offset value = original program coordinate - new program coordinate

G01 X25.0 Y25.0 Z35.0 // machining to new program coordinate X25.0 Y25.0 Z35.0

41 G94/G95： Feed Unit Setup

Command Form

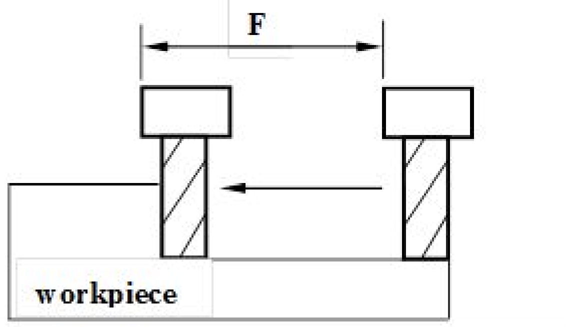
G94 F_;

G95 F_;

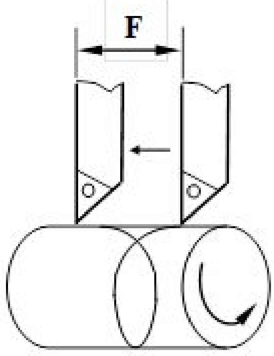
41.1 Description

This command set up the feed unit of F_function (tool moving distance per unit time or per revolution). G94 is feed per minute, unit: mm/min, inch/min; G95 is feed per revolution, unit: mm/rev, inch/rev.

41.2 Figure



G94. feed per minute(mm/min or inch/min)



G95. feed per revolution(mm/rev or inch/rev)